

Actividad - Proyecto práctico

Actividad en grupos de 4 alumnos. Nombres en orden alfabético (de apellidos):

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- **Alumno 2:** Larios Pérez, José María
- **Alumno 3:** Fernández Gordo, Javier
- **Alumno 4:** Carvajal Benítez, Pablo

Entorno: SpaceInvaders-v0 · **Algoritmo:** DQN (Stable-Baselines3) · **Requisito mínimo:** media de recompensa en modo test por encima del umbral indicado en el listado del grupo.

Repositorio del proyecto: https://github.com/pablocarvajal12/08_MIAR_GRUPO22

Proyecto 10 - DQN con Stable-Baselines3

Entrenamiento de un agente DQN para SpaceInvaders-v0 usando Stable-Baselines3.

```
In [ ]: %pip install gym==0.17.3
        %pip install git+https://github.com/Kojoley/atari-py.git
        %pip install stable-baselines3==0.10.0
```

```

Collecting gym==0.17.3
  Downloading gym-0.17.3.tar.gz (1.6 MB)
    _____ 0.0/1.6 MB ? eta -:-:--
    _____ 1.6/1.6 MB 88.4 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from gym==0.17.3) (1.16.3)
Requirement already satisfied: numpy>=1.10.4 in /usr/local/lib/python3.12/dist-packages (from gym==0.17.3) (2.0.2)
Collecting pygame<=1.5.0,>=1.4.0 (from gym==0.17.3)
  Downloading pygame-1.5.0-py2.py3-none-any.whl.metadata (7.6 kB)
Collecting cloudpickle<1.7.0,>=1.2.0 (from gym==0.17.3)
  Downloading cloudpickle-1.6.0-py3-none-any.whl.metadata (4.3 kB)
Requirement already satisfied: future in /usr/local/lib/python3.12/dist-packages (from pygame<=1.5.0,>=1.4.0->gym==0.17.3) (1.0.0)
Downloading cloudpickle-1.6.0-py3-none-any.whl (23 kB)
Downloading pygame-1.5.0-py2.py3-none-any.whl (1.0 MB)
    _____ 1.0/1.0 MB 75.4 MB/s eta 0:00:00
Building wheels for collected packages: gym
  Building wheel for gym (setup.py) ... done
  Created wheel for gym: filename=gym-0.17.3-py3-none-any.whl size=1654617 sha256=e505e11a63ce15147a25682a91ca848f40e094af3e06a672427a198fd55b5bee
  Stored in directory: /root/.cache/pip/wheels/99/7e/03/69cdeb432380e05bc9a56ff68451d8063e41aa06db508f4be5
Successfully built gym
Installing collected packages: pygame, cloudpickle, gym
  Attempting uninstall: cloudpickle
    Found existing installation: cloudpickle 3.1.2
    Uninstalling cloudpickle-3.1.2:
      Successfully uninstalled cloudpickle-3.1.2
  Attempting uninstall: gym
    Found existing installation: gym 0.25.2
    Uninstalling gym-0.25.2:
      Successfully uninstalled gym-0.25.2
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
distributed 2026.1.1 requires cloudpickle>=3.0.0, but you have cloudpickle 1.6.0 which is incompatible.
dask 2026.1.1 requires cloudpickle>=3.0.0, but you have cloudpickle 1.6.0 which is incompatible.
bigframes 2.42.0 requires cloudpickle>=2.0.0, but you have cloudpickle 1.6.0 which is incompatible.
Successfully installed cloudpickle-1.6.0 gym-0.17.3 pygame-1.5.0
Collecting git+https://github.com/Kojoley/atari-py.git
  Cloning https://github.com/Kojoley/atari-py.git to /tmp/pip-req-build-1xfjf4dk
  Running command git clone --filter=blob:none --quiet https://github.com/Kojoley/atari-py.git /tmp/pip-req-build-1xfjf4dk
  Resolved https://github.com/Kojoley/atari-py.git to commit 86a1e05c0a95e9e6233c3a413521fdb34ca8a089

```

```
Preparing metadata (setup.py) ... done
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from atari-py==1.2.2) (2.0.2)
Building wheels for collected packages: atari-py
  Building wheel for atari-py (setup.py) ... done
  Created wheel for atari-py: filename=atari_py-1.2.2-cp312-cp312-linux_x86_64.whl size=4739377 sha256=a79d5e0f149fe2bb755f5ad44f5c9cc3c5edcab3b52457c6529ccde07384945e
  Stored in directory: /tmp/pip-ephem-wheel-cache-zwyyxcx/wheels/cb/1f/56/35b393898f74ae3d978063738e8eeb4b3d7357d0d7eeec6b9f
Successfully built atari-py
Installing collected packages: atari-py
Successfully installed atari-py-1.2.2
Collecting stable-baselines3==0.10.0
  Downloading stable_baselines3-0.10.0-py3-none-any.whl.metadata (3.5 kB)
Requirement already satisfied: gym>=0.17 in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (0.17.3)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (2.0.2)
Requirement already satisfied: torch>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (2.11.0+cu128)
Requirement already satisfied: cloudpickle in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (1.6.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from stable-baselines3==0.10.0) (3.10.0)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from gym>=0.17->stable-baselines3==0.10.0) (1.16.3)
Requirement already satisfied: pygame<=1.5.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from gym>=0.17->stable-baselines3==0.10.0) (1.5.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.29.4)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (4.15.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (2025.3.0)
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cuspars,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.1)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-
```

baselines3==0.10.0) (12.9.7)
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (9.19.0.56)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (2.28.9)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.4.5)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.4.0->stable-baselines3==0.10.0) (3.6.0)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.4.1)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.90)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (11.3.3.83)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (1.13.1.3)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.90)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (11.7.3.90)
Requirement already satisfied: nvidia-cusparselt-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.5.8.93)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.93)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.93)

Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch>=1.4.0->stable-baselines3==0.10.0) (12.8.90)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (4.63.0)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (1.5.0)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (26.2)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (3.3.2)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->stable-baselines3==0.10.0) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->stable-baselines3==0.10.0) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->stable-baselines3==0.10.0) (2026.2)

Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings<13,>=12.9.4->torch>=1.4.0->stable-baselines3==0.10.0) (1.5.5)

Requirement already satisfied: future in /usr/local/lib/python3.12/dist-packages (from pygame<=1.5.0,>=1.4.0->gym>=0.17->stable-baselines3==0.10.0) (1.0.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->stable-baselines3==0.10.0) (1.17.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.4.0->stable-baselines3==0.10.0) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.4.0->stable-baselines3==0.10.0) (3.0.3)

Downloading stable_baselines3-0.10.0-py3-none-any.whl (145 kB)

146.0/146.0 kB 13.1 MB/s eta 0:00:00

Installing collected packages: stable-baselines3

Successfully installed stable-baselines3-0.10.0

```
In [2]: import gym
        from stable_baselines3 import DQN
```

```
ENV_ID = "SpaceInvaders-v0"
env = gym.make(ENV_ID)

print("Gym OK")
print("SB3 DQN OK")
```

Gym OK

SB3 DQN OK

```
In [ ]: from pathlib import Path
import json
import time
from collections import deque

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import gym
import torch as th
import torch.nn as nn

from google.colab import drive

from stable_baselines3 import DQN
from stable_baselines3.common.env_util import make_atari_env
from stable_baselines3.common.vec_env import VecFrameStack
from stable_baselines3.common.callbacks import BaseCallback
from stable_baselines3.common.torch_layers import BaseFeaturesExtractor
```

```
In [ ]: drive.mount('/content/gdrive')

drive_root = '/content/gdrive/My Drive/08_MIAR/sb3/conlogs'
PROJECT_NAME = 'sb3_1_spaceinvaders_v0'

OUTPUT_DIR = Path(drive_root) / 'datos' / PROJECT_NAME
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

# Modelos
MODEL_LATEST = OUTPUT_DIR / 'dqn_spaceinvaders_v0_sb3_latest.zip'
POLICY_LATEST = OUTPUT_DIR / 'dqn_spaceinvaders_v0_policy_latest.pt'
```

```

# Logs completos propios
LOG_FILENAME = OUTPUT_DIR / 'dqn_SpaceInvaders-v0_sb3_log.json'
CSV_LOG_FILENAME = OUTPUT_DIR / 'dqn_SpaceInvaders-v0_sb3_log.csv'

# Logs oficiales de SB3
TENSORBOARD_DIR = OUTPUT_DIR / 'tensorboard'
MONITOR_DIR = OUTPUT_DIR / 'monitor'
TENSORBOARD_DIR.mkdir(parents=True, exist_ok=True)
MONITOR_DIR.mkdir(parents=True, exist_ok=True)

print('OUTPUT_DIR:', OUTPUT_DIR)
print('MODEL_LATEST:', MODEL_LATEST)
print('POLICY_LATEST:', POLICY_LATEST)
print('LOG_FILENAME:', LOG_FILENAME)
print('CSV_LOG_FILENAME:', CSV_LOG_FILENAME)
print('TENSORBOARD_DIR:', TENSORBOARD_DIR)
print('MONITOR_DIR:', MONITOR_DIR)

```

Mounted at /content/gdrive

OUTPUT_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0

MODEL_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_sb3_latest.zip

POLICY_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_policy_latest.pt

LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json

CSV_LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv

TENSORBOARD_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/tensorboard

MONITOR_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/monitor

```

In [ ]: ENV_ID = 'SpaceInvaders-v0'
        SEED = 42

        TOTAL_TIMESTEPS = 10_000_000

        LEARNING_RATE = 1e-4
        BUFFER_SIZE = 100_000
        LEARNING_STARTS = 50_000
        BATCH_SIZE = 32
        GAMMA = 0.99
        TRAIN_FREQ = 4
        TARGET_UPDATE_INTERVAL = 10_000

```

```

EXPLORATION_INITIAL_EPS = 1.0
EXPLORATION_FINAL_EPS = 0.001
EXPLORATION_FRACTION = 0.30

CONSECUTIVE_THRESHOLD = 20.0
TARGET_CONSECUTIVE_EPISODES = 100

SAVE_FREQ_STEPS = 100_000
PRINT_EVERY_EPISODES = 10

LOAD_MODEL = False

```

```

In [ ]: test_env = gym.make(ENV_ID)
print('Entorno creado correctamente:', ENV_ID)
print('Action space:', test_env.action_space)
print('Observation space:', test_env.observation_space)
test_env.close()

```

Entorno creado correctamente: SpaceInvaders-v0

Action space: Discrete(6)

Observation space: Box(0, 255, (210, 160, 3), uint8)

```

In [ ]: def reset_env_compat(env):
    result = env.reset()
    if isinstance(result, tuple):
        return result[0]
    return result

raw_env = gym.make(ENV_ID)
obs = reset_env_compat(raw_env)

try:
    frame_rgb = raw_env.render(mode='rgb_array')
except TypeError:
    frame_rgb = raw_env.render()

plt.figure(figsize=(6, 6))
plt.imshow(frame_rgb)
plt.title('Imagen original del entorno')
plt.axis('off')
plt.show()

```

```
raw_env.close()
```

Imagen original del entorno



```
In [ ]: from stable_baselines3.common.monitor import Monitor
from stable_baselines3.common.atari_wrappers import AtariWrapper
from stable_baselines3.common.vec_env import DummyVecEnv

class StrictRewardClipWrapper(gym.RewardWrapper):
    def _init_(self, env):
        super()._init_(env)
```

```

def reward(self, reward):
    return float(np.clip(reward, -1.0, 1.0))

def create_env():
    env = gym.make(ENV_ID)
    env = StrictRewardClipWrapper(env)

    monitor_dir = OUTPUT_DIR / 'monitor'
    monitor_dir.mkdir(parents=True, exist_ok=True)
    env = Monitor(env, str(monitor_dir))

    env = AtariWrapper(env, clip_reward=False)

    env = DummyVecEnv([lambda: env])
    env = VecFrameStack(env, n_stack=4)
    return env

env = create_env()
print('Observation space SB3:', env.observation_space)
print('Action space SB3:', env.action_space)

```

Observation space SB3: Box(0, 255, (84, 84, 4), uint8)
 Action space SB3: Discrete(6)

```

In [ ]: obs = env.reset()
        print('Shape observación:', obs.shape)

        obs0 = obs[0]

        if obs0.shape[0] in [1, 3, 4, 12]:
            frames = obs0
        else:
            frames = np.transpose(obs0, (2, 0, 1))

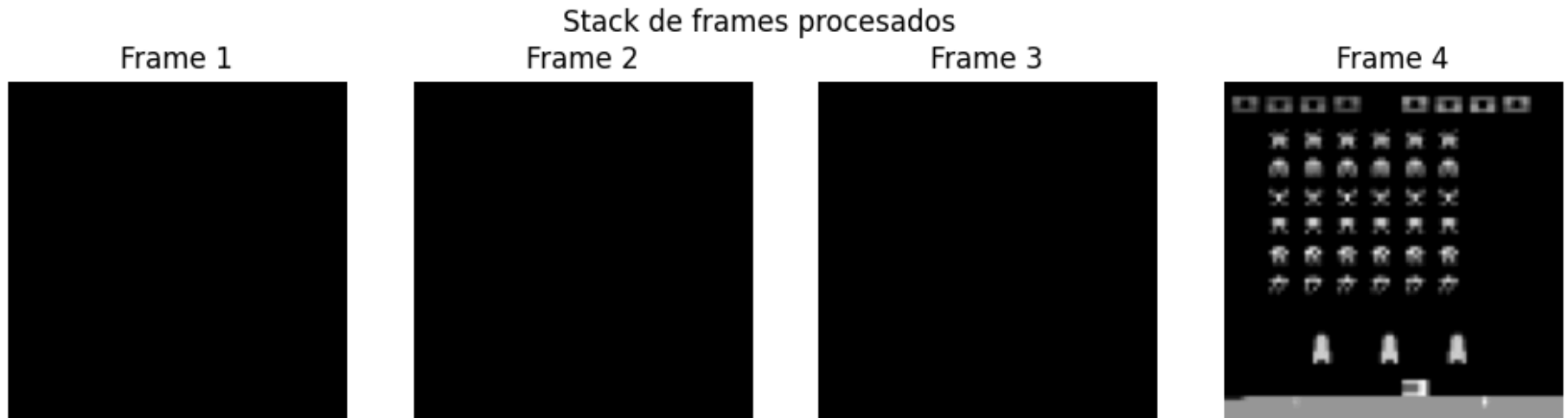
        num_frames = min(4, frames.shape[0])

        plt.figure(figsize=(12, 3))
        for i in range(num_frames):
            plt.subplot(1, num_frames, i + 1)
            plt.imshow(frames[i], cmap='gray')

```

```
plt.title(f'Frame {i + 1}')
plt.axis('off')
plt.suptitle('Stack de frames procesados')
plt.show()
```

Shape observación: (1, 84, 84, 4)



```
In [ ]: # Entrada: 4 x 84 x 84
# Conv2D 32 filtros 8x8 stride 4
# Conv2D 64 filtros 4x4 stride 2
# Conv2D 64 filtros 3x3 stride 1
# Capa densa de 512 características

class CustomCNN84(BaseFeaturesExtractor):
    def __init__(self, observation_space, features_dim=512):
        super().__init__(observation_space, features_dim)

        n_input_channels = observation_space.shape[0]

        self.cnn = nn.Sequential(
            nn.Conv2d(n_input_channels, 32, kernel_size=8, stride=4),
            nn.ReLU(),
            nn.Conv2d(32, 64, kernel_size=4, stride=2),
            nn.ReLU(),
            nn.Conv2d(64, 64, kernel_size=3, stride=1),
            nn.ReLU(),
```

```

        nn.Flatten()
    )

    with th.no_grad():
        sample = th.as_tensor(observation_space.sample()[None]).float()
        n_flatten = self.cnn(sample).shape[1]

    self.linear = nn.Sequential(
        nn.Linear(n_flatten, features_dim),
        nn.ReLU()
    )

    def forward(self, observations):
        return self.linear(self.cnn(observations))

policy_kwargs = dict(
    features_extractor_class=CustomCNN84,
    features_extractor_kwargs=dict(features_dim=512),
    net_arch=[]
)

```

```

In [ ]: def save_policy_checkpoint(model, path):
    path = Path(path)
    path.parent.mkdir(parents=True, exist_ok=True)

    th.save(
        {
            "policy_state_dict": model.policy.state_dict(),
            "optimizer_state_dict": model.policy.optimizer.state_dict(),
            "num_timesteps": model.num_timesteps,
            "exploration_rate": getattr(model, "exploration_rate", None),
        },
        path
    )

```

```

In [ ]: # Logger

class SB3TrainingLogger:
    def __init__(self, json_path, csv_path):
        self.json_path = Path(json_path)

```

```
self.csv_path = Path(csv_path)
self.json_path.parent.mkdir(parents=True, exist_ok=True)
self.csv_path.parent.mkdir(parents=True, exist_ok=True)

if self.json_path.exists():
    with open(self.json_path, 'r') as f:
        self.data = json.load(f)
    print('Log JSON previo cargado:', self.json_path)
else:
    self.data = {
        'episode': [],
        'total_steps': [],
        'episode_reward': [],
        'clipped_episode_reward': [],
        'consecutive_clipped_over_20': [],
        'max_consecutive_clipped_over_20': [],
        'avg_clip50': [],
        'duration': [],
        'exploration_rate': []
    }

def add_episode(self, episode, total_steps, reward, clipped_reward,
               consecutive, max_consecutive, avg_clip50, duration,
               exploration_rate):
    row = {
        'episode': int(episode),
        'total_steps': int(total_steps),
        'episode_reward': float(reward),
        'clipped_episode_reward': float(clipped_reward),
        'consecutive_clipped_over_20': int(consecutive),
        'max_consecutive_clipped_over_20': int(max_consecutive),
        'avg_clip50': float(avg_clip50),
        'duration': float(duration),
        'exploration_rate': float(exploration_rate) if exploration_rate is not None else np.nan
    }

    for key, value in row.items():
        self.data[key].append(value)

    pd.DataFrame([row]).to_csv(
        self.csv_path,
```

```

        mode='a',
        header=not self.csv_path.exists(),
        index=False
    )

    def save_json(self):
        with open(self.json_path, 'w') as f:
            json.dump(self.data, f, indent=2)

    def __len__(self):
        return len(self.data.get('episode', []))

class ConsecutiveRewardCallback(BaseCallback):
    def __init__(self, training_logger, verbose=0):
        super().__init__(verbose)
        self.training_logger = training_logger
        self.episode = len(training_logger)
        self.consecutive = 0
        self.max_consecutive = 0
        self.recent_clipped_rewards = deque(maxlen=50)
        self.start_time = time.time()
        self.last_json_save_step = 0

        previous_clipped = training_logger.data.get('clipped_episode_reward', [])[-50:]
        for value in previous_clipped:
            self.recent_clipped_rewards.append(float(value))

        previous_consec = training_logger.data.get('consecutive_clipped_over_20', [])
        if len(previous_consec) > 0:
            self.consecutive = int(previous_consec[-1])

        previous_max = training_logger.data.get('max_consecutive_clipped_over_20', [])
        if len(previous_max) > 0:
            self.max_consecutive = int(max(previous_max))

    def _on_step(self) -> bool:
        infos = self.locals.get('infos', [])

        for info in infos:
            if 'episode' not in info:

```

```
        continue

    ep_reward = float(info['episode']['r'])

    clipped_reward = ep_reward

    duration = time.time() - self.start_time
    self.start_time = time.time()

    self.recent_clipped_rewards.append(clipped_reward)

    if clipped_reward > CONSECUTIVE_THRESHOLD:
        self.consecutive += 1
    else:
        self.consecutive = 0

    if self.consecutive > self.max_consecutive:
        self.max_consecutive = self.consecutive

    avg_clip50 = (
        float(np.mean(self.recent_clipped_rewards))
        if len(self.recent_clipped_rewards) > 0
        else 0.0
    )

    exploration_rate = getattr(self.model, 'exploration_rate', np.nan)

    self.training_logger.add_episode(
        episode=self.episode,
        total_steps=self.num_timesteps,
        reward=ep_reward,
        clipped_reward=clipped_reward,
        consecutive=self.consecutive,
        max_consecutive=self.max_consecutive,
        avg_clip50=avg_clip50,
        duration=duration,
        exploration_rate=exploration_rate
    )

    # Métricas personalizadas para TensorBoard/SB3.
    self.logger.record('custom/clipped_episode_reward', clipped_reward)
```

```

self.logger.record('custom/consecutive_clipped_over_20', self.consecutive)
self.logger.record('custom/max_consecutive_clipped_over_20', self.max_consecutive)
self.logger.record('custom/avg_clip50', avg_clip50)

if self.episode % PRINT_EVERY_EPISODES == 0:
    print(
        f'EP={self.episode:,} | '
        f'STEP={self.num_timesteps:,} | '
        f'clip={clipped_reward:.0f} | '
        f'consec={self.consecutive} | '
        f'max_consec={self.max_consecutive} | '
        f'avg_clip50={avg_clip50:.2f}'
    )

    self.episode += 1

if self.max_consecutive >= TARGET_CONSECUTIVE_EPISODES:
    print('Objetivo conseguido: 100 episodios consecutivos superando el umbral.')
    self.training_logger.save_json()
    return False

if self.num_timesteps - self.last_json_save_step >= SAVE_FREQ_STEPS:
    self.training_logger.save_json()
    self.last_json_save_step = self.num_timesteps

return True

def _on_training_end(self):
    self.training_logger.save_json()
    print('Entrenamiento finalizado.')
    print('max_consec alcanzado:', self.max_consecutive)
    print('Log JSON:', LOG_FILENAME)
    print('Log CSV:', CSV_LOG_FILENAME)

```

```

In [ ]: if LOAD_MODEL and MODEL_LATEST.exists():
        print("Cargando modelo desde:", MODEL_LATEST)

        model = DQN.load(
            str(MODEL_LATEST),
            env=env,

```

```
        device="auto",
        tensorboard_log=str(TENSORBOARD_DIR)
    )

    print("Modelo cargado correctamente.")

else:
    print("Creando modelo nuevo.")

    model = DQN(
        policy="CnnPolicy",
        env=env,
        learning_rate=LEARNING_RATE,
        buffer_size=BUFFER_SIZE,
        learning_starts=LEARNING_STARTS,
        batch_size=BATCH_SIZE,
        gamma=GAMMA,
        train_freq=TRAIN_FREQ,
        target_update_interval=TARGET_UPDATE_INTERVAL,
        exploration_initial_eps=EXPLORATION_INITIAL_EPS,
        exploration_final_eps=EXPLORATION_FINAL_EPS,
        exploration_fraction=EXPLORATION_FRACTION,
        policy_kwargs=policy_kwargs,
        tensorboard_log=str(TENSORBOARD_DIR),
        verbose=1,
        seed=SEED,
        device="auto"
    )

    print("Modelo nuevo creado correctamente.")

print("Política usada:")
print(model.policy)
```

```
Creando modelo nuevo.
Using cuda device
Wrapping the env in a VecTransposeImage.
Modelo nuevo creado correctamente.
Política usada:
CnnPolicy(
  (q_net): QNetwork(
    (features_extractor): CustomCNN84(
      (cnn): Sequential(
        (0): Conv2d(4, 32, kernel_size=(8, 8), stride=(4, 4))
        (1): ReLU()
        (2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))
        (3): ReLU()
        (4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))
        (5): ReLU()
        (6): Flatten(start_dim=1, end_dim=-1)
      )
      (linear): Sequential(
        (0): Linear(in_features=3136, out_features=512, bias=True)
        (1): ReLU()
      )
    )
  )
  (q_net): Sequential(
    (0): Linear(in_features=512, out_features=6, bias=True)
  )
)
(q_net_target): QNetwork(
  (features_extractor): CustomCNN84(
    (cnn): Sequential(
      (0): Conv2d(4, 32, kernel_size=(8, 8), stride=(4, 4))
      (1): ReLU()
      (2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))
      (3): ReLU()
      (4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))
      (5): ReLU()
      (6): Flatten(start_dim=1, end_dim=-1)
    )
    (linear): Sequential(
      (0): Linear(in_features=3136, out_features=512, bias=True)
      (1): ReLU()
    )
  )
)
```

```
)  
(q_net): Sequential(  
  (0): Linear(in_features=512, out_features=6, bias=True)  
)  
)  
)
```

```
In [ ]: training_logger = SB3TrainingLogger(  
        json_path=LOG_FILENAME,  
        csv_path=CSV_LOG_FILENAME  
)  
  
callback = ConsecutiveRewardCallback(  
    training_logger=training_logger,  
    verbose=1  
)  
  
model.learn(  
    total_timesteps=TOTAL_TIMESTEPS,  
    callback=callback,  
    reset_num_timesteps=not LOAD_MODEL,  
    tb_log_name='DQN_SpaceInvaders_v0',  
    log_interval=100  
)
```

Se han truncado las últimas 5000 líneas del flujo de salida.

episodes	15800	
fps	256	
time_elapsed	3686	
total timesteps	947076	
train/		
learning_rate	0.0001	
loss	0.191	
n_updates	224268	

EP=5,310 | STEP=949,259 | clip=21 | consec=2 | max_consec=6 | avg_clip50=15.40
 EP=5,320 | STEP=950,999 | clip=6 | consec=0 | max_consec=6 | avg_clip50=14.70
 EP=5,330 | STEP=952,847 | clip=4 | consec=0 | max_consec=6 | avg_clip50=14.42

custom/		
avg_clip50	14.5	
clipped_episode_r...	5	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	812	
ep_rew_mean	14.7	
exploration rate	0.682	
time/		
episodes	15900	
fps	256	
time_elapsed	3713	
total timesteps	953603	
train/		
learning_rate	0.0001	
loss	0.451	
n_updates	225900	

EP=5,340 | STEP=954,887 | clip=22 | consec=1 | max_consec=6 | avg_clip50=15.28
 EP=5,350 | STEP=957,168 | clip=11 | consec=0 | max_consec=6 | avg_clip50=15.88
 EP=5,360 | STEP=959,445 | clip=9 | consec=0 | max_consec=6 | avg_clip50=15.98

custom/		
avg_clip50	16.4	
clipped_episode_r...	6	
consecutive_clipp...	0	

max_consecutive_c...	6
rollout/	
ep_len_mean	837
ep_rew_mean	15.6
exploration rate	0.68
time/	
episodes	16000
fps	256
time_elapsed	3744
total timesteps	961181
train/	
learning_rate	0.0001
loss	0.173
n_updates	227795

EP=5,370	STEP=961,331	clip=8	consec=0	max_consec=6	avg_clip50=16.48
EP=5,380	STEP=963,309	clip=5	consec=0	max_consec=6	avg_clip50=16.66
EP=5,390	STEP=965,362	clip=15	consec=0	max_consec=6	avg_clip50=17.02
EP=5,400	STEP=967,588	clip=27	consec=2	max_consec=6	avg_clip50=16.84

custom/	
avg_clip50	16.6
clipped_episode_r...	7
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	872
ep_rew_mean	16.3
exploration rate	0.678
time/	
episodes	16100
fps	256
time_elapsed	3773
total timesteps	968230
train/	
learning_rate	0.0001
loss	0.232
n_updates	229557

EP=5,410	STEP=969,334	clip=7	consec=0	max_consec=6	avg_clip50=15.60
EP=5,420	STEP=971,136	clip=6	consec=0	max_consec=6	avg_clip50=15.54

EP=5,430 | STEP=973,211 | clip=28 | consec=1 | max_consec=6 | avg_clip50=15.86

custom/		
avg_clip50	14.4	
clipped_episode_r...	5	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	844	
ep_rew_mean	15.9	
exploration rate	0.676	
time/		
episodes	16200	
fps	256	
time_elapsed	3797	
total timesteps	974153	
train/		
learning_rate	0.0001	
loss	0.373	
n_updates	231038	

EP=5,440 | STEP=974,581 | clip=5 | consec=0 | max_consec=6 | avg_clip50=14.28
 EP=5,450 | STEP=976,300 | clip=8 | consec=0 | max_consec=6 | avg_clip50=13.52
 EP=5,460 | STEP=978,335 | clip=13 | consec=0 | max_consec=6 | avg_clip50=14.56
 EP=5,470 | STEP=980,387 | clip=23 | consec=1 | max_consec=6 | avg_clip50=14.88

custom/		
avg_clip50	14.9	
clipped_episode_r...	23	
consecutive_clipp...	1	
max_consecutive_c...	6	
rollout/		
ep_len_mean	809	
ep_rew_mean	15.2	
exploration rate	0.674	
time/		
episodes	16300	
fps	256	
time_elapsed	3824	
total timesteps	980455	
train/		

	learning_rate		0.0001	
	loss		0.0354	
	n_updates		232613	

EP=5,480		STEP=982,627		clip=11		consec=0		max_consec=6		avg_clip50=15.10
EP=5,490		STEP=984,404		clip=3		consec=0		max_consec=6		avg_clip50=15.76
EP=5,500		STEP=986,445		clip=22		consec=1		max_consec=6		avg_clip50=16.02

	custom/			
	avg_clip50		15.8	
	clipped_episode_r...		8	
	consecutive_clipp...		0	
	max_consecutive_c...		6	
	rollout/			
	ep_len_mean		802	
	ep_rew_mean		14.9	
	exploration rate		0.671	
	time/			
	episodes		16400	
	fps		256	
	time_elapsed		3852	
	total timesteps		987257	
	train/			
	learning_rate		0.0001	
	loss		0.459	
	n_updates		234314	

EP=5,510		STEP=988,287		clip=9		consec=0		max_consec=6		avg_clip50=15.26
EP=5,520		STEP=989,891		clip=7		consec=0		max_consec=6		avg_clip50=14.44
EP=5,530		STEP=992,288		clip=13		consec=0		max_consec=6		avg_clip50=14.66

	custom/			
	avg_clip50		14.8	
	clipped_episode_r...		19	
	consecutive_clipp...		0	
	max_consecutive_c...		6	
	rollout/			
	ep_len_mean		835	
	ep_rew_mean		15.4	
	exploration rate		0.669	
	time/			

	episodes		16500	
	fps		256	
	time_elapsed		3881	
	total timesteps		993994	
	train/			
	learning_rate		0.0001	
	loss		0.094	
	n_updates		235998	

```

-----
EP=5,540 | STEP=994,730 | clip=39 | consec=2 | max_consec=6 | avg_clip50=15.76
EP=5,550 | STEP=996,802 | clip=21 | consec=1 | max_consec=6 | avg_clip50=15.92
EP=5,560 | STEP=998,706 | clip=22 | consec=1 | max_consec=6 | avg_clip50=16.06
EP=5,570 | STEP=1,000,771 | clip=20 | consec=0 | max_consec=6 | avg_clip50=17.06

```

	custom/			
	avg_clip50		16.7	
	clipped_episode_r...		11	
	consecutive_clipp...		0	
	max_consecutive_c...		6	
	rollout/			
	ep_len_mean		856	
	ep_rew_mean		15.7	
	exploration rate		0.667	
	time/			
	episodes		16600	
	fps		256	
	time_elapsed		3910	
	total timesteps		1001128	
	train/			
	learning_rate		0.0001	
	loss		0.449	
	n_updates		237781	

```

-----
EP=5,580 | STEP=1,002,683 | clip=23 | consec=1 | max_consec=6 | avg_clip50=16.22
EP=5,590 | STEP=1,004,618 | clip=19 | consec=0 | max_consec=6 | avg_clip50=15.62
EP=5,600 | STEP=1,006,644 | clip=35 | consec=1 | max_consec=6 | avg_clip50=15.64

```

	custom/			
	avg_clip50		16.3	
	clipped_episode_r...		27	
	consecutive_clipp...		1	

max_consecutive_c...	6	
rollout/		
ep_len_mean	865	
ep_rew_mean	16	
exploration rate	0.664	
time/		
episodes	16700	
fps	255	
time_elapsed	3939	
total timesteps	1008240	
train/		
learning_rate	0.0001	
loss	0.0348	
n_updates	239559	

EP=5,610	STEP=1,008,819	clip=18	consec=0	max_consec=6	avg_clip50=16.10
EP=5,620	STEP=1,010,915	clip=24	consec=1	max_consec=6	avg_clip50=15.76
EP=5,630	STEP=1,013,216	clip=2	consec=0	max_consec=6	avg_clip50=16.66
EP=5,640	STEP=1,015,452	clip=4	consec=0	max_consec=6	avg_clip50=16.40

custom/		
avg_clip50	16.4	
clipped_episode_r...	4	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	874	
ep_rew_mean	16	
exploration rate	0.662	
time/		
episodes	16800	
fps	255	
time_elapsed	3969	
total timesteps	1015474	
train/		
learning_rate	0.0001	
loss	0.0497	
n_updates	241368	

EP=5,650	STEP=1,017,162	clip=7	consec=0	max_consec=6	avg_clip50=15.50
EP=5,660	STEP=1,018,997	clip=19	consec=0	max_consec=6	avg_clip50=14.94

EP=5,670 | STEP=1,020,963 | clip=26 | consec=1 | max_consec=6 | avg_clip50=15.12

custom/		
avg_clip50	14.6	
clipped_episode_r...	17	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	860	
ep_rew_mean	15.6	
exploration rate	0.66	
time/		
episodes	16900	
fps	255	
time_elapsed	3995	
total timesteps	1021755	
train/		
learning_rate	0.0001	
loss	0.0864	
n_updates	242938	

EP=5,680 | STEP=1,023,025 | clip=30 | consec=2 | max_consec=6 | avg_clip50=14.74

EP=5,690 | STEP=1,025,258 | clip=9 | consec=0 | max_consec=6 | avg_clip50=15.32

EP=5,700 | STEP=1,027,459 | clip=23 | consec=1 | max_consec=6 | avg_clip50=16.62

custom/		
avg_clip50	17.3	
clipped_episode_r...	11	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	871	
ep_rew_mean	16	
exploration rate	0.657	
time/		
episodes	17000	
fps	255	
time_elapsed	4025	
total timesteps	1029055	
train/		
learning_rate	0.0001	

loss	0.0418				
n_updates	244763				

EP=5,710	STEP=1,029,397	clip=11	consec=0	max_consec=6	avg_clip50=16.92
EP=5,720	STEP=1,031,590	clip=11	consec=0	max_consec=6	avg_clip50=16.78
EP=5,730	STEP=1,033,719	clip=4	consec=0	max_consec=6	avg_clip50=16.82
EP=5,740	STEP=1,036,268	clip=23	consec=1	max_consec=6	avg_clip50=17.60

custom/					
avg_clip50	17.7				
clipped_episode_r...	26				
consecutive_clipp...	2				
max_consecutive_c...	6				
rollout/					
ep_len_mean	884				
ep_rew_mean	16.7				
exploration rate	0.655				
time/					
episodes	17100				
fps	255				
time_elapsed	4056				
total timesteps	1036725				
train/					
learning_rate	0.0001				
loss	0.0306				
n_updates	246681				

EP=5,750	STEP=1,038,444	clip=18	consec=0	max_consec=6	avg_clip50=17.26
EP=5,760	STEP=1,040,374	clip=3	consec=0	max_consec=6	avg_clip50=17.02
EP=5,770	STEP=1,042,662	clip=16	consec=0	max_consec=6	avg_clip50=17.82

custom/					
avg_clip50	17.6				
clipped_episode_r...	20				
consecutive_clipp...	0				
max_consecutive_c...	6				
rollout/					
ep_len_mean	917				
ep_rew_mean	17.3				
exploration rate	0.652				
time/					

	episodes	17200	
	fps	255	
	time_elapsed	4085	
	total timesteps	1044014	
	train/		
	learning_rate	0.0001	
	loss	0.153	
	n_updates	248503	

```

-----
EP=5,780 | STEP=1,044,729 | clip=16 | consec=0 | max_consec=6 | avg_clip50=17.54
EP=5,790 | STEP=1,046,266 | clip=13 | consec=0 | max_consec=6 | avg_clip50=15.46
EP=5,800 | STEP=1,048,106 | clip=13 | consec=0 | max_consec=6 | avg_clip50=14.56
EP=5,810 | STEP=1,050,724 | clip=18 | consec=0 | max_consec=6 | avg_clip50=16.04

```

	custom/		
	avg_clip50	16	
	clipped_episode_r...	18	
	consecutive_clipp...	0	
	max_consecutive_c...	6	
	rollout/		
	ep_len_mean	896	
	ep_rew_mean	16.5	
	exploration rate	0.65	
	time/		
	episodes	17300	
	fps	255	
	time_elapsed	4113	
	total timesteps	1050724	
	train/		
	learning_rate	0.0001	
	loss	0.0635	
	n_updates	250180	

```

-----
EP=5,820 | STEP=1,052,568 | clip=18 | consec=0 | max_consec=6 | avg_clip50=15.04
EP=5,830 | STEP=1,054,673 | clip=13 | consec=0 | max_consec=6 | avg_clip50=15.58
EP=5,840 | STEP=1,056,943 | clip=20 | consec=0 | max_consec=6 | avg_clip50=16.80

```

	custom/		
	avg_clip50	17.3	
	clipped_episode_r...	12	
	consecutive_clipp...	0	

max_consecutive_c...	6
rollout/	
ep_len_mean	870
ep_rew_mean	16.1
exploration rate	0.648
time/	
episodes	17400
fps	255
time_elapsed	4142
total timesteps	1057688
train/	
learning_rate	0.0001
loss	0.0692
n_updates	251921

EP=5,850	STEP=1,058,782	clip=15	consec=0	max_consec=6	avg_clip50=17.42
EP=5,860	STEP=1,061,065	clip=34	consec=3	max_consec=6	avg_clip50=17.42
EP=5,870	STEP=1,063,575	clip=25	consec=1	max_consec=6	avg_clip50=18.54

custom/	
avg_clip50	17.8
clipped_episode_r...	8
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	872
ep_rew_mean	16.7
exploration rate	0.645
time/	
episodes	17500
fps	255
time_elapsed	4171
total timesteps	1064969
train/	
learning_rate	0.0001
loss	0.338
n_updates	253742

EP=5,880	STEP=1,065,454	clip=30	consec=1	max_consec=6	avg_clip50=17.98
EP=5,890	STEP=1,067,673	clip=18	consec=0	max_consec=6	avg_clip50=17.74
EP=5,900	STEP=1,069,785	clip=13	consec=0	max_consec=6	avg_clip50=18.22

EP=5,910 | STEP=1,071,995 | clip=20 | consec=0 | max_consec=6 | avg_clip50=17.50

custom/		
avg_clip50	17.5	
clipped_episode_r...	20	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	897	
ep_rew_mean	17.5	
exploration rate	0.643	
time/		
episodes	17600	
fps	255	
time_elapsed	4201	
total timesteps	1072397	
train/		
learning_rate	0.0001	
loss	0.045	
n_updates	255599	

EP=5,920 | STEP=1,074,365 | clip=24 | consec=2 | max_consec=6 | avg_clip50=17.18

EP=5,930 | STEP=1,076,747 | clip=20 | consec=0 | max_consec=6 | avg_clip50=18.26

EP=5,940 | STEP=1,079,385 | clip=9 | consec=0 | max_consec=6 | avg_clip50=19.24

custom/		
avg_clip50	19.1	
clipped_episode_r...	22	
consecutive_clipp...	1	
max_consecutive_c...	6	
rollout/		
ep_len_mean	946	
ep_rew_mean	18.5	
exploration rate	0.64	
time/		
episodes	17700	
fps	255	
time_elapsed	4233	
total timesteps	1080329	
train/		
learning_rate	0.0001	

loss	0.0468
n_updates	257582

```
-----
EP=5,950 | STEP=1,081,446 | clip=15 | consec=0 | max_consec=6 | avg_clip50=19.34
EP=5,960 | STEP=1,083,234 | clip=16 | consec=0 | max_consec=6 | avg_clip50=18.48
EP=5,970 | STEP=1,085,640 | clip=27 | consec=1 | max_consec=6 | avg_clip50=19.02
```

custom/	
avg_clip50	18.7
clipped_episode_r...	6
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	947
ep_rew_mean	18.6
exploration rate	0.638
time/	
episodes	17800
fps	255
time_elapsed	4264
total timesteps	1087664
train/	
learning_rate	0.0001
loss	0.0505
n_updates	259415

```
-----
EP=5,980 | STEP=1,087,776 | clip=4 | consec=0 | max_consec=6 | avg_clip50=18.34
EP=5,990 | STEP=1,090,275 | clip=25 | consec=3 | max_consec=6 | avg_clip50=18.26
EP=6,000 | STEP=1,092,256 | clip=32 | consec=1 | max_consec=6 | avg_clip50=17.84
EP=6,010 | STEP=1,094,526 | clip=29 | consec=1 | max_consec=6 | avg_clip50=18.80
```

custom/	
avg_clip50	18.7
clipped_episode_r...	6
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	940
ep_rew_mean	18.4
exploration rate	0.635
time/	

episodes	17900	
fps	254	
time_elapsed	4294	
total timesteps	1095042	
train/		
learning_rate	0.0001	
loss	0.281	
n_updates	261260	

```
-----
EP=6,020 | STEP=1,096,522 | clip=15 | consec=0 | max_consec=6 | avg_clip50=17.70
EP=6,030 | STEP=1,098,934 | clip=12 | consec=0 | max_consec=6 | avg_clip50=18.34
EP=6,040 | STEP=1,100,613 | clip=7 | consec=0 | max_consec=6 | avg_clip50=16.60
-----
```

custom/		
avg_clip50	16.8	
clipped_episode_r...	11	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	890	
ep_rew_mean	17.4	
exploration rate	0.633	
time/		
episodes	18000	
fps	254	
time_elapsed	4322	
total timesteps	1101789	
train/		
learning_rate	0.0001	
loss	0.253	
n_updates	262947	

```
-----
EP=6,050 | STEP=1,102,609 | clip=22 | consec=2 | max_consec=6 | avg_clip50=16.78
EP=6,060 | STEP=1,104,913 | clip=14 | consec=0 | max_consec=6 | avg_clip50=16.82
EP=6,070 | STEP=1,107,065 | clip=28 | consec=2 | max_consec=6 | avg_clip50=17.26
EP=6,080 | STEP=1,109,320 | clip=29 | consec=3 | max_consec=6 | avg_clip50=17.00
-----
```

custom/		
avg_clip50	17	
clipped_episode_r...	29	
consecutive_clipp...	3	

	max_consecutive_c...	6	
	rollout/		
	ep_len_mean	906	
	ep_rew_mean	17.7	
	exploration rate	0.631	
	time/		
	episodes	18100	
	fps	254	
	time_elapsed	4353	
	total timesteps	1109385	
	train/		
	learning_rate	0.0001	
	loss	0.292	
	n_updates	264846	

EP=6,090		STEP=1,111,199		clip=19		consec=0		max_consec=6		avg_clip50=17.88
EP=6,100		STEP=1,113,394		clip=3		consec=0		max_consec=6		avg_clip50=18.26
EP=6,110		STEP=1,115,490		clip=18		consec=0		max_consec=6		avg_clip50=17.92

	custom/		
	avg_clip50	18.3	
	clipped_episode_r...	29	
	consecutive_clipp...	2	
	max_consecutive_c...	6	
	rollout/		
	ep_len_mean	900	
	ep_rew_mean	17.8	
	exploration rate	0.628	
	time/		
	episodes	18200	
	fps	254	
	time_elapsed	4383	
	total timesteps	1116694	
	train/		
	learning_rate	0.0001	
	loss	0.048	
	n_updates	266673	

EP=6,120		STEP=1,117,799		clip=14		consec=0		max_consec=6		avg_clip50=18.02
EP=6,130		STEP=1,120,302		clip=33		consec=1		max_consec=6		avg_clip50=18.18
EP=6,140		STEP=1,122,275		clip=15		consec=0		max_consec=6		avg_clip50=18.54

custom/		
avg_clip50	17.6	
clipped_episode_r...	17	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	912	
ep_rew_mean	18.2	
exploration rate	0.626	
time/		
episodes	18300	
fps	254	
time_elapsed	4413	
total timesteps	1123835	
train/		
learning_rate	0.0001	
loss	0.033	
n_updates	268458	

```

EP=6,150 | STEP=1,124,176 | clip=28 | consec=1 | max_consec=6 | avg_clip50=17.96
EP=6,160 | STEP=1,126,550 | clip=28 | consec=3 | max_consec=6 | avg_clip50=18.66
EP=6,170 | STEP=1,128,790 | clip=4 | consec=0 | max_consec=6 | avg_clip50=18.96
EP=6,180 | STEP=1,130,906 | clip=12 | consec=0 | max_consec=6 | avg_clip50=18.34

```

custom/		
avg_clip50	18.1	
clipped_episode_r...	7	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	913	
ep_rew_mean	18.4	
exploration rate	0.623	
time/		
episodes	18400	
fps	254	
time_elapsed	4444	
total timesteps	1131342	
train/		
learning_rate	0.0001	

loss	0.0517
n_updates	270335

```

-----
EP=6,190 | STEP=1,133,497 | clip=23 | consec=1 | max_consec=6 | avg_clip50=19.22
EP=6,200 | STEP=1,135,401 | clip=12 | consec=0 | max_consec=6 | avg_clip50=19.02
EP=6,210 | STEP=1,137,184 | clip=15 | consec=0 | max_consec=6 | avg_clip50=18.06

```

custom/	
avg_clip50	17.4
clipped_episode_r...	24
consecutive_clipp...	1
max_consecutive_c...	6
rollout/	
ep_len_mean	904
ep_rew_mean	18.1
exploration rate	0.621
time/	
episodes	18500
fps	254
time_elapsed	4474
total timesteps	1138696
train/	
learning_rate	0.0001
loss	0.0618
n_updates	272173

```

-----
EP=6,220 | STEP=1,139,352 | clip=29 | consec=1 | max_consec=6 | avg_clip50=17.68
EP=6,230 | STEP=1,141,447 | clip=20 | consec=0 | max_consec=6 | avg_clip50=17.56
EP=6,240 | STEP=1,143,078 | clip=7 | consec=0 | max_consec=6 | avg_clip50=15.54
EP=6,250 | STEP=1,145,086 | clip=18 | consec=0 | max_consec=6 | avg_clip50=15.74

```

custom/	
avg_clip50	15.7
clipped_episode_r...	18
consecutive_clipp...	0
max_consecutive_c...	6
rollout/	
ep_len_mean	882
ep_rew_mean	17.4
exploration rate	0.619
time/	

episodes	18600	
fps	254	
time_elapsed	4502	
total timesteps	1145222	
train/		
learning_rate	0.0001	
loss	0.0192	
n_updates	273805	

```
-----
EP=6,260 | STEP=1,147,504 | clip=10 | consec=0 | max_consec=6 | avg_clip50=16.62
EP=6,270 | STEP=1,149,806 | clip=33 | consec=2 | max_consec=6 | avg_clip50=17.04
EP=6,280 | STEP=1,152,034 | clip=23 | consec=1 | max_consec=6 | avg_clip50=18.02
-----
```

custom/		
avg_clip50	18	
clipped_episode_r...	6	
consecutive_clipp...	0	
max_consecutive_c...	6	
rollout/		
ep_len_mean	890	
ep_rew_mean	17.8	
exploration rate	0.616	
time/		
episodes	18700	
fps	254	
time_elapsed	4533	
total timesteps	1152822	
train/		
learning_rate	0.0001	
loss	0.359	
n_updates	275705	

```
-----
EP=6,290 | STEP=1,154,052 | clip=27 | consec=3 | max_consec=6 | avg_clip50=19.08
EP=6,300 | STEP=1,156,344 | clip=14 | consec=0 | max_consec=6 | avg_clip50=20.08
EP=6,310 | STEP=1,158,360 | clip=25 | consec=3 | max_consec=6 | avg_clip50=19.28
-----
```

custom/		
avg_clip50	19.9	
clipped_episode_r...	15	
consecutive_clipp...	0	
max_consecutive_c...	8	

rollout/	
ep_len_mean	905
ep_rew_mean	18.3
exploration rate	0.614
time/	
episodes	18800
fps	254
time_elapsed	4565
total timesteps	1160408
train/	
learning_rate	0.0001
loss	0.0449
n_updates	277601

```

-----
EP=6,320 | STEP=1,160,683 | clip=8 | consec=0 | max_consec=8 | avg_clip50=19.08
EP=6,330 | STEP=1,162,650 | clip=12 | consec=0 | max_consec=8 | avg_clip50=17.74
EP=6,340 | STEP=1,164,874 | clip=11 | consec=0 | max_consec=8 | avg_clip50=17.78
EP=6,350 | STEP=1,167,504 | clip=29 | consec=3 | max_consec=8 | avg_clip50=18.14
-----

```

custom/	
avg_clip50	18.2
clipped_episode_r...	19
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	936
ep_rew_mean	19.1
exploration rate	0.611
time/	
episodes	18900
fps	254
time_elapsed	4596
total timesteps	1167961
train/	
learning_rate	0.0001
loss	0.246
n_updates	279490

```

-----
EP=6,360 | STEP=1,169,457 | clip=20 | consec=0 | max_consec=8 | avg_clip50=18.14
EP=6,370 | STEP=1,171,742 | clip=7 | consec=0 | max_consec=8 | avg_clip50=18.16
EP=6,380 | STEP=1,173,943 | clip=31 | consec=2 | max_consec=8 | avg_clip50=19.22
-----

```

custom/		
avg_clip50	19.2	
clipped_episode_r...	22	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	923	
ep_rew_mean	18.4	
exploration rate	0.609	
time/		
episodes	19000	
fps	253	
time_elapsed	4626	
total timesteps	1175053	
train/		
learning_rate	0.0001	
loss	0.091	
n_updates	281263	

```

EP=6,390 | STEP=1,175,680 | clip=13 | consec=0 | max_consec=8 | avg_clip50=18.52
EP=6,400 | STEP=1,177,973 | clip=20 | consec=0 | max_consec=8 | avg_clip50=17.60
EP=6,410 | STEP=1,180,425 | clip=25 | consec=3 | max_consec=8 | avg_clip50=18.94
EP=6,420 | STEP=1,182,533 | clip=19 | consec=0 | max_consec=8 | avg_clip50=18.42

```

custom/		
avg_clip50	18.3	
clipped_episode_r...	7	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	918	
ep_rew_mean	18.2	
exploration rate	0.606	
time/		
episodes	19100	
fps	253	
time_elapsed	4657	
total timesteps	1182660	
train/		
learning_rate	0.0001	

loss	0.407
n_updates	283164

```
-----
EP=6,430 | STEP=1,184,929 | clip=5 | consec=0 | max_consec=8 | avg_clip50=18.44
EP=6,440 | STEP=1,187,102 | clip=16 | consec=0 | max_consec=8 | avg_clip50=19.34
EP=6,450 | STEP=1,188,962 | clip=17 | consec=0 | max_consec=8 | avg_clip50=18.56
-----
```

custom/	
avg_clip50	17.7
clipped_episode_r...	8
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	909
ep_rew_mean	18
exploration rate	0.604
time/	
episodes	19200
fps	253
time_elapsed	4687
total timesteps	1190049
train/	
learning_rate	0.0001
loss	0.0818
n_updates	285012

```
-----
EP=6,460 | STEP=1,191,000 | clip=33 | consec=1 | max_consec=8 | avg_clip50=17.28
EP=6,470 | STEP=1,193,027 | clip=24 | consec=2 | max_consec=8 | avg_clip50=17.16
EP=6,480 | STEP=1,195,201 | clip=24 | consec=1 | max_consec=8 | avg_clip50=16.32
-----
```

custom/	
avg_clip50	15.9
clipped_episode_r...	28
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	915
ep_rew_mean	17.6
exploration rate	0.601
time/	
episodes	19300

fps	253
time_elapsed	4717
total timesteps	1197291
train/	
learning_rate	0.0001
loss	0.152
n_updates	286822

```

-----
EP=6,490 | STEP=1,197,653 | clip=27 | consec=4 | max_consec=8 | avg_clip50=16.32
EP=6,500 | STEP=1,199,836 | clip=9 | consec=0 | max_consec=8 | avg_clip50=17.08
EP=6,510 | STEP=1,202,027 | clip=18 | consec=0 | max_consec=8 | avg_clip50=17.80
EP=6,520 | STEP=1,204,186 | clip=16 | consec=0 | max_consec=8 | avg_clip50=18.12

```

custom/	
avg_clip50	17.9
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	916
ep_rew_mean	17.7
exploration rate	0.599
time/	
episodes	19400
fps	253
time_elapsed	4747
total timesteps	1204582
train/	
learning_rate	0.0001
loss	0.0469
n_updates	288645

```

-----
EP=6,530 | STEP=1,206,294 | clip=8 | consec=0 | max_consec=8 | avg_clip50=18.64
EP=6,540 | STEP=1,208,278 | clip=16 | consec=0 | max_consec=8 | avg_clip50=18.30
EP=6,550 | STEP=1,210,940 | clip=12 | consec=0 | max_consec=8 | avg_clip50=19.10

```

custom/	
avg_clip50	19.1
clipped_episode_r...	20
consecutive_clipp...	0
max_consecutive_c...	8

rollout/	
ep_len_mean	924
ep_rew_mean	18.3
exploration rate	0.596
time/	
episodes	19500
fps	253
time_elapsed	4779
total timesteps	1212131
train/	
learning_rate	0.0001
loss	0.124
n_updates	290532

```

-----
EP=6,560 | STEP=1,213,091 | clip=14 | consec=0 | max_consec=8 | avg_clip50=18.70
EP=6,570 | STEP=1,215,322 | clip=3 | consec=0 | max_consec=8 | avg_clip50=18.98
EP=6,580 | STEP=1,217,433 | clip=21 | consec=1 | max_consec=8 | avg_clip50=18.64
EP=6,590 | STEP=1,219,470 | clip=9 | consec=0 | max_consec=8 | avg_clip50=18.88
-----

```

custom/	
avg_clip50	18.9
clipped_episode_r...	9
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	920
ep_rew_mean	18.6
exploration rate	0.594
time/	
episodes	19600
fps	253
time_elapsed	4809
total timesteps	1219502
train/	
learning_rate	0.0001
loss	0.0568
n_updates	292375

```

-----
EP=6,600 | STEP=1,221,527 | clip=22 | consec=1 | max_consec=8 | avg_clip50=17.70
EP=6,610 | STEP=1,223,531 | clip=22 | consec=1 | max_consec=8 | avg_clip50=17.74
EP=6,620 | STEP=1,225,471 | clip=24 | consec=1 | max_consec=8 | avg_clip50=16.96
-----

```

custom/		
avg_clip50	17.6	
clipped_episode_r...	31	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	914	
ep_rew_mean	18.1	
exploration rate	0.591	
time/		
episodes	19700	
fps	253	
time_elapsed	4840	
total timesteps	1227008	
train/		
learning_rate	0.0001	
loss	0.074	
n_updates	294251	

EP=6,630 | STEP=1,227,996 | clip=13 | consec=0 | max_consec=8 | avg_clip50=17.60
 EP=6,640 | STEP=1,229,906 | clip=24 | consec=2 | max_consec=8 | avg_clip50=17.48
 EP=6,650 | STEP=1,232,131 | clip=5 | consec=0 | max_consec=8 | avg_clip50=17.74

custom/		
avg_clip50	17.8	
clipped_episode_r...	13	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	885	
ep_rew_mean	17.7	
exploration rate	0.589	
time/		
episodes	19800	
fps	253	
time_elapsed	4869	
total timesteps	1233835	
train/		
learning_rate	0.0001	
loss	0.0275	

n_updates		295958			

EP=6,660	STEP=1,234,021	clip=16	consec=0	max_consec=8	avg_clip50=17.64
EP=6,670	STEP=1,235,752	clip=22	consec=1	max_consec=8	avg_clip50=17.50
EP=6,680	STEP=1,238,326	clip=30	consec=1	max_consec=8	avg_clip50=18.20
EP=6,690	STEP=1,240,463	clip=18	consec=0	max_consec=8	avg_clip50=18.46

custom/					
avg_clip50		18.7			
clipped_episode_r...		20			
consecutive_clipp...		0			
max_consecutive_c...		8			
rollout/					
ep_len_mean		893			
ep_rew_mean		18.3			
exploration rate		0.587			
time/					
episodes		19900			
fps		253			
time_elapsed		4901			
total timesteps		1241432			
train/					
learning_rate		0.0001			
loss		0.656			
n_updates		297857			

EP=6,700	STEP=1,243,077	clip=29	consec=2	max_consec=8	avg_clip50=19.88
EP=6,710	STEP=1,244,824	clip=11	consec=0	max_consec=8	avg_clip50=19.22
EP=6,720	STEP=1,247,018	clip=30	consec=4	max_consec=8	avg_clip50=19.92

custom/					
avg_clip50		19.9			
clipped_episode_r...		33			
consecutive_clipp...		1			
max_consecutive_c...		8			
rollout/					
ep_len_mean		901			
ep_rew_mean		18.8			
exploration rate		0.584			
time/					
episodes		20000			

fps	253
time_elapsed	4932
total timesteps	1249019
train/	
learning_rate	0.0001
loss	0.149
n_updates	299754

```

-----
EP=6,730 | STEP=1,249,432 | clip=6 | consec=0 | max_consec=8 | avg_clip50=19.54
EP=6,740 | STEP=1,251,780 | clip=23 | consec=1 | max_consec=8 | avg_clip50=19.54
EP=6,750 | STEP=1,253,662 | clip=13 | consec=0 | max_consec=8 | avg_clip50=18.20
EP=6,760 | STEP=1,256,202 | clip=13 | consec=0 | max_consec=8 | avg_clip50=19.50

```

custom/	
avg_clip50	19.8
clipped_episode_r...	22
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	935
ep_rew_mean	19.4
exploration rate	0.582
time/	
episodes	20100
fps	253
time_elapsed	4963
total timesteps	1256575
train/	
learning_rate	0.0001
loss	0.0285
n_updates	301643

```

-----
EP=6,770 | STEP=1,258,695 | clip=34 | consec=2 | max_consec=8 | avg_clip50=20.28
EP=6,780 | STEP=1,260,828 | clip=18 | consec=0 | max_consec=8 | avg_clip50=19.60
EP=6,790 | STEP=1,263,391 | clip=24 | consec=2 | max_consec=8 | avg_clip50=20.12

```

custom/	
avg_clip50	19.9
clipped_episode_r...	14
consecutive_clipp...	0
max_consecutive_c...	8

rollout/	
ep_len_mean	951
ep_rew_mean	19.4
exploration rate	0.579
time/	
episodes	20200
fps	253
time_elapsed	4997
total timesteps	1264902
train/	
learning_rate	0.0001
loss	0.0289
n_updates	303725

```

-----
EP=6,800 | STEP=1,265,625 | clip=31 | consec=1 | max_consec=8 | avg_clip50=20.26
EP=6,810 | STEP=1,268,221 | clip=11 | consec=0 | max_consec=8 | avg_clip50=20.64
EP=6,820 | STEP=1,270,716 | clip=29 | consec=2 | max_consec=8 | avg_clip50=20.28
EP=6,830 | STEP=1,273,280 | clip=21 | consec=2 | max_consec=8 | avg_clip50=21.36
-----

```

custom/	
avg_clip50	21.4
clipped_episode_r...	21
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	997
ep_rew_mean	20.5
exploration rate	0.576
time/	
episodes	20300
fps	253
time_elapsed	5031
total timesteps	1273302
train/	
learning_rate	0.0001
loss	0.0217
n_updates	305825

```

-----
EP=6,840 | STEP=1,275,719 | clip=33 | consec=1 | max_consec=8 | avg_clip50=21.46
EP=6,850 | STEP=1,277,816 | clip=11 | consec=0 | max_consec=8 | avg_clip50=21.42
EP=6,860 | STEP=1,280,244 | clip=31 | consec=2 | max_consec=8 | avg_clip50=21.10
-----

```

custom/		
avg_clip50	21.2	
clipped_episode_r...	6	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	1e+03	
ep_rew_mean	20.6	
exploration rate	0.573	
time/		
episodes	20400	
fps	252	
time_elapsed	5064	
total timesteps	1281241	
train/		
learning_rate	0.0001	
loss	0.117	
n_updates	307810	

EP=6,870 | STEP=1,282,199 | clip=18 | consec=0 | max_consec=8 | avg_clip50=20.16
 EP=6,880 | STEP=1,284,526 | clip=24 | consec=2 | max_consec=8 | avg_clip50=19.58
 EP=6,890 | STEP=1,287,010 | clip=31 | consec=1 | max_consec=8 | avg_clip50=19.36

custom/		
avg_clip50	19.2	
clipped_episode_r...	8	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	987	
ep_rew_mean	20.4	
exploration rate	0.571	
time/		
episodes	20500	
fps	252	
time_elapsed	5095	
total timesteps	1288789	
train/		
learning_rate	0.0001	
loss	0.0647	

n_updates		309697			

EP=6,900	STEP=1,289,209	clip=41	consec=1	max_consec=8	avg_clip50=19.52
EP=6,910	STEP=1,291,424	clip=21	consec=2	max_consec=8	avg_clip50=18.94
EP=6,920	STEP=1,293,863	clip=20	consec=0	max_consec=8	avg_clip50=20.42
EP=6,930	STEP=1,296,351	clip=8	consec=0	max_consec=8	avg_clip50=20.68

custom/					
avg_clip50		20.5			
clipped_episode_r...		5			
consecutive_clipp...		0			
max_consecutive_c...		8			
rollout/					
ep_len_mean		971			
ep_rew_mean		20.1			
exploration rate		0.568			
time/					
episodes		20600			
fps		252			
time_elapsed		5128			
total timesteps		1296745			
train/					
learning_rate		0.0001			
loss		0.0504			
n_updates		311686			

EP=6,940	STEP=1,298,024	clip=7	consec=0	max_consec=8	avg_clip50=19.00
EP=6,950	STEP=1,300,120	clip=10	consec=0	max_consec=8	avg_clip50=18.96
EP=6,960	STEP=1,302,795	clip=26	consec=2	max_consec=8	avg_clip50=20.26

custom/					
avg_clip50		19.7			
clipped_episode_r...		10			
consecutive_clipp...		0			
max_consecutive_c...		8			
rollout/					
ep_len_mean		961			
ep_rew_mean		20			
exploration rate		0.566			
time/					
episodes		20700			

fps	252
time_elapsed	5159
total timesteps	1304238
train/	
learning_rate	0.0001
loss	0.0448
n_updates	313559

```

-----
EP=6,970 | STEP=1,305,339 | clip=27 | consec=2 | max_consec=8 | avg_clip50=20.00
EP=6,980 | STEP=1,307,773 | clip=33 | consec=1 | max_consec=8 | avg_clip50=19.80
EP=6,990 | STEP=1,310,439 | clip=11 | consec=0 | max_consec=8 | avg_clip50=21.90
EP=7,000 | STEP=1,312,277 | clip=3 | consec=0 | max_consec=8 | avg_clip50=21.24

```

custom/	
avg_clip50	21.2
clipped_episode_r...	3
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	968
ep_rew_mean	20.1
exploration rate	0.563
time/	
episodes	20800
fps	252
time_elapsed	5192
total timesteps	1312277
train/	
learning_rate	0.0001
loss	0.0452
n_updates	315569

```

-----
EP=7,010 | STEP=1,314,436 | clip=3 | consec=0 | max_consec=8 | avg_clip50=19.68
EP=7,020 | STEP=1,316,712 | clip=30 | consec=1 | max_consec=8 | avg_clip50=19.26
EP=7,030 | STEP=1,318,991 | clip=27 | consec=1 | max_consec=8 | avg_clip50=18.88

```

custom/	
avg_clip50	18.6
clipped_episode_r...	24
consecutive_clipp...	1
max_consecutive_c...	8

rollout/	
ep_len_mean	956
ep_rew_mean	19.5
exploration rate	0.56
time/	
episodes	20900
fps	252
time_elapsed	5225
total timesteps	1319924
train/	
learning_rate	0.0001
loss	0.368
n_updates	317480

EP=7,040	STEP=1,321,444	clip=20	consec=0	max_consec=8	avg_clip50=18.54
EP=7,050	STEP=1,323,765	clip=28	consec=1	max_consec=8	avg_clip50=19.74
EP=7,060	STEP=1,325,879	clip=9	consec=0	max_consec=8	avg_clip50=20.10

custom/	
avg_clip50	19.6
clipped_episode_r...	30
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	958
ep_rew_mean	19.7
exploration rate	0.558
time/	
episodes	21000
fps	252
time_elapsed	5257
total timesteps	1327626
train/	
learning_rate	0.0001
loss	0.0866
n_updates	319406

EP=7,070	STEP=1,327,763	clip=2	consec=0	max_consec=8	avg_clip50=19.26
EP=7,080	STEP=1,330,342	clip=15	consec=0	max_consec=8	avg_clip50=19.80
EP=7,090	STEP=1,332,298	clip=4	consec=0	max_consec=8	avg_clip50=18.84
EP=7,100	STEP=1,334,684	clip=19	consec=0	max_consec=8	avg_clip50=19.12

custom/		
avg_clip50	19	
clipped_episode_r...	26	
consecutive_clipp...	2	
max_consecutive_c...	8	
rollout/		
ep_len_mean	941	
ep_rew_mean	19.4	
exploration rate	0.555	
time/		
episodes	21100	
fps	252	
time_elapsed	5289	
total timesteps	1335371	
train/		
learning_rate	0.0001	
loss	0.0684	
n_updates	321342	

EP=7,110 | STEP=1,337,115 | clip=26 | consec=1 | max_consec=8 | avg_clip50=19.76
 EP=7,120 | STEP=1,339,534 | clip=21 | consec=1 | max_consec=8 | avg_clip50=20.96
 EP=7,130 | STEP=1,341,612 | clip=18 | consec=0 | max_consec=8 | avg_clip50=19.90

custom/		
avg_clip50	20	
clipped_episode_r...	9	
consecutive_clipp...	0	
max_consecutive_c...	8	
rollout/		
ep_len_mean	941	
ep_rew_mean	19.8	
exploration rate	0.553	
time/		
episodes	21200	
fps	252	
time_elapsed	5320	
total timesteps	1343079	
train/		
learning_rate	0.0001	
loss	0.0669	

n_updates		323269			

EP=7,140	STEP=1,343,594	clip=18	consec=0	max_consec=8	avg_clip50=20.14
EP=7,150	STEP=1,345,866	clip=9	consec=0	max_consec=8	avg_clip50=19.40
EP=7,160	STEP=1,348,358	clip=33	consec=1	max_consec=8	avg_clip50=19.46
EP=7,170	STEP=1,350,920	clip=29	consec=1	max_consec=8	avg_clip50=19.78

custom/					
avg_clip50		20.1			
clipped_episode_r...		21			
consecutive_clipp...		2			
max_consecutive_c...		8			
rollout/					
ep_len_mean		969			
ep_rew_mean		20.3			
exploration rate		0.55			
time/					
episodes		21300			
fps		252			
time_elapsed		5354			
total timesteps		1351189			
train/					
learning_rate		0.0001			
loss		0.0582			
n_updates		325297			

EP=7,180	STEP=1,353,178	clip=32	consec=1	max_consec=8	avg_clip50=20.50
EP=7,190	STEP=1,355,628	clip=16	consec=0	max_consec=8	avg_clip50=21.40
EP=7,200	STEP=1,357,920	clip=13	consec=0	max_consec=8	avg_clip50=21.66

custom/					
avg_clip50		21.8			
clipped_episode_r...		14			
consecutive_clipp...		0			
max_consecutive_c...		8			
rollout/					
ep_len_mean		984			
ep_rew_mean		20.7			
exploration rate		0.547			
time/					
episodes		21400			

fps	252
time_elapsed	5388
total timesteps	1359464
train/	
learning_rate	0.0001
loss	0.199
n_updates	327365

```

-----
EP=7,210 | STEP=1,360,568 | clip=25 | consec=1 | max_consec=8 | avg_clip50=21.92
EP=7,220 | STEP=1,362,806 | clip=25 | consec=1 | max_consec=8 | avg_clip50=21.22
EP=7,230 | STEP=1,365,070 | clip=25 | consec=3 | max_consec=8 | avg_clip50=21.54
-----

```

custom/	
avg_clip50	22
clipped_episode_r...	30
consecutive_clipp...	3
max_consecutive_c...	8
rollout/	
ep_len_mean	1.01e+03
ep_rew_mean	21.7
exploration rate	0.545
time/	
episodes	21500
fps	252
time_elapsed	5423
total timesteps	1367820
train/	
learning_rate	0.0001
loss	0.471
n_updates	329454

```

-----
EP=7,240 | STEP=1,367,940 | clip=31 | consec=4 | max_consec=8 | avg_clip50=22.32
EP=7,250 | STEP=1,370,105 | clip=17 | consec=0 | max_consec=8 | avg_clip50=21.70
EP=7,260 | STEP=1,373,261 | clip=33 | consec=1 | max_consec=8 | avg_clip50=22.44
EP=7,270 | STEP=1,376,259 | clip=23 | consec=1 | max_consec=8 | avg_clip50=23.90
-----

```

custom/	
avg_clip50	24
clipped_episode_r...	24
consecutive_clipp...	2
max_consecutive_c...	8

rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	22.3
exploration rate	0.541
time/	
episodes	21600
fps	252
time_elapsed	5461
total timesteps	1377072
train/	
learning_rate	0.0001
loss	0.0522
n_updates	331767

```
-----
EP=7,280 | STEP=1,378,373 | clip=21 | consec=2 | max_consec=8 | avg_clip50=23.36
EP=7,290 | STEP=1,380,879 | clip=29 | consec=1 | max_consec=8 | avg_clip50=22.66
EP=7,300 | STEP=1,383,256 | clip=31 | consec=1 | max_consec=8 | avg_clip50=23.06
-----
```

custom/	
avg_clip50	22.2
clipped_episode_r...	5
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	1.05e+03
ep_rew_mean	22.2
exploration rate	0.539
time/	
episodes	21700
fps	252
time_elapsed	5494
total timesteps	1385250
train/	
learning_rate	0.0001
loss	0.0588
n_updates	333812

```
-----
EP=7,310 | STEP=1,385,683 | clip=6 | consec=0 | max_consec=8 | avg_clip50=21.72
EP=7,320 | STEP=1,388,241 | clip=33 | consec=4 | max_consec=8 | avg_clip50=21.30
EP=7,330 | STEP=1,390,940 | clip=22 | consec=8 | max_consec=8 | avg_clip50=23.08
EP=7,340 | STEP=1,393,377 | clip=10 | consec=0 | max_consec=8 | avg_clip50=22.54
-----
```

custom/		
avg_clip50	22.9	
clipped_episode_r...	26	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	1.07e+03	
ep_rew_mean	22.7	
exploration rate	0.536	
time/		
episodes	21800	
fps	252	
time_elapsed	5530	
total timesteps	1393879	
train/		
learning_rate	0.0001	
loss	0.0602	
n_updates	335969	

EP=7,350 | STEP=1,395,530 | clip=13 | consec=0 | max_consec=8 | avg_clip50=22.18
 EP=7,360 | STEP=1,397,880 | clip=4 | consec=0 | max_consec=8 | avg_clip50=21.92
 EP=7,370 | STEP=1,399,652 | clip=9 | consec=0 | max_consec=8 | avg_clip50=20.36

custom/		
avg_clip50	18.9	
clipped_episode_r...	26	
consecutive_clipp...	1	
max_consecutive_c...	8	
rollout/		
ep_len_mean	977	
ep_rew_mean	20.7	
exploration rate	0.534	
time/		
episodes	21900	
fps	251	
time_elapsed	5559	
total timesteps	1400668	
train/		
learning_rate	0.0001	
loss	0.0578	

n_updates		337666			

EP=7,380	STEP=1,401,566	clip=18	consec=0	max_consec=8	avg_clip50=17.84
EP=7,390	STEP=1,403,899	clip=8	consec=0	max_consec=8	avg_clip50=17.70
EP=7,400	STEP=1,405,769	clip=23	consec=3	max_consec=8	avg_clip50=17.22

custom/					
avg_clip50		18.1			
clipped_episode_r...		33			
consecutive_clipp...		5			
max_consecutive_c...		8			
rollout/					
ep_len_mean		957			
ep_rew_mean		20			
exploration rate		0.531			
time/					
episodes		22000			
fps		251			
time_elapsed		5592			
total timesteps		1408638			
train/					
learning_rate		0.0001			
loss		0.0752			
n_updates		339659			

EP=7,410	STEP=1,408,697	clip=30	consec=6	max_consec=8	avg_clip50=18.60
EP=7,420	STEP=1,410,818	clip=7	consec=0	max_consec=8	avg_clip50=19.02
EP=7,430	STEP=1,413,297	clip=20	consec=0	max_consec=8	avg_clip50=20.36
EP=7,440	STEP=1,415,874	clip=10	consec=0	max_consec=8	avg_clip50=21.44

custom/					
avg_clip50		22.1			
clipped_episode_r...		29			
consecutive_clipp...		3			
max_consecutive_c...		8			
rollout/					
ep_len_mean		954			
ep_rew_mean		19.8			
exploration rate		0.528			
time/					
episodes		22100			

	fps		251	
	time_elapsed		5627	
	total timesteps		1416862	
	train/			
	learning_rate		0.0001	
	loss		0.0702	
	n_updates		341715	

```

-----
EP=7,450 | STEP=1,418,070 | clip=22 | consec=2 | max_consec=8 | avg_clip50=22.38
EP=7,460 | STEP=1,419,940 | clip=12 | consec=0 | max_consec=8 | avg_clip50=20.36
EP=7,470 | STEP=1,422,296 | clip=27 | consec=1 | max_consec=8 | avg_clip50=20.84
-----

```

	custom/			
	avg_clip50		20.1	
	clipped_episode_r...		11	
	consecutive_clipp...		0	
	max_consecutive_c...		8	
	rollout/			
	ep_len_mean		957	
	ep_rew_mean		20.1	
	exploration rate		0.526	
	time/			
	episodes		22200	
	fps		251	
	time_elapsed		5657	
	total timesteps		1424030	
	train/			
	learning_rate		0.0001	
	loss		0.0652	
	n_updates		343507	

```

-----
EP=7,480 | STEP=1,424,636 | clip=10 | consec=0 | max_consec=8 | avg_clip50=20.34
EP=7,490 | STEP=1,427,062 | clip=27 | consec=1 | max_consec=8 | avg_clip50=19.68
EP=7,500 | STEP=1,429,302 | clip=23 | consec=2 | max_consec=8 | avg_clip50=19.76
EP=7,510 | STEP=1,431,321 | clip=13 | consec=0 | max_consec=8 | avg_clip50=19.98
-----

```

	custom/			
	avg_clip50		20	
	clipped_episode_r...		17	
	consecutive_clipp...		0	
	max_consecutive_c...		8	

rollout/	
ep_len_mean	952
ep_rew_mean	20.1
exploration rate	0.523
time/	
episodes	22300
fps	251
time_elapsed	5688
total timesteps	1431650
train/	
learning_rate	0.0001
loss	0.0724
n_updates	345412

```
-----
EP=7,520 | STEP=1,433,158 | clip=11 | consec=0 | max_consec=8 | avg_clip50=18.60
EP=7,530 | STEP=1,435,246 | clip=18 | consec=0 | max_consec=8 | avg_clip50=18.22
EP=7,540 | STEP=1,437,970 | clip=24 | consec=1 | max_consec=8 | avg_clip50=19.24
-----
```

custom/	
avg_clip50	19.2
clipped_episode_r...	21
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	920
ep_rew_mean	19.2
exploration rate	0.521
time/	
episodes	22400
fps	251
time_elapsed	5719
total timesteps	1439002
train/	
learning_rate	0.0001
loss	0.0565
n_updates	347250

```
-----
EP=7,550 | STEP=1,439,968 | clip=25 | consec=3 | max_consec=8 | avg_clip50=18.84
EP=7,560 | STEP=1,442,435 | clip=30 | consec=2 | max_consec=8 | avg_clip50=19.72
EP=7,570 | STEP=1,444,769 | clip=20 | consec=0 | max_consec=8 | avg_clip50=20.80
-----
```

custom/	
avg_clip50	22.5
clipped_episode_r...	33
consecutive_clipp...	2
max_consecutive_c...	8
rollout/	
ep_len_mean	961
ep_rew_mean	20.3
exploration rate	0.518
time/	
episodes	22500
fps	251
time_elapsed	5755
total timesteps	1447551
train/	
learning_rate	0.0001
loss	0.11
n_updates	349387

EP=7,580	STEP=1,447,635	clip=21	consec=3	max_consec=8	avg_clip50=22.56
EP=7,590	STEP=1,449,801	clip=26	consec=1	max_consec=8	avg_clip50=21.20
EP=7,600	STEP=1,452,045	clip=22	consec=3	max_consec=8	avg_clip50=21.38
EP=7,610	STEP=1,454,608	clip=26	consec=4	max_consec=8	avg_clip50=21.34

custom/	
avg_clip50	22.2
clipped_episode_r...	60
consecutive_clipp...	6
max_consecutive_c...	8
rollout/	
ep_len_mean	997
ep_rew_mean	21.1
exploration rate	0.515
time/	
episodes	22600
fps	251
time_elapsed	5789
total timesteps	1455748
train/	
learning_rate	0.0001
loss	0.0602

n_updates		351436			

EP=7,620	STEP=1,457,338	clip=27	consec=1	max_consec=8	avg_clip50=22.52
EP=7,630	STEP=1,459,530	clip=14	consec=0	max_consec=8	avg_clip50=21.02
EP=7,640	STEP=1,462,028	clip=9	consec=0	max_consec=8	avg_clip50=21.72

custom/					
avg_clip50		22.1			
clipped_episode_r...		11			
consecutive_clipp...		0			
max_consecutive_c...		8			
rollout/					
ep_len_mean		1.02e+03			
ep_rew_mean		21.6			
exploration rate		0.513			
time/					
episodes		22700			
fps		251			
time_elapsed		5822			
total timesteps		1463473			
train/					
learning_rate		0.0001			
loss		0.0628			
n_updates		353368			

EP=7,650	STEP=1,464,191	clip=17	consec=0	max_consec=8	avg_clip50=21.66
EP=7,660	STEP=1,466,938	clip=15	consec=0	max_consec=8	avg_clip50=22.52
EP=7,670	STEP=1,469,267	clip=28	consec=1	max_consec=8	avg_clip50=21.64
EP=7,680	STEP=1,471,484	clip=26	consec=1	max_consec=8	avg_clip50=21.82

custom/					
avg_clip50		21.5			
clipped_episode_r...		5			
consecutive_clipp...		0			
max_consecutive_c...		8			
rollout/					
ep_len_mean		997			
ep_rew_mean		21.4			
exploration rate		0.51			
time/					
episodes		22800			

	fps		251	
	time_elapsed		5856	
	total timesteps		1471584	
	train/			
	learning_rate		0.0001	
	loss		0.211	
	n_updates		355395	

```
-----
EP=7,690 | STEP=1,473,617 | clip=30 | consec=1 | max_consec=8 | avg_clip50=21.00
EP=7,700 | STEP=1,476,387 | clip=7 | consec=0 | max_consec=8 | avg_clip50=22.26
EP=7,710 | STEP=1,478,713 | clip=22 | consec=2 | max_consec=8 | avg_clip50=21.54
-----
```

	custom/			
	avg_clip50		21.6	
	clipped_episode_r...		29	
	consecutive_clipp...		1	
	max_consecutive_c...		8	
	rollout/			
	ep_len_mean		1e+03	
	ep_rew_mean		21.8	
	exploration rate		0.507	
	time/			
	episodes		22900	
	fps		251	
	time_elapsed		5892	
	total timesteps		1480146	
	train/			
	learning_rate		0.0001	
	loss		0.0586	
	n_updates		357536	

```
-----
EP=7,720 | STEP=1,481,274 | clip=15 | consec=0 | max_consec=8 | avg_clip50=21.68
EP=7,730 | STEP=1,483,863 | clip=32 | consec=1 | max_consec=8 | avg_clip50=22.68
EP=7,740 | STEP=1,486,168 | clip=25 | consec=1 | max_consec=8 | avg_clip50=23.48
-----
```

	custom/			
	avg_clip50		22	
	clipped_episode_r...		17	
	consecutive_clipp...		0	
	max_consecutive_c...		8	
	rollout/			

ep_len_mean	1.01e+03
ep_rew_mean	22.2
exploration rate	0.504
time/	
episodes	23000
fps	251
time_elapsed	5925
total timesteps	1488125
train/	
learning_rate	0.0001
loss	0.0773
n_updates	359531

```

-----
EP=7,750 | STEP=1,488,155 | clip=7 | consec=0 | max_consec=8 | avg_clip50=22.02
EP=7,760 | STEP=1,490,679 | clip=31 | consec=2 | max_consec=8 | avg_clip50=22.22
EP=7,770 | STEP=1,493,240 | clip=30 | consec=1 | max_consec=8 | avg_clip50=22.80
EP=7,780 | STEP=1,495,652 | clip=24 | consec=2 | max_consec=8 | avg_clip50=22.40
-----

```

custom/	
avg_clip50	22.6
clipped_episode_r...	23
consecutive_clipp...	5
max_consecutive_c...	8
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	22.7
exploration rate	0.502
time/	
episodes	23100
fps	251
time_elapsed	5960
total timesteps	1496531
train/	
learning_rate	0.0001
loss	0.189
n_updates	361632

```

-----
EP=7,790 | STEP=1,498,393 | clip=34 | consec=3 | max_consec=8 | avg_clip50=22.68
EP=7,800 | STEP=1,500,952 | clip=33 | consec=1 | max_consec=8 | avg_clip50=23.12
EP=7,810 | STEP=1,503,871 | clip=29 | consec=3 | max_consec=8 | avg_clip50=23.74
-----

```

custom/	
avg_clip50	23
clipped_episode_r...	11
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	22.9
exploration rate	0.499
time/	
episodes	23200
fps	251
time_elapsed	5997
total timesteps	1505602
train/	
learning_rate	0.0001
loss	0.121
n_updates	363900

EP=7,820	STEP=1,505,994	clip=16	consec=0	max_consec=8	avg_clip50=22.86
EP=7,830	STEP=1,508,443	clip=15	consec=0	max_consec=8	avg_clip50=22.90
EP=7,840	STEP=1,510,508	clip=11	consec=0	max_consec=8	avg_clip50=21.82
EP=7,850	STEP=1,512,948	clip=17	consec=0	max_consec=8	avg_clip50=21.96

custom/	
avg_clip50	22.1
clipped_episode_r...	27
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	22.6
exploration rate	0.496
time/	
episodes	23300
fps	250
time_elapsed	6029
total timesteps	1513254
train/	
learning_rate	0.0001
loss	0.057

n_updates		365813			

EP=7,860	STEP=1,515,192	clip=18	consec=0	max_consec=8	avg_clip50=21.20
EP=7,870	STEP=1,517,575	clip=21	consec=1	max_consec=8	avg_clip50=21.42
EP=7,880	STEP=1,520,147	clip=25	consec=3	max_consec=8	avg_clip50=21.46

custom/					
avg_clip50		22			
clipped_episode_r...		21			
consecutive_clipp...		2			
max_consecutive_c...		8			
rollout/					
ep_len_mean		1.02e+03			
ep_rew_mean		22.2			
exploration rate		0.493			
time/					
episodes		23400			
fps		250			
time_elapsed		6063			
total timesteps		1521406			
train/					
learning_rate		0.0001			
loss		0.324			
n_updates		367851			

EP=7,890	STEP=1,522,695	clip=26	consec=1	max_consec=8	avg_clip50=22.26
EP=7,900	STEP=1,524,852	clip=14	consec=0	max_consec=8	avg_clip50=21.62
EP=7,910	STEP=1,527,016	clip=19	consec=0	max_consec=8	avg_clip50=21.34

custom/					
avg_clip50		21.6			
clipped_episode_r...		12			
consecutive_clipp...		0			
max_consecutive_c...		8			
rollout/					
ep_len_mean		976			
ep_rew_mean		21.4			
exploration rate		0.491			
time/					
episodes		23500			
fps		250			

	time_elapsed	6095	
	total timesteps	1529057	
	train/		
	learning_rate	0.0001	
	loss	0.131	
	n_updates	369764	

```

-----
EP=7,920 | STEP=1,529,311 | clip=30 | consec=1 | max_consec=8 | avg_clip50=21.76
EP=7,930 | STEP=1,531,655 | clip=19 | consec=0 | max_consec=8 | avg_clip50=21.66
EP=7,940 | STEP=1,534,380 | clip=14 | consec=0 | max_consec=8 | avg_clip50=22.18
EP=7,950 | STEP=1,536,630 | clip=32 | consec=1 | max_consec=8 | avg_clip50=22.80

```

	custom/		
	avg_clip50	22.6	
	clipped_episode_r...	29	
	consecutive_clipp...	3	
	max_consecutive_c...	8	
	rollout/		
	ep_len_mean	1e+03	
	ep_rew_mean	22.2	
	exploration rate	0.488	
	time/		
	episodes	23600	
	fps	250	
	time_elapsed	6129	
	total timesteps	1537240	
	train/		
	learning_rate	0.0001	
	loss	0.0471	
	n_updates	371809	

```

-----
EP=7,960 | STEP=1,539,200 | clip=17 | consec=0 | max_consec=8 | avg_clip50=23.08
EP=7,970 | STEP=1,541,832 | clip=24 | consec=3 | max_consec=8 | avg_clip50=23.10
EP=7,980 | STEP=1,544,830 | clip=20 | consec=0 | max_consec=8 | avg_clip50=23.76

```

	custom/		
	avg_clip50	22.4	
	clipped_episode_r...	21	
	consecutive_clipp...	2	
	max_consecutive_c...	8	
	rollout/		

ep_len_mean	1.01e+03
ep_rew_mean	22.1
exploration rate	0.485
time/	
episodes	23700
fps	250
time_elapsed	6165
total timesteps	1546038
train/	
learning_rate	0.0001
loss	0.525
n_updates	374009

```

-----
EP=7,990 | STEP=1,546,934 | clip=34 | consec=2 | max_consec=8 | avg_clip50=22.40
EP=8,000 | STEP=1,549,523 | clip=29 | consec=3 | max_consec=8 | avg_clip50=22.86
EP=8,010 | STEP=1,552,192 | clip=32 | consec=1 | max_consec=8 | avg_clip50=23.02
-----

```

custom/	
avg_clip50	21.9
clipped_episode_r...	13
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	22.6
exploration rate	0.482
time/	
episodes	23800
fps	250
time_elapsed	6199
total timesteps	1554223
train/	
learning_rate	0.0001
loss	0.18
n_updates	376055

```

-----
EP=8,020 | STEP=1,554,269 | clip=27 | consec=1 | max_consec=8 | avg_clip50=21.94
EP=8,030 | STEP=1,556,123 | clip=9 | consec=0 | max_consec=8 | avg_clip50=19.86
EP=8,040 | STEP=1,557,960 | clip=23 | consec=2 | max_consec=8 | avg_clip50=19.50
EP=8,050 | STEP=1,560,184 | clip=9 | consec=0 | max_consec=8 | avg_clip50=18.76
-----

```

custom/	
avg_clip50	17.9
clipped_episode_r...	11
consecutive_clipp...	0
max_consecutive_c...	8
rollout/	
ep_len_mean	977
ep_rew_mean	20.7
exploration rate	0.48
time/	
episodes	23900
fps	250
time_elapsed	6228
total timesteps	1560985
train/	
learning_rate	0.0001
loss	0.0461
n_updates	377746

EP=8,060 | STEP=1,562,188 | clip=16 | consec=0 | max_consec=8 | avg_clip50=17.56
 EP=8,070 | STEP=1,565,171 | clip=12 | consec=0 | max_consec=8 | avg_clip50=19.58
 EP=8,080 | STEP=1,567,421 | clip=6 | consec=0 | max_consec=8 | avg_clip50=20.82

custom/	
avg_clip50	21.6
clipped_episode_r...	35
consecutive_clipp...	1
max_consecutive_c...	8
rollout/	
ep_len_mean	959
ep_rew_mean	20.7
exploration rate	0.477
time/	
episodes	24000
fps	250
time_elapsed	6262
total timesteps	1569244
train/	
learning_rate	0.0001
loss	0.0187
n_updates	379810

```

-----
EP=8,090 | STEP=1,569,755 | clip=27 | consec=3 | max_consec=8 | avg_clip50=21.84
EP=8,100 | STEP=1,572,806 | clip=21 | consec=13 | max_consec=13 | avg_clip50=24.04
EP=8,110 | STEP=1,575,301 | clip=29 | consec=5 | max_consec=13 | avg_clip50=25.22
EP=8,120 | STEP=1,577,646 | clip=33 | consec=3 | max_consec=13 | avg_clip50=23.78
-----

```

```

| custom/
|   avg_clip50          | 24.1 |
|   clipped_episode_r... | 21   |
|   consecutive_clipp... | 7    |
|   max_consecutive_c... | 13   |
| rollout/
|   ep_len_mean         | 994  |
|   ep_rew_mean         | 22   |
|   exploration rate    | 0.474 |
| time/
|   episodes            | 24100 |
|   fps                 | 250   |
|   time_elapsed        | 6302  |
|   total timesteps     | 1578896 |
| train/
|   learning_rate       | 0.0001 |
|   loss                | 0.0673 |
|   n_updates           | 382223 |
-----

```

```

EP=8,130 | STEP=1,580,040 | clip=22 | consec=1 | max_consec=13 | avg_clip50=23.78
EP=8,140 | STEP=1,582,072 | clip=8  | consec=0 | max_consec=13 | avg_clip50=22.74
EP=8,150 | STEP=1,584,396 | clip=19 | consec=0 | max_consec=13 | avg_clip50=20.94
-----

```

```

| custom/
|   avg_clip50          | 22.2 |
|   clipped_episode_r... | 30   |
|   consecutive_clipp... | 6    |
|   max_consecutive_c... | 13   |
| rollout/
|   ep_len_mean         | 1.04e+03 |
|   ep_rew_mean         | 23.3   |
|   exploration rate    | 0.472  |
| time/
|   episodes            | 24200  |
|   fps                 | 250    |
-----

```

	time_elapsed		6336	
	total timesteps		1586722	
	train/			
	learning_rate		0.0001	
	loss		0.0905	
	n_updates		384180	

EP=8,160		STEP=1,587,281		clip=19		consec=0		max_consec=13		avg_clip50=22.04
EP=8,170		STEP=1,589,643		clip=17		consec=0		max_consec=13		avg_clip50=22.50
EP=8,180		STEP=1,592,429		clip=31		consec=2		max_consec=13		avg_clip50=23.12
EP=8,190		STEP=1,595,106		clip=28		consec=3		max_consec=13		avg_clip50=24.94

	custom/			
	avg_clip50		25.2	
	clipped_episode_r...		26	
	consecutive_clipp...		1	
	max_consecutive_c...		13	
	rollout/			
	ep_len_mean		1.05e+03	
	ep_rew_mean		23.6	
	exploration rate		0.469	
	time/			
	episodes		24300	
	fps		250	
	time_elapsed		6374	
	total timesteps		1595642	
	train/			
	learning_rate		0.0001	
	loss		0.0455	
	n_updates		386410	

EP=8,200		STEP=1,597,828		clip=26		consec=6		max_consec=13		avg_clip50=25.44
EP=8,210		STEP=1,600,332		clip=22		consec=4		max_consec=13		avg_clip50=24.38
EP=8,220		STEP=1,602,887		clip=29		consec=3		max_consec=13		avg_clip50=24.10

	custom/			
	avg_clip50		23.7	
	clipped_episode_r...		17	
	consecutive_clipp...		0	
	max_consecutive_c...		13	
	rollout/			

ep_len_mean	1.05e+03
ep_rew_mean	23.1
exploration rate	0.466
time/	
episodes	24400
fps	250
time_elapsed	6412
total timesteps	1604789
train/	
learning_rate	0.0001
loss	0.0396
n_updates	388697

EP=8,230	STEP=1,605,260	clip=16	consec=0	max_consec=13	avg_clip50=23.36
EP=8,240	STEP=1,607,707	clip=29	consec=2	max_consec=13	avg_clip50=22.60
EP=8,250	STEP=1,609,845	clip=25	consec=2	max_consec=13	avg_clip50=21.78
EP=8,260	STEP=1,612,250	clip=10	consec=0	max_consec=13	avg_clip50=21.16

custom/	
avg_clip50	21
clipped_episode_r...	12
consecutive_clipp...	0
max_consecutive_c...	13
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	22.6
exploration rate	0.463
time/	
episodes	24500
fps	250
time_elapsed	6444
total timesteps	1612556
train/	
learning_rate	0.0001
loss	0.0737
n_updates	390638

EP=8,270	STEP=1,614,311	clip=26	consec=2	max_consec=13	avg_clip50=20.16
EP=8,280	STEP=1,617,022	clip=33	consec=1	max_consec=13	avg_clip50=20.66
EP=8,290	STEP=1,619,470	clip=24	consec=5	max_consec=13	avg_clip50=20.78

custom/	
avg_clip50	21.1
clipped_episode_r...	32
consecutive_clipp...	2
max_consecutive_c...	13
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	21.7
exploration rate	0.46
time/	
episodes	24600
fps	250
time_elapsed	6480
total timesteps	1621090
train/	
learning_rate	0.0001
loss	0.372
n_updates	392772

```

-----
EP=8,300 | STEP=1,622,033 | clip=28 | consec=1 | max_consec=13 | avg_clip50=21.38
EP=8,310 | STEP=1,624,545 | clip=15 | consec=0 | max_consec=13 | avg_clip50=21.72
EP=8,320 | STEP=1,626,761 | clip=19 | consec=0 | max_consec=13 | avg_clip50=22.38
EP=8,330 | STEP=1,629,443 | clip=27 | consec=5 | max_consec=13 | avg_clip50=22.68

```

custom/	
avg_clip50	22.7
clipped_episode_r...	23
consecutive_clipp...	6
max_consecutive_c...	13
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	21.8
exploration rate	0.457
time/	
episodes	24700
fps	250
time_elapsed	6515
total timesteps	1629677
train/	
learning_rate	0.0001
loss	0.346

n_updates		394919			

EP=8,340	STEP=1,631,987	clip=7	consec=0	max_consec=13	avg_clip50=22.70
EP=8,350	STEP=1,634,415	clip=31	consec=2	max_consec=13	avg_clip50=22.60
EP=8,360	STEP=1,636,571	clip=32	consec=1	max_consec=13	avg_clip50=22.16

custom/					
avg_clip50		22			
clipped_episode_r...		24			
consecutive_clipp...		1			
max_consecutive_c...		13			
rollout/					
ep_len_mean		1.02e+03			
ep_rew_mean		22.1			
exploration rate		0.455			
time/					
episodes		24800			
fps		250			
time_elapsed		6548			
total timesteps		1637475			
train/					
learning_rate		0.0001			
loss		0.186			
n_updates		396868			

EP=8,370	STEP=1,638,876	clip=23	consec=1	max_consec=13	avg_clip50=22.44
EP=8,380	STEP=1,640,940	clip=16	consec=0	max_consec=13	avg_clip50=21.20
EP=8,390	STEP=1,643,846	clip=33	consec=2	max_consec=13	avg_clip50=22.20

custom/					
avg_clip50		21.9			
clipped_episode_r...		24			
consecutive_clipp...		2			
max_consecutive_c...		13			
rollout/					
ep_len_mean		1.01e+03			
ep_rew_mean		22.2			
exploration rate		0.452			
time/					
episodes		24900			
fps		249			

	time_elapsed	6583	
	total timesteps	1645841	
	train/		
	learning_rate	0.0001	
	loss	0.0542	
	n_updates	398960	

```

-----
EP=8,400 | STEP=1,645,987 | clip=10 | consec=0 | max_consec=13 | avg_clip50=21.52
EP=8,410 | STEP=1,648,423 | clip=22 | consec=1 | max_consec=13 | avg_clip50=22.00
EP=8,420 | STEP=1,651,204 | clip=24 | consec=3 | max_consec=13 | avg_clip50=22.74
EP=8,430 | STEP=1,653,347 | clip=26 | consec=1 | max_consec=13 | avg_clip50=22.90

```

	custom/		
	avg_clip50	22.8	
	clipped_episode_r...	23	
	consecutive_clipp...	2	
	max_consecutive_c...	13	
	rollout/		
	ep_len_mean	1e+03	
	ep_rew_mean	21.9	
	exploration rate	0.449	
	time/		
	episodes	25000	
	fps	249	
	time_elapsed	6618	
	total timesteps	1654219	
	train/		
	learning_rate	0.0001	
	loss	0.498	
	n_updates	401054	

```

-----
EP=8,440 | STEP=1,655,591 | clip=6 | consec=0 | max_consec=13 | avg_clip50=21.48
EP=8,450 | STEP=1,658,165 | clip=21 | consec=5 | max_consec=13 | avg_clip50=22.52
EP=8,460 | STEP=1,660,690 | clip=19 | consec=0 | max_consec=13 | avg_clip50=22.72

```

	custom/		
	avg_clip50	23.2	
	clipped_episode_r...	33	
	consecutive_clipp...	4	
	max_consecutive_c...	13	
	rollout/		

ep_len_mean	1.03e+03
ep_rew_mean	22.7
exploration rate	0.446
time/	
episodes	25100
fps	249
time_elapsed	6654
total timesteps	1663016
train/	
learning_rate	0.0001
loss	0.265
n_updates	403253

```

-----
EP=8,470 | STEP=1,663,653 | clip=22 | consec=7 | max_consec=13 | avg_clip50=23.14
EP=8,480 | STEP=1,666,202 | clip=25 | consec=4 | max_consec=13 | avg_clip50=24.08
EP=8,490 | STEP=1,668,888 | clip=4 | consec=0 | max_consec=13 | avg_clip50=24.80
EP=8,500 | STEP=1,671,613 | clip=26 | consec=3 | max_consec=13 | avg_clip50=25.00
-----

```

custom/	
avg_clip50	25.1
clipped_episode_r...	32
consecutive_clipp...	5
max_consecutive_c...	13
rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	23.9
exploration rate	0.443
time/	
episodes	25200
fps	249
time_elapsed	6692
total timesteps	1672179
train/	
learning_rate	0.0001
loss	0.0416
n_updates	405544

```

-----
EP=8,510 | STEP=1,674,059 | clip=23 | consec=2 | max_consec=13 | avg_clip50=24.98
EP=8,520 | STEP=1,677,072 | clip=33 | consec=3 | max_consec=13 | avg_clip50=25.04
EP=8,530 | STEP=1,679,625 | clip=35 | consec=1 | max_consec=13 | avg_clip50=25.06
-----

```

custom/	
avg_clip50	24.7
clipped_episode_r...	34
consecutive_clipp...	3
max_consecutive_c...	13
rollout/	
ep_len_mean	1.1e+03
ep_rew_mean	24.7
exploration rate	0.44
time/	
episodes	25300
fps	249
time_elapsed	6731
total timesteps	1681554
train/	
learning_rate	0.0001
loss	0.0687
n_updates	407888

EP=8,540	STEP=1,682,195	clip=32	consec=1	max_consec=13	avg_clip50=25.16
EP=8,550	STEP=1,684,526	clip=23	consec=1	max_consec=13	avg_clip50=24.62
EP=8,560	STEP=1,686,863	clip=24	consec=4	max_consec=13	avg_clip50=24.74
EP=8,570	STEP=1,689,342	clip=9	consec=0	max_consec=13	avg_clip50=23.94

custom/	
avg_clip50	23.8
clipped_episode_r...	19
consecutive_clipp...	0
max_consecutive_c...	13
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	24.5
exploration rate	0.437
time/	
episodes	25400
fps	249
time_elapsed	6767
total timesteps	1690179
train/	
learning_rate	0.0001
loss	0.321

n_updates		410044			

EP=8,580	STEP=1,691,924	clip=29	consec=3	max_consec=13	avg_clip50=23.60
EP=8,590	STEP=1,694,575	clip=33	consec=3	max_consec=13	avg_clip50=23.74
EP=8,600	STEP=1,697,107	clip=13	consec=0	max_consec=13	avg_clip50=23.98

custom/					
avg_clip50		24.2			
clipped_episode_r...		20			
consecutive_clipp...		0			
max_consecutive_c...		13			
rollout/					
ep_len_mean		1.07e+03			
ep_rew_mean		24.3			
exploration rate		0.434			
time/					
episodes		25500			
fps		249			
time_elapsed		6805			
total timesteps		1699097			
train/					
learning_rate		0.0001			
loss		0.532			
n_updates		412274			

EP=8,610	STEP=1,699,519	clip=9	consec=0	max_consec=13	avg_clip50=23.56
EP=8,620	STEP=1,702,295	clip=10	consec=0	max_consec=13	avg_clip50=24.00
EP=8,630	STEP=1,705,409	clip=11	consec=0	max_consec=13	avg_clip50=25.12
EP=8,640	STEP=1,708,096	clip=19	consec=0	max_consec=13	avg_clip50=25.34

custom/					
avg_clip50		25.5			
clipped_episode_r...		31			
consecutive_clipp...		2			
max_consecutive_c...		13			
rollout/					
ep_len_mean		1.08e+03			
ep_rew_mean		24.5			
exploration rate		0.431			
time/					
episodes		25600			

	fps		249	
	time_elapsed		6846	
	total timesteps		1708953	
	train/			
	learning_rate		0.0001	
	loss		0.0369	
	n_updates		414738	

```
-----
EP=8,650 | STEP=1,710,672 | clip=17 | consec=0 | max_consec=13 | avg_clip50=25.34
EP=8,660 | STEP=1,713,616 | clip=25 | consec=2 | max_consec=13 | avg_clip50=25.88
EP=8,670 | STEP=1,715,434 | clip=14 | consec=0 | max_consec=13 | avg_clip50=23.94
-----
```

	custom/			
	avg_clip50		22.6	
	clipped_episode_r...		31	
	consecutive_clipp...		1	
	max_consecutive_c...		13	
	rollout/			
	ep_len_mean		1.08e+03	
	ep_rew_mean		23.9	
	exploration rate		0.428	
	time/			
	episodes		25700	
	fps		249	
	time_elapsed		6878	
	total timesteps		1716679	
	train/			
	learning_rate		0.0001	
	loss		0.119	
	n_updates		416669	

```
-----
EP=8,680 | STEP=1,718,099 | clip=32 | consec=2 | max_consec=13 | avg_clip50=23.08
EP=8,690 | STEP=1,720,985 | clip=32 | consec=4 | max_consec=13 | avg_clip50=23.12
EP=8,700 | STEP=1,723,726 | clip=35 | consec=1 | max_consec=13 | avg_clip50=23.42
EP=8,710 | STEP=1,726,418 | clip=33 | consec=1 | max_consec=13 | avg_clip50=23.56
-----
```

	custom/			
	avg_clip50		23.6	
	clipped_episode_r...		33	
	consecutive_clipp...		1	
	max_consecutive_c...		13	

rollout/	
ep_len_mean	1.12e+03
ep_rew_mean	24.7
exploration rate	0.425
time/	
episodes	25800
fps	249
time_elapsed	6920
total timesteps	1726418
train/	
learning_rate	0.0001
loss	0.14
n_updates	419104

```

-----
EP=8,720 | STEP=1,729,250 | clip=21 | consec=3 | max_consec=13 | avg_clip50=25.92
EP=8,730 | STEP=1,731,594 | clip=20 | consec=0 | max_consec=13 | avg_clip50=25.36
EP=8,740 | STEP=1,734,029 | clip=34 | consec=1 | max_consec=13 | avg_clip50=24.78
-----

```

custom/	
avg_clip50	24.5
clipped_episode_r...	34
consecutive_clipp...	1
max_consecutive_c...	13
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	23.7
exploration rate	0.422
time/	
episodes	25900
fps	249
time_elapsed	6957
total timesteps	1735415
train/	
learning_rate	0.0001
loss	0.0459
n_updates	421353

```

-----
EP=8,750 | STEP=1,736,711 | clip=15 | consec=0 | max_consec=13 | avg_clip50=24.54
EP=8,760 | STEP=1,739,475 | clip=31 | consec=7 | max_consec=13 | avg_clip50=24.86
EP=8,770 | STEP=1,742,628 | clip=35 | consec=2 | max_consec=14 | avg_clip50=25.28
-----

```

custom/	
avg_clip50	26.3
clipped_episode_r...	51
consecutive_clipp...	4
max_consecutive_c...	14
rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	25.9
exploration rate	0.419
time/	
episodes	26000
fps	249
time_elapsed	6998
total timesteps	1745338
train/	
learning_rate	0.0001
loss	0.0447
n_updates	423834

EP=8,780	STEP=1,745,513	clip=21	consec=5	max_consec=14	avg_clip50=26.32
EP=8,790	STEP=1,748,070	clip=27	consec=1	max_consec=14	avg_clip50=26.22
EP=8,800	STEP=1,750,671	clip=30	consec=2	max_consec=14	avg_clip50=26.40
EP=8,810	STEP=1,753,284	clip=30	consec=4	max_consec=14	avg_clip50=25.86

custom/	
avg_clip50	25.8
clipped_episode_r...	28
consecutive_clipp...	6
max_consecutive_c...	14
rollout/	
ep_len_mean	1.12e+03
ep_rew_mean	25.3
exploration rate	0.416
time/	
episodes	26100
fps	249
time_elapsed	7034
total timesteps	1754025
train/	
learning_rate	0.0001
loss	0.0542

n_updates		426006			

EP=8,820	STEP=1,755,785	clip=33	consec=5	max_consec=14	avg_clip50=24.94
EP=8,830	STEP=1,758,185	clip=25	consec=2	max_consec=14	avg_clip50=24.34
EP=8,840	STEP=1,760,305	clip=20	consec=0	max_consec=14	avg_clip50=23.78

custom/					
avg_clip50		23.3			
clipped_episode_r...		26			
consecutive_clipp...		1			
max_consecutive_c...		14			
rollout/					
ep_len_mean		1.09e+03			
ep_rew_mean		24.8			
exploration rate		0.413			
time/					
episodes		26200			
fps		249			
time_elapsed		7068			
total timesteps		1762077			
train/					
learning_rate		0.0001			
loss		0.0376			
n_updates		428019			

EP=8,850	STEP=1,762,956	clip=39	consec=2	max_consec=14	avg_clip50=23.68
EP=8,860	STEP=1,765,143	clip=10	consec=0	max_consec=14	avg_clip50=22.66
EP=8,870	STEP=1,767,401	clip=16	consec=0	max_consec=14	avg_clip50=21.88
EP=8,880	STEP=1,769,560	clip=22	consec=5	max_consec=14	avg_clip50=21.00

custom/					
avg_clip50		21			
clipped_episode_r...		22			
consecutive_clipp...		5			
max_consecutive_c...		14			
rollout/					
ep_len_mean		1.01e+03			
ep_rew_mean		22.7			
exploration rate		0.411			
time/					
episodes		26300			

	fps		249	
	time_elapsed		7101	
	total timesteps		1769703	
	train/			
	learning_rate		0.0001	
	loss		0.0363	
	n_updates		429925	

```
-----
EP=8,890 | STEP=1,772,576 | clip=32 | consec=4 | max_consec=14 | avg_clip50=23.10
EP=8,900 | STEP=1,775,576 | clip=31 | consec=4 | max_consec=14 | avg_clip50=24.08
EP=8,910 | STEP=1,778,145 | clip=30 | consec=3 | max_consec=14 | avg_clip50=25.18
-----
```

	custom/			
	avg_clip50		25.8	
	clipped_episode_r...		17	
	consecutive_clipp...		0	
	max_consecutive_c...		14	
	rollout/			
	ep_len_mean		1.05e+03	
	ep_rew_mean		24	
	exploration rate		0.407	
	time/			
	episodes		26400	
	fps		249	
	time_elapsed		7142	
	total timesteps		1779692	
	train/			
	learning_rate		0.0001	
	loss		0.106	
	n_updates		432422	

```
-----
EP=8,920 | STEP=1,781,097 | clip=24 | consec=5 | max_consec=14 | avg_clip50=26.18
EP=8,930 | STEP=1,783,729 | clip=19 | consec=0 | max_consec=14 | avg_clip50=26.90
EP=8,940 | STEP=1,786,137 | clip=15 | consec=0 | max_consec=14 | avg_clip50=25.24
EP=8,950 | STEP=1,789,116 | clip=33 | consec=6 | max_consec=14 | avg_clip50=25.04
-----
```

	custom/			
	avg_clip50		25	
	clipped_episode_r...		33	
	consecutive_clipp...		6	
	max_consecutive_c...		14	

rollout/	
ep_len_mean	1.09e+03
ep_rew_mean	24.6
exploration rate	0.404
time/	
episodes	26500
fps	249
time_elapsed	7181
total timesteps	1789116
train/	
learning_rate	0.0001
loss	0.0529
n_updates	434778

```
-----
EP=8,960 | STEP=1,791,664 | clip=30 | consec=4 | max_consec=14 | avg_clip50=25.20
EP=8,970 | STEP=1,794,386 | clip=19 | consec=0 | max_consec=14 | avg_clip50=24.88
EP=8,980 | STEP=1,796,916 | clip=28 | consec=1 | max_consec=14 | avg_clip50=24.32
-----
```

custom/	
avg_clip50	24.5
clipped_episode_r...	22
consecutive_clipp...	5
max_consecutive_c...	14
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	25.3
exploration rate	0.401
time/	
episodes	26600
fps	249
time_elapsed	7218
total timesteps	1797992
train/	
learning_rate	0.0001
loss	0.034
n_updates	436997

```
-----
EP=8,990 | STEP=1,799,953 | clip=25 | consec=11 | max_consec=14 | avg_clip50=25.74
EP=9,000 | STEP=1,802,644 | clip=21 | consec=1 | max_consec=17 | avg_clip50=25.40
EP=9,010 | STEP=1,805,141 | clip=22 | consec=1 | max_consec=17 | avg_clip50=25.30
-----
```

custom/	
avg_clip50	25.9
clipped_episode_r...	23
consecutive_clipp...	2
max_consecutive_c...	17
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	25.5
exploration rate	0.398
time/	
episodes	26700
fps	249
time_elapsed	7258
total timesteps	1807467
train/	
learning_rate	0.0001
loss	0.512
n_updates	439366

EP=9,020	STEP=1,807,916	clip=16	consec=0	max_consec=17	avg_clip50=25.54
EP=9,030	STEP=1,810,763	clip=35	consec=6	max_consec=17	avg_clip50=26.84
EP=9,040	STEP=1,813,735	clip=29	consec=4	max_consec=17	avg_clip50=26.52
EP=9,050	STEP=1,816,445	clip=32	consec=2	max_consec=17	avg_clip50=26.46

custom/	
avg_clip50	26.5
clipped_episode_r...	21
consecutive_clipp...	4
max_consecutive_c...	17
rollout/	
ep_len_mean	1.14e+03
ep_rew_mean	26
exploration rate	0.395
time/	
episodes	26800
fps	248
time_elapsed	7298
total timesteps	1817330
train/	
learning_rate	0.0001
loss	0.0771

n_updates		441832			
EP=9,060	STEP=1,819,178	clip=7	consec=0	max_consec=17	avg_clip50=26.60
EP=9,070	STEP=1,822,337	clip=27	consec=3	max_consec=17	avg_clip50=27.50
EP=9,080	STEP=1,824,843	clip=16	consec=0	max_consec=17	avg_clip50=27.30
custom/					
avg_clip50		26.9			
clipped_episode_r...		35			
consecutive_clipp...		1			
max_consecutive_c...		17			
rollout/					
ep_len_mean		1.15e+03			
ep_rew_mean		26.9			
exploration rate		0.392			
time/					
episodes		26900			
fps		248			
time_elapsed		7338			
total timesteps		1827042			
train/					
learning_rate		0.0001			
loss		0.553			
n_updates		444260			
EP=9,090	STEP=1,827,580	clip=13	consec=0	max_consec=17	avg_clip50=26.82
EP=9,100	STEP=1,830,155	clip=26	consec=7	max_consec=17	avg_clip50=26.52
EP=9,110	STEP=1,832,878	clip=34	consec=6	max_consec=17	avg_clip50=26.92
EP=9,120	STEP=1,835,107	clip=31	consec=3	max_consec=17	avg_clip50=24.72
custom/					
avg_clip50		24.2			
clipped_episode_r...		10			
consecutive_clipp...		0			
max_consecutive_c...		17			
rollout/					
ep_len_mean		1.12e+03			
ep_rew_mean		25.9			
exploration rate		0.389			
time/					
episodes		27000			

	fps		248	
	time_elapsed		7373	
	total timesteps		1835276	
	train/			
	learning_rate		0.0001	
	loss		0.108	
	n_updates		446318	

```
-----
EP=9,130 | STEP=1,837,787 | clip=33 | consec=5 | max_consec=17 | avg_clip50=24.72
EP=9,140 | STEP=1,840,137 | clip=12 | consec=0 | max_consec=17 | avg_clip50=23.82
EP=9,150 | STEP=1,842,329 | clip=28 | consec=1 | max_consec=17 | avg_clip50=22.88
-----
```

	custom/			
	avg_clip50		23.2	
	clipped_episode_r...		32	
	consecutive_clipp...		6	
	max_consecutive_c...		17	
	rollout/			
	ep_len_mean		1.09e+03	
	ep_rew_mean		24.9	
	exploration rate		0.386	
	time/			
	episodes		27100	
	fps		248	
	time_elapsed		7410	
	total timesteps		1844074	
	train/			
	learning_rate		0.0001	
	loss		0.086	
	n_updates		448518	

```
-----
EP=9,160 | STEP=1,845,282 | clip=32 | consec=2 | max_consec=17 | avg_clip50=22.92
EP=9,170 | STEP=1,847,808 | clip=32 | consec=1 | max_consec=17 | avg_clip50=23.78
EP=9,180 | STEP=1,850,296 | clip=18 | consec=0 | max_consec=17 | avg_clip50=23.40
-----
```

	custom/			
	avg_clip50		24	
	clipped_episode_r...		28	
	consecutive_clipp...		1	
	max_consecutive_c...		17	
	rollout/			

ep_len_mean	1.06e+03
ep_rew_mean	23.9
exploration rate	0.383
time/	
episodes	27200
fps	248
time_elapsed	7447
total timesteps	1853067
train/	
learning_rate	0.0001
loss	0.402
n_updates	450766

EP=9,190	STEP=1,853,215	clip=42	consec=2	max_consec=17	avg_clip50=24.64
EP=9,200	STEP=1,855,878	clip=34	consec=1	max_consec=17	avg_clip50=25.56
EP=9,210	STEP=1,858,565	clip=31	consec=6	max_consec=17	avg_clip50=25.88
EP=9,220	STEP=1,861,344	clip=69	consec=1	max_consec=17	avg_clip50=26.96

custom/	
avg_clip50	26.9
clipped_episode_r...	35
consecutive_clipp...	1
max_consecutive_c...	17
rollout/	
ep_len_mean	1.08e+03
ep_rew_mean	25.1
exploration rate	0.38
time/	
episodes	27300
fps	248
time_elapsed	7486
total timesteps	1862391
train/	
learning_rate	0.0001
loss	0.074
n_updates	453097

EP=9,230	STEP=1,863,560	clip=30	consec=1	max_consec=17	avg_clip50=26.40
EP=9,240	STEP=1,865,885	clip=29	consec=1	max_consec=17	avg_clip50=25.44
EP=9,250	STEP=1,868,274	clip=25	consec=3	max_consec=17	avg_clip50=25.16

custom/	
avg_clip50	23.9
clipped_episode_r...	31
consecutive_clipp...	1
max_consecutive_c...	17
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	24.9
exploration rate	0.377
time/	
episodes	27400
fps	248
time_elapsed	7520
total timesteps	1870344
train/	
learning_rate	0.0001
loss	0.0718
n_updates	455085

```

-----
EP=9,260 | STEP=1,870,592 | clip=23 | consec=2 | max_consec=17 | avg_clip50=23.76
EP=9,270 | STEP=1,872,962 | clip=21 | consec=5 | max_consec=17 | avg_clip50=22.62
EP=9,280 | STEP=1,875,419 | clip=23 | consec=2 | max_consec=17 | avg_clip50=22.76
EP=9,290 | STEP=1,878,032 | clip=26 | consec=1 | max_consec=17 | avg_clip50=23.16

```

custom/	
avg_clip50	24.1
clipped_episode_r...	28
consecutive_clipp...	4
max_consecutive_c...	17
rollout/	
ep_len_mean	1.04e+03
ep_rew_mean	24.6
exploration rate	0.374
time/	
episodes	27500
fps	248
time_elapsed	7557
total timesteps	1879068
train/	
learning_rate	0.0001
loss	0.0602

n_updates		457266			

EP=9,300	STEP=1,880,776	clip=34	consec=5	max_consec=17	avg_clip50=24.08
EP=9,310	STEP=1,883,202	clip=13	consec=0	max_consec=17	avg_clip50=24.42
EP=9,320	STEP=1,886,222	clip=28	consec=5	max_consec=17	avg_clip50=25.52

custom/		
avg_clip50		26.9
clipped_episode_r...		31
consecutive_clipp...		14
max_consecutive_c...		17
rollout/		
ep_len_mean		1.07e+03
ep_rew_mean		24.9
exploration rate		0.371
time/		
episodes		27600
fps		248
time_elapsed		7598
total timesteps		1888908
train/		
learning_rate		0.0001
loss		0.0222
n_updates		459726

EP=9,330	STEP=1,889,135	clip=31	consec=15	max_consec=17	avg_clip50=27.10
EP=9,340	STEP=1,891,942	clip=26	consec=25	max_consec=25	avg_clip50=27.96
EP=9,350	STEP=1,894,480	clip=16	consec=0	max_consec=25	avg_clip50=27.36
EP=9,360	STEP=1,897,087	clip=35	consec=5	max_consec=25	avg_clip50=27.92

custom/		
avg_clip50		27.6
clipped_episode_r...		29
consecutive_clipp...		3
max_consecutive_c...		25
rollout/		
ep_len_mean		1.11e+03
ep_rew_mean		26.3
exploration rate		0.368
time/		
episodes		27700

	fps		248	
	time_elapsed		7637	
	total timesteps		1898421	
	train/			
	learning_rate		0.0001	
	loss		0.0968	
	n_updates		462105	

```
-----
EP=9,370 | STEP=1,899,813 | clip=22 | consec=9 | max_consec=25 | avg_clip50=27.46
EP=9,380 | STEP=1,902,401 | clip=20 | consec=0 | max_consec=25 | avg_clip50=26.80
EP=9,390 | STEP=1,904,959 | clip=30 | consec=3 | max_consec=25 | avg_clip50=26.12
-----
```

	custom/			
	avg_clip50		25.5	
	clipped_episode_r...		14	
	consecutive_clipp...		0	
	max_consecutive_c...		25	
	rollout/			
	ep_len_mean		1.11e+03	
	ep_rew_mean		26.6	
	exploration rate		0.365	
	time/			
	episodes		27800	
	fps		248	
	time_elapsed		7674	
	total timesteps		1907057	
	train/			
	learning_rate		0.0001	
	loss		0.172	
	n_updates		464264	

```
-----
EP=9,400 | STEP=1,907,501 | clip=34 | consec=2 | max_consec=25 | avg_clip50=26.06
EP=9,410 | STEP=1,910,068 | clip=38 | consec=1 | max_consec=25 | avg_clip50=25.80
EP=9,420 | STEP=1,912,417 | clip=34 | consec=4 | max_consec=25 | avg_clip50=24.92
EP=9,430 | STEP=1,915,113 | clip=33 | consec=6 | max_consec=25 | avg_clip50=24.74
-----
```

	custom/			
	avg_clip50		24.6	
	clipped_episode_r...		10	
	consecutive_clipp...		0	
	max_consecutive_c...		25	

rollout/	
ep_len_mean	1.07e+03
ep_rew_mean	25.5
exploration rate	0.362
time/	
episodes	27900
fps	248
time_elapsed	7710
total timesteps	1915765
train/	
learning_rate	0.0001
loss	0.13
n_updates	466441

```

-----
EP=9,440 | STEP=1,917,083 | clip=18 | consec=0 | max_consec=25 | avg_clip50=23.44
EP=9,450 | STEP=1,919,456 | clip=19 | consec=0 | max_consec=25 | avg_clip50=22.92
EP=9,460 | STEP=1,922,167 | clip=32 | consec=10 | max_consec=25 | avg_clip50=23.62
-----

```

custom/	
avg_clip50	25.3
clipped_episode_r...	28
consecutive_clipp...	17
max_consecutive_c...	25
rollout/	
ep_len_mean	1.06e+03
ep_rew_mean	25
exploration rate	0.359
time/	
episodes	28000
fps	248
time_elapsed	7747
total timesteps	1924479
train/	
learning_rate	0.0001
loss	0.779
n_updates	468619

```

-----
EP=9,470 | STEP=1,925,231 | clip=24 | consec=20 | max_consec=25 | avg_clip50=25.30
EP=9,480 | STEP=1,927,445 | clip=16 | consec=0 | max_consec=25 | avg_clip50=24.60
EP=9,490 | STEP=1,930,076 | clip=23 | consec=1 | max_consec=25 | avg_clip50=25.80
EP=9,500 | STEP=1,932,886 | clip=24 | consec=3 | max_consec=25 | avg_clip50=26.48
-----

```

custom/		
avg_clip50	26.3	
clipped_episode_r...	33	
consecutive_clipp...	1	
max_consecutive_c...	25	
rollout/		
ep_len_mean	1.07e+03	
ep_rew_mean	24.8	
exploration rate	0.356	
time/		
episodes	28100	
fps	248	
time_elapsed	7785	
total timesteps	1933528	
train/		
learning_rate	0.0001	
loss	0.491	
n_updates	470881	

EP=9,510 | STEP=1,935,992 | clip=32 | consec=9 | max_consec=25 | avg_clip50=26.88
 EP=9,520 | STEP=1,938,461 | clip=22 | consec=4 | max_consec=25 | avg_clip50=25.72
 EP=9,530 | STEP=1,941,598 | clip=22 | consec=14 | max_consec=25 | avg_clip50=27.20

custom/		
avg_clip50	28	
clipped_episode_r...	24	
consecutive_clipp...	3	
max_consecutive_c...	25	
rollout/		
ep_len_mean	1.14e+03	
ep_rew_mean	26.9	
exploration rate	0.353	
time/		
episodes	28200	
fps	248	
time_elapsed	7830	
total timesteps	1944272	
train/		
learning_rate	0.0001	
loss	0.102	

n_updates		473567			

EP=9,540	STEP=1,944,552	clip=29	consec=5	max_consec=25	avg_clip50=28.28
EP=9,550	STEP=1,946,942	clip=34	consec=2	max_consec=25	avg_clip50=27.76
EP=9,560	STEP=1,949,645	clip=26	consec=4	max_consec=25	avg_clip50=26.66
EP=9,570	STEP=1,952,422	clip=19	consec=0	max_consec=25	avg_clip50=27.02

custom/		
avg_clip50		26.9
clipped_episode_r...		31
consecutive_clipp...		3
max_consecutive_c...		25
rollout/		
ep_len_mean		1.14e+03
ep_rew_mean		26.6
exploration rate		0.35
time/		
episodes		28300
fps		248
time_elapsed		7869
total timesteps		1953352
train/		
learning_rate		0.0001
loss		0.0894
n_updates		475837

EP=9,580	STEP=1,955,623	clip=30	consec=10	max_consec=25	avg_clip50=27.62
EP=9,590	STEP=1,958,331	clip=21	consec=1	max_consec=25	avg_clip50=26.64
EP=9,600	STEP=1,961,152	clip=22	consec=2	max_consec=25	avg_clip50=27.90

custom/		
avg_clip50		27.7
clipped_episode_r...		15
consecutive_clipp...		0
max_consecutive_c...		25
rollout/		
ep_len_mean		1.15e+03
ep_rew_mean		27.2
exploration rate		0.346
time/		
episodes		28400

fps	248
time_elapsed	7909
total timesteps	1962927
train/	
learning_rate	0.0001
loss	0.0799
n_updates	478231

```

-----
EP=9,610 | STEP=1,963,535 | clip=35 | consec=2 | max_consec=25 | avg_clip50=27.72
EP=9,620 | STEP=1,966,779 | clip=24 | consec=12 | max_consec=25 | avg_clip50=28.58
EP=9,630 | STEP=1,969,308 | clip=27 | consec=5 | max_consec=25 | avg_clip50=27.34
EP=9,640 | STEP=1,972,303 | clip=34 | consec=5 | max_consec=25 | avg_clip50=27.96

```

custom/	
avg_clip50	27.9
clipped_episode_r...	33
consecutive_clipp...	8
max_consecutive_c...	25
rollout/	
ep_len_mean	1.17e+03
ep_rew_mean	27.6
exploration rate	0.343
time/	
episodes	28500
fps	248
time_elapsed	7953
total timesteps	1973494
train/	
learning_rate	0.0001
loss	0.0341
n_updates	480873

```

-----
EP=9,650 | STEP=1,975,035 | clip=27 | consec=3 | max_consec=25 | avg_clip50=27.46
EP=9,660 | STEP=1,977,733 | clip=25 | consec=5 | max_consec=25 | avg_clip50=27.80
EP=9,670 | STEP=1,980,637 | clip=32 | consec=8 | max_consec=25 | avg_clip50=27.08

```

custom/	
avg_clip50	27.4
clipped_episode_r...	30
consecutive_clipp...	3
max_consecutive_c...	25

rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	27.3
exploration rate	0.34
time/	
episodes	28600
fps	248
time_elapsed	7992
total timesteps	1982768
train/	
learning_rate	0.0001
loss	0.06
n_updates	483191

```

-----
EP=9,680 | STEP=1,983,149 | clip=16 | consec=0 | max_consec=25 | avg_clip50=26.98
EP=9,690 | STEP=1,985,699 | clip=21 | consec=5 | max_consec=25 | avg_clip50=26.14
EP=9,700 | STEP=1,988,293 | clip=30 | consec=7 | max_consec=25 | avg_clip50=25.70
EP=9,710 | STEP=1,990,927 | clip=32 | consec=1 | max_consec=25 | avg_clip50=25.54
-----

```

custom/	
avg_clip50	25.8
clipped_episode_r...	27
consecutive_clipp...	4
max_consecutive_c...	25
rollout/	
ep_len_mean	1.14e+03
ep_rew_mean	26.6
exploration rate	0.337
time/	
episodes	28700
fps	248
time_elapsed	8030
total timesteps	1991755
train/	
learning_rate	0.0001
loss	0.0605
n_updates	485438

```

-----
EP=9,720 | STEP=1,993,840 | clip=4 | consec=0 | max_consec=25 | avg_clip50=25.56
EP=9,730 | STEP=1,996,597 | clip=21 | consec=8 | max_consec=25 | avg_clip50=26.28
EP=9,740 | STEP=1,999,171 | clip=35 | consec=3 | max_consec=25 | avg_clip50=26.58
-----

```

custom/		
avg_clip50	27.1	
clipped_episode_r...	28	
consecutive_clipp...	1	
max_consecutive_c...	25	
rollout/		
ep_len_mean	1.11e+03	
ep_rew_mean	26.3	
exploration rate	0.334	
time/		
episodes	28800	
fps	247	
time_elapsed	8071	
total timesteps	2001218	
train/		
learning_rate	0.0001	
loss	0.545	
n_updates	487804	

```

EP=9,750 | STEP=2,001,680 | clip=30 | consec=4 | max_consec=25 | avg_clip50=26.90
EP=9,760 | STEP=2,003,920 | clip=13 | consec=0 | max_consec=25 | avg_clip50=26.26
EP=9,770 | STEP=2,007,040 | clip=35 | consec=1 | max_consec=25 | avg_clip50=26.58
EP=9,780 | STEP=2,010,137 | clip=19 | consec=0 | max_consec=25 | avg_clip50=26.60

```

custom/		
avg_clip50	26.5	
clipped_episode_r...	29	
consecutive_clipp...	1	
max_consecutive_c...	25	
rollout/		
ep_len_mean	1.12e+03	
ep_rew_mean	26.5	
exploration rate	0.33	
time/		
episodes	28900	
fps	247	
time_elapsed	8113	
total timesteps	2011211	
train/		
learning_rate	0.0001	

loss	0.0476
n_updates	490302

EP=9,790	STEP=2,012,842	clip=33	consec=4	max_consec=25	avg_clip50=26.78
EP=9,800	STEP=2,015,708	clip=31	consec=9	max_consec=25	avg_clip50=27.48
EP=9,810	STEP=2,018,397	clip=24	consec=5	max_consec=25	avg_clip50=28.82

custom/	
avg_clip50	28.8
clipped_episode_r...	27
consecutive_clipp...	6
max_consecutive_c...	25
rollout/	
ep_len_mean	1.13e+03
ep_rew_mean	27.4
exploration rate	0.327
time/	
episodes	29000
fps	247
time_elapsed	8155
total timesteps	2020927
train/	
learning_rate	0.0001
loss	0.285
n_updates	492731

EP=9,820	STEP=2,021,089	clip=18	consec=0	max_consec=25	avg_clip50=28.48
EP=9,830	STEP=2,024,176	clip=30	consec=4	max_consec=25	avg_clip50=28.42
EP=9,840	STEP=2,026,765	clip=32	consec=9	max_consec=25	avg_clip50=28.86
EP=9,850	STEP=2,029,979	clip=32	consec=2	max_consec=25	avg_clip50=29.28

custom/	
avg_clip50	28.8
clipped_episode_r...	26
consecutive_clipp...	5
max_consecutive_c...	25
rollout/	
ep_len_mean	1.18e+03
ep_rew_mean	28.5
exploration rate	0.324
time/	

episodes	29100	
fps	247	
time_elapsed	8197	
total timesteps	2030835	
train/		
learning_rate	0.0001	
loss	0.0512	
n_updates	495208	

```
-----
EP=9,860 | STEP=2,033,078 | clip=18 | consec=0 | max_consec=25 | avg_clip50=29.32
EP=9,870 | STEP=2,035,767 | clip=23 | consec=3 | max_consec=25 | avg_clip50=29.06
EP=9,880 | STEP=2,038,344 | clip=19 | consec=0 | max_consec=25 | avg_clip50=28.70
-----
```

custom/		
avg_clip50	28.9	
clipped_episode_r...	33	
consecutive_clipp...	6	
max_consecutive_c...	25	
rollout/		
ep_len_mean	1.18e+03	
ep_rew_mean	28.8	
exploration rate	0.321	
time/		
episodes	29200	
fps	247	
time_elapsed	8238	
total timesteps	2040532	
train/		
learning_rate	0.0001	
loss	0.0854	
n_updates	497632	

```
-----
EP=9,890 | STEP=2,041,145 | clip=25 | consec=9 | max_consec=25 | avg_clip50=28.56
EP=9,900 | STEP=2,043,779 | clip=18 | consec=0 | max_consec=25 | avg_clip50=27.68
EP=9,910 | STEP=2,046,782 | clip=23 | consec=6 | max_consec=25 | avg_clip50=27.64
EP=9,920 | STEP=2,049,786 | clip=32 | consec=16 | max_consec=25 | avg_clip50=28.40
-----
```

custom/		
avg_clip50	28.5	
clipped_episode_r...	33	
consecutive_clipp...	20	

	max_consecutive_c...	25	
	rollout/		
	ep_len_mean	1.19e+03	
	ep_rew_mean	28.7	
	exploration rate	0.317	
	time/		
	episodes	29300	
	fps	247	
	time_elapsed	8282	
	total timesteps	2051174	
	train/		
	learning_rate	0.0001	
	loss	0.249	
	n_updates	500293	

EP=9,930		STEP=2,052,896		clip=32		consec=26		max_consec=26		avg_clip50=29.50
EP=9,940		STEP=2,055,698		clip=35		consec=4		max_consec=31		avg_clip50=29.60
EP=9,950		STEP=2,058,534		clip=36		consec=3		max_consec=31		avg_clip50=29.74

	custom/		
	avg_clip50	29.5	
	clipped_episode_r...	20	
	consecutive_clipp...	0	
	max_consecutive_c...	31	
	rollout/		
	ep_len_mean	1.17e+03	
	ep_rew_mean	28.5	
	exploration rate	0.314	
	time/		
	episodes	29400	
	fps	247	
	time_elapsed	8324	
	total timesteps	2061032	
	train/		
	learning_rate	0.0001	
	loss	0.0812	
	n_updates	502757	

EP=9,960		STEP=2,061,362		clip=34		consec=1		max_consec=31		avg_clip50=29.70
EP=9,970		STEP=2,064,197		clip=35		consec=6		max_consec=31		avg_clip50=29.46
EP=9,980		STEP=2,067,114		clip=40		consec=6		max_consec=31		avg_clip50=29.12

EP=9,990 | STEP=2,069,789 | clip=29 | consec=6 | max_consec=31 | avg_clip50=29.08

custom/		
avg_clip50	29.1	
clipped_episode_r...	40	
consecutive_clipp...	2	
max_consecutive_c...	31	
rollout/		
ep_len_mean	1.2e+03	
ep_rew_mean	29.4	
exploration rate	0.31	
time/		
episodes	29500	
fps	247	
time_elapsed	8369	
total timesteps	2071510	
train/		
learning_rate	0.0001	
loss	0.0507	
n_updates	505377	

EP=10,000 | STEP=2,072,894 | clip=30 | consec=7 | max_consec=31 | avg_clip50=29.72
 EP=10,010 | STEP=2,076,302 | clip=33 | consec=17 | max_consec=31 | avg_clip50=30.52
 EP=10,020 | STEP=2,079,073 | clip=35 | consec=5 | max_consec=31 | avg_clip50=30.30
 EP=10,030 | STEP=2,081,705 | clip=24 | consec=5 | max_consec=31 | avg_clip50=29.30

custom/		
avg_clip50	29.3	
clipped_episode_r...	30	
consecutive_clipp...	6	
max_consecutive_c...	31	
rollout/		
ep_len_mean	1.2e+03	
ep_rew_mean	29.2	
exploration rate	0.307	
time/		
episodes	29600	
fps	247	
time_elapsed	8413	
total timesteps	2082053	
train/		

learning_rate	0.0001
loss	0.267
n_updates	508013

```

-----
EP=10,040 | STEP=2,084,429 | clip=30 | consec=3 | max_consec=31 | avg_clip50=29.02
EP=10,050 | STEP=2,086,652 | clip=32 | consec=1 | max_consec=31 | avg_clip50=26.86
EP=10,060 | STEP=2,089,329 | clip=6 | consec=0 | max_consec=31 | avg_clip50=25.74

```

custom/	
avg_clip50	26.5
clipped_episode_r...	33
consecutive_clipp...	6
max_consecutive_c...	31
rollout/	
ep_len_mean	1.17e+03
ep_rew_mean	28.4
exploration rate	0.304
time/	
episodes	29700
fps	247
time_elapsed	8452
total timesteps	2091271
train/	
learning_rate	0.0001
loss	0.0438
n_updates	510317

```

-----
EP=10,070 | STEP=2,092,255 | clip=27 | consec=2 | max_consec=31 | avg_clip50=26.16
EP=10,080 | STEP=2,095,240 | clip=27 | consec=3 | max_consec=31 | avg_clip50=26.76
EP=10,090 | STEP=2,098,018 | clip=31 | consec=4 | max_consec=31 | avg_clip50=27.00
EP=10,100 | STEP=2,101,341 | clip=47 | consec=8 | max_consec=31 | avg_clip50=29.40

```

custom/	
avg_clip50	29
clipped_episode_r...	33
consecutive_clipp...	1
max_consecutive_c...	31
rollout/	
ep_len_mean	1.18e+03
ep_rew_mean	27.9
exploration rate	0.3

time/	
episodes	29800
fps	247
time_elapsed	8497
total timesteps	2101772
train/	
learning_rate	0.0001
loss	0.0583
n_updates	512942

```
-----
EP=10,110 | STEP=2,103,911 | clip=25 | consec=2 | max_consec=31 | avg_clip50=28.86
EP=10,120 | STEP=2,106,759 | clip=10 | consec=0 | max_consec=31 | avg_clip50=28.72
EP=10,130 | STEP=2,109,616 | clip=25 | consec=1 | max_consec=31 | avg_clip50=28.84
-----
```

custom/	
avg_clip50	29.2
clipped_episode_r...	33
consecutive_clipp...	8
max_consecutive_c...	31
rollout/	
ep_len_mean	1.17e+03
ep_rew_mean	28.1
exploration rate	0.297
time/	
episodes	29900
fps	247
time_elapsed	8538
total timesteps	2111591
train/	
learning_rate	0.0001
loss	0.0582
n_updates	515397

```
-----
EP=10,140 | STEP=2,112,664 | clip=35 | consec=11 | max_consec=31 | avg_clip50=29.28
EP=10,150 | STEP=2,115,817 | clip=33 | consec=1 | max_consec=31 | avg_clip50=29.10
EP=10,160 | STEP=2,118,894 | clip=24 | consec=4 | max_consec=31 | avg_clip50=30.48
EP=10,170 | STEP=2,121,606 | clip=24 | consec=9 | max_consec=31 | avg_clip50=30.08
-----
```

custom/	
avg_clip50	29.6
clipped_episode_r...	21

consecutive_clipp...	1
max_consecutive_c...	31
rollout/	
ep_len_mean	1.2e+03
ep_rew_mean	29.1
exploration rate	0.293
time/	
episodes	30000
fps	247
time_elapsed	8583
total timesteps	2122195
train/	
learning_rate	0.0001
loss	0.0869
n_updates	518048

EP=10,180	STEP=2,124,110	clip=24	consec=2	max_consec=31	avg_clip50=29.24
EP=10,190	STEP=2,127,349	clip=21	consec=12	max_consec=31	avg_clip50=29.58
EP=10,200	STEP=2,129,936	clip=19	consec=0	max_consec=31	avg_clip50=28.66

custom/	
avg_clip50	28.7
clipped_episode_r...	35
consecutive_clipp...	2
max_consecutive_c...	31
rollout/	
ep_len_mean	1.19e+03
ep_rew_mean	29
exploration rate	0.29
time/	
episodes	30100
fps	247
time_elapsed	8624
total timesteps	2131950
train/	
learning_rate	0.0001
loss	0.349
n_updates	520487

EP=10,210	STEP=2,132,860	clip=35	consec=5	max_consec=31	avg_clip50=28.14
EP=10,220	STEP=2,135,718	clip=16	consec=0	max_consec=31	avg_clip50=28.34

```

EP=10,230 | STEP=2,138,880 | clip=29 | consec=10 | max_consec=31 | avg_clip50=29.82
EP=10,240 | STEP=2,141,532 | clip=12 | consec=0 | max_consec=31 | avg_clip50=28.46

```

```

-----
| custom/                |      |
|   avg_clip50            | 28.2  |
|   clipped_episode_r...  | 36    |
|   consecutive_clipp...  | 1     |
|   max_consecutive_c...  | 31    |
| rollout/               |      |
|   ep_len_mean           | 1.2e+03 |
|   ep_rew_mean           | 29.1   |
|   exploration rate      | 0.287  |
| time/                  |      |
|   episodes              | 30200  |
|   fps                   | 247    |
|   time_elapsed          | 8667   |
|   total timesteps       | 2142054 |
| train/                 |      |
|   learning_rate         | 0.0001 |
|   loss                  | 0.0917 |
|   n_updates             | 523013 |

```

```

EP=10,250 | STEP=2,144,298 | clip=22 | consec=1 | max_consec=31 | avg_clip50=28.60
EP=10,260 | STEP=2,147,371 | clip=34 | consec=4 | max_consec=31 | avg_clip50=29.08
EP=10,270 | STEP=2,149,855 | clip=29 | consec=1 | max_consec=31 | avg_clip50=28.46

```

```

-----
| custom/                |      |
|   avg_clip50            | 28.2  |
|   clipped_episode_r...  | 33    |
|   consecutive_clipp...  | 10    |
|   max_consecutive_c...  | 31    |
| rollout/               |      |
|   ep_len_mean           | 1.19e+03 |
|   ep_rew_mean           | 29     |
|   exploration rate      | 0.283  |
| time/                  |      |
|   episodes              | 30300  |
|   fps                   | 247    |
|   time_elapsed          | 8711   |
|   total timesteps       | 2152553 |
| train/                 |      |

```

learning_rate	0.0001				
loss	0.0271				
n_updates	525638				

EP=10,280	STEP=2,152,765	clip=22	consec=11	max_consec=31	avg_clip50=28.08
EP=10,290	STEP=2,155,267	clip=21	consec=1	max_consec=31	avg_clip50=28.18
EP=10,300	STEP=2,157,726	clip=13	consec=0	max_consec=31	avg_clip50=28.02
EP=10,310	STEP=2,160,169	clip=32	consec=4	max_consec=31	avg_clip50=26.70

custom/					
avg_clip50	26.7				
clipped_episode_r...	24				
consecutive_clipp...	1				
max_consecutive_c...	31				
rollout/					
ep_len_mean	1.12e+03				
ep_rew_mean	27.4				
exploration rate	0.28				
time/					
episodes	30400				
fps	247				
time_elapsed	8749				
total timesteps	2161288				
train/					
learning_rate	0.0001				
loss	0.0413				
n_updates	527821				

EP=10,320	STEP=2,162,402	clip=20	consec=0	max_consec=31	avg_clip50=26.20
EP=10,330	STEP=2,164,749	clip=34	consec=3	max_consec=31	avg_clip50=25.02
EP=10,340	STEP=2,167,510	clip=22	consec=1	max_consec=31	avg_clip50=25.30

custom/					
avg_clip50	24.8				
clipped_episode_r...	32				
consecutive_clipp...	2				
max_consecutive_c...	31				
rollout/					
ep_len_mean	1.07e+03				
ep_rew_mean	26.5				
exploration rate	0.277				

time/	
episodes	30500
fps	246
time_elapsed	8785
total timesteps	2169750
train/	
learning_rate	0.0001
loss	0.0543
n_updates	529937

```
-----
EP=10,350 | STEP=2,169,967 | clip=33 | consec=3 | max_consec=31 | avg_clip50=25.18
EP=10,360 | STEP=2,172,601 | clip=25 | consec=13 | max_consec=31 | avg_clip50=26.12
EP=10,370 | STEP=2,175,223 | clip=20 | consec=0 | max_consec=31 | avg_clip50=27.00
EP=10,380 | STEP=2,177,352 | clip=9 | consec=0 | max_consec=31 | avg_clip50=26.04
-----
```

custom/	
avg_clip50	26.1
clipped_episode_r...	35
consecutive_clipp...	3
max_consecutive_c...	31
rollout/	
ep_len_mean	1.02e+03
ep_rew_mean	25.5
exploration rate	0.274
time/	
episodes	30600
fps	246
time_elapsed	8823
total timesteps	2178773
train/	
learning_rate	0.0001
loss	0.0409
n_updates	532193

```
-----
EP=10,390 | STEP=2,180,101 | clip=32 | consec=1 | max_consec=31 | avg_clip50=26.42
EP=10,400 | STEP=2,182,530 | clip=11 | consec=0 | max_consec=31 | avg_clip50=25.94
EP=10,410 | STEP=2,185,106 | clip=33 | consec=3 | max_consec=31 | avg_clip50=25.26
EP=10,420 | STEP=2,187,662 | clip=22 | consec=13 | max_consec=31 | avg_clip50=25.46
-----
```

custom/	
avg_clip50	25.1

	clipped_episode_r...	32	
	consecutive_clipp...	1	
	max_consecutive_c...	31	
	rollout/		
	ep_len_mean	1.05e+03	
	ep_rew_mean	26.1	
	exploration rate	0.271	
	time/		
	episodes	30700	
	fps	246	
	time_elapsed	8864	
	total timesteps	2188339	
	train/		
	learning_rate	0.0001	
	loss	0.283	
	n_updates	534584	

EP=10,430		STEP=2,190,319		clip=27		consec=9		max_consec=31		avg_clip50=26.88
EP=10,440		STEP=2,192,665		clip=28		consec=2		max_consec=31		avg_clip50=25.48
EP=10,450		STEP=2,195,059		clip=25		consec=1		max_consec=31		avg_clip50=25.46

	custom/		
	avg_clip50	25.3	
	clipped_episode_r...	24	
	consecutive_clipp...	2	
	max_consecutive_c...	31	
	rollout/		
	ep_len_mean	1.03e+03	
	ep_rew_mean	25.1	
	exploration rate	0.269	
	time/		
	episodes	30800	
	fps	246	
	time_elapsed	8900	
	total timesteps	2196668	
	train/		
	learning_rate	0.0001	
	loss	0.448	
	n_updates	536666	

EP=10,460		STEP=2,197,404		clip=32		consec=5		max_consec=31		avg_clip50=25.14
-----------	--	----------------	--	---------	--	----------	--	---------------	--	------------------

EP=10,470	STEP=2,200,006	clip=15	consec=0	max_consec=31	avg_clip50=24.68
EP=10,480	STEP=2,202,513	clip=33	consec=3	max_consec=31	avg_clip50=24.24
EP=10,490	STEP=2,205,047	clip=27	consec=1	max_consec=31	avg_clip50=24.92

custom/		
avg_clip50	24.8	
clipped_episode_r...	31	
consecutive_clipp...	4	
max_consecutive_c...	31	
rollout/		
ep_len_mean	1.04e+03	
ep_rew_mean	25.4	
exploration rate	0.265	
time/		
episodes	30900	
fps	246	
time_elapsed	8941	
total timesteps	2206178	
train/		
learning_rate	0.0001	
loss	0.417	
n_updates	539044	

EP=10,500	STEP=2,207,831	clip=31	consec=1	max_consec=31	avg_clip50=26.04
EP=10,510	STEP=2,210,753	clip=33	consec=5	max_consec=31	avg_clip50=27.14
EP=10,520	STEP=2,213,418	clip=23	consec=1	max_consec=31	avg_clip50=27.28

custom/		
avg_clip50	27.8	
clipped_episode_r...	9	
consecutive_clipp...	0	
max_consecutive_c...	31	
rollout/		
ep_len_mean	1.07e+03	
ep_rew_mean	25.9	
exploration rate	0.262	
time/		
episodes	31000	
fps	246	
time_elapsed	8982	
total timesteps	2215664	

train/	
learning_rate	0.0001
loss	0.313
n_updates	541415

EP=10,530	STEP=2,215,934	clip=33	consec=1	max_consec=31	avg_clip50=27.78
EP=10,540	STEP=2,218,972	clip=34	consec=1	max_consec=31	avg_clip50=29.10
EP=10,550	STEP=2,221,792	clip=10	consec=0	max_consec=31	avg_clip50=28.88
EP=10,560	STEP=2,224,549	clip=31	consec=6	max_consec=31	avg_clip50=28.72

custom/	
avg_clip50	29.6
clipped_episode_r...	60
consecutive_clipp...	11
max_consecutive_c...	31
rollout/	
ep_len_mean	1.15e+03
ep_rew_mean	28.4
exploration rate	0.259
time/	
episodes	31100
fps	246
time_elapsed	9028
total timesteps	2226567
train/	
learning_rate	0.0001
loss	0.085
n_updates	544141

EP=10,570	STEP=2,228,099	clip=33	consec=16	max_consec=31	avg_clip50=30.46
EP=10,580	STEP=2,230,846	clip=30	consec=26	max_consec=31	avg_clip50=30.98
EP=10,590	STEP=2,233,852	clip=29	consec=36	max_consec=36	avg_clip50=30.70
EP=10,600	STEP=2,236,486	clip=15	consec=0	max_consec=36	avg_clip50=30.62

custom/	
avg_clip50	30.7
clipped_episode_r...	34
consecutive_clipp...	2
max_consecutive_c...	36
rollout/	
ep_len_mean	1.19e+03

ep_rew_mean	29.8
exploration rate	0.255
time/	
episodes	31200
fps	246
time_elapsed	9073
total timesteps	2237217
train/	
learning_rate	0.0001
loss	0.026
n_updates	546804

```

-----
EP=10,610 | STEP=2,239,113 | clip=35 | consec=1 | max_consec=36 | avg_clip50=30.48
EP=10,620 | STEP=2,241,989 | clip=32 | consec=7 | max_consec=36 | avg_clip50=29.20
EP=10,630 | STEP=2,245,076 | clip=24 | consec=5 | max_consec=36 | avg_clip50=29.48
-----

```

custom/	
avg_clip50	28.8
clipped_episode_r...	25
consecutive_clipp...	3
max_consecutive_c...	36
rollout/	
ep_len_mean	1.2e+03
ep_rew_mean	29.8
exploration rate	0.252
time/	
episodes	31300
fps	246
time_elapsed	9117
total timesteps	2247468
train/	
learning_rate	0.0001
loss	0.0637
n_updates	549366

```

-----
EP=10,640 | STEP=2,247,788 | clip=34 | consec=4 | max_consec=36 | avg_clip50=28.88
EP=10,650 | STEP=2,250,932 | clip=32 | consec=14 | max_consec=36 | avg_clip50=29.72
EP=10,660 | STEP=2,253,518 | clip=28 | consec=1 | max_consec=36 | avg_clip50=29.44
EP=10,670 | STEP=2,256,160 | clip=42 | consec=2 | max_consec=36 | avg_clip50=29.02
-----

```

custom/	
---------	--

avg_clip50	28.8
clipped_episode_r...	33
consecutive_clipp...	5
max_consecutive_c...	36
rollout/	
ep_len_mean	1.16e+03
ep_rew_mean	29.1
exploration rate	0.248
time/	
episodes	31400
fps	246
time_elapsed	9159
total timesteps	2257305
train/	
learning_rate	0.0001
loss	0.321
n_updates	551826

EP=10,680	STEP=2,258,833	clip=32	consec=2	max_consec=36	avg_clip50=28.22
EP=10,690	STEP=2,261,971	clip=64	consec=3	max_consec=36	avg_clip50=28.78
EP=10,700	STEP=2,265,122	clip=33	consec=13	max_consec=36	avg_clip50=28.78

custom/	
avg_clip50	29.4
clipped_episode_r...	33
consecutive_clipp...	3
max_consecutive_c...	36
rollout/	
ep_len_mean	1.2e+03
ep_rew_mean	29.6
exploration rate	0.245
time/	
episodes	31500
fps	246
time_elapsed	9203
total timesteps	2267647
train/	
learning_rate	0.0001
loss	0.0582
n_updates	554411

EP=10,710	STEP=2,268,070	clip=27	consec=5	max_consec=36	avg_clip50=29.66
EP=10,720	STEP=2,270,476	clip=34	consec=1	max_consec=36	avg_clip50=29.56
EP=10,730	STEP=2,273,531	clip=35	consec=11	max_consec=36	avg_clip50=30.42
EP=10,740	STEP=2,276,238	clip=33	consec=5	max_consec=36	avg_clip50=29.84

custom/		
avg_clip50	29.8	
clipped_episode_r...	23	
consecutive_clipp...	8	
max_consecutive_c...	36	
rollout/		
ep_len_mean	1.18e+03	
ep_rew_mean	29.3	
exploration rate	0.242	
time/		
episodes	31600	
fps	246	
time_elapsed	9244	
total timesteps	2277260	
train/		
learning_rate	0.0001	
loss	0.0365	
n_updates	556814	

EP=10,750	STEP=2,279,135	clip=34	consec=15	max_consec=36	avg_clip50=29.58
EP=10,760	STEP=2,281,923	clip=32	consec=4	max_consec=36	avg_clip50=29.16
EP=10,770	STEP=2,285,183	clip=31	consec=14	max_consec=36	avg_clip50=30.48
EP=10,780	STEP=2,287,637	clip=32	consec=1	max_consec=36	avg_clip50=29.58

custom/		
avg_clip50	29.6	
clipped_episode_r...	32	
consecutive_clipp...	1	
max_consecutive_c...	36	
rollout/		
ep_len_mean	1.2e+03	
ep_rew_mean	30	
exploration rate	0.238	
time/		
episodes	31700	
fps	246	

	time_elapsed	9288	
	total timesteps	2287637	
	train/		
	learning_rate	0.0001	
	loss	0.0342	
	n_updates	559409	

```
-----
EP=10,790 | STEP=2,290,723 | clip=38 | consec=11 | max_consec=36 | avg_clip50=30.32
EP=10,800 | STEP=2,293,992 | clip=34 | consec=1 | max_consec=36 | avg_clip50=30.92
EP=10,810 | STEP=2,296,753 | clip=27 | consec=2 | max_consec=36 | avg_clip50=30.94
-----
```

	custom/		
	avg_clip50	30.6	
	clipped_episode_r...	32	
	consecutive_clipp...	7	
	max_consecutive_c...	36	
	rollout/		
	ep_len_mean	1.19e+03	
	ep_rew_mean	30.2	
	exploration rate	0.235	
	time/		
	episodes	31800	
	fps	246	
	time_elapsed	9332	
	total timesteps	2298028	
	train/		
	learning_rate	0.0001	
	loss	0.0554	
	n_updates	562006	

```
-----
EP=10,820 | STEP=2,299,351 | clip=26 | consec=2 | max_consec=36 | avg_clip50=29.98
EP=10,830 | STEP=2,301,705 | clip=1 | consec=0 | max_consec=36 | avg_clip50=29.34
EP=10,840 | STEP=2,303,846 | clip=19 | consec=0 | max_consec=36 | avg_clip50=27.52
-----
```

	custom/		
	avg_clip50	27	
	clipped_episode_r...	33	
	consecutive_clipp...	9	
	max_consecutive_c...	36	
	rollout/		
	ep_len_mean	1.15e+03	

ep_rew_mean	29
exploration rate	0.232
time/	
episodes	31900
fps	246
time_elapsed	9369
total timesteps	2306653
train/	
learning_rate	0.0001
loss	0.048
n_updates	564163

```

-----
EP=10,850 | STEP=2,306,751 | clip=29 | consec=10 | max_consec=36 | avg_clip50=26.94
EP=10,860 | STEP=2,309,441 | clip=33 | consec=5 | max_consec=36 | avg_clip50=26.74
EP=10,870 | STEP=2,312,221 | clip=31 | consec=2 | max_consec=36 | avg_clip50=26.82
EP=10,880 | STEP=2,315,324 | clip=29 | consec=2 | max_consec=36 | avg_clip50=28.12
-----

```

custom/	
avg_clip50	28.3
clipped_episode_r...	28
consecutive_clipp...	1
max_consecutive_c...	36
rollout/	
ep_len_mean	1.14e+03
ep_rew_mean	28.3
exploration rate	0.229
time/	
episodes	32000
fps	246
time_elapsed	9411
total timesteps	2316503
train/	
learning_rate	0.0001
loss	0.074
n_updates	566625

```

-----
EP=10,890 | STEP=2,318,147 | clip=33 | consec=7 | max_consec=36 | avg_clip50=29.58
EP=10,900 | STEP=2,320,933 | clip=19 | consec=0 | max_consec=36 | avg_clip50=29.54
EP=10,910 | STEP=2,324,234 | clip=30 | consec=7 | max_consec=36 | avg_clip50=30.38
EP=10,920 | STEP=2,327,851 | clip=73 | consec=2 | max_consec=36 | avg_clip50=32.52
-----

```

custom/	
avg_clip50	33.1
clipped_episode_r...	35
consecutive_clipp...	3
max_consecutive_c...	36
rollout/	
ep_len_mean	1.18e+03
ep_rew_mean	29.7
exploration rate	0.225
time/	
episodes	32100
fps	246
time_elapsed	9461
total timesteps	2328206
train/	
learning_rate	0.0001
loss	0.0357
n_updates	569551

EP=10,930	STEP=2,331,203	clip=22	consec=12	max_consec=36	avg_clip50=33.06
EP=10,940	STEP=2,333,994	clip=29	consec=6	max_consec=36	avg_clip50=33.04
EP=10,950	STEP=2,336,898	clip=34	consec=5	max_consec=36	avg_clip50=33.00

custom/	
avg_clip50	32.4
clipped_episode_r...	22
consecutive_clipp...	11
max_consecutive_c...	36
rollout/	
ep_len_mean	1.26e+03
ep_rew_mean	31.4
exploration rate	0.221
time/	
episodes	32200
fps	246
time_elapsed	9507
total timesteps	2338822
train/	
learning_rate	0.0001
loss	0.341
n_updates	572205

```

-----
EP=10,960 | STEP=2,340,325 | clip=62 | consec=1 | max_consec=36 | avg_clip50=33.18
EP=10,970 | STEP=2,343,423 | clip=29 | consec=11 | max_consec=36 | avg_clip50=31.54
EP=10,980 | STEP=2,346,371 | clip=31 | consec=4 | max_consec=36 | avg_clip50=31.04
EP=10,990 | STEP=2,349,145 | clip=33 | consec=2 | max_consec=36 | avg_clip50=30.94
-----

```

```

| custom/
|   avg_clip50          | 30.8
|   clipped_episode_r... | 32
|   consecutive_clipp... | 1
|   max_consecutive_c... | 36
| rollout/
|   ep_len_mean         | 1.29e+03
|   ep_rew_mean         | 32
|   exploration rate    | 0.217
| time/
|   episodes            | 32300
|   fps                 | 245
|   time_elapsed        | 9555
|   total timesteps     | 2350299
| train/
|   learning_rate       | 0.0001
|   loss                | 0.0426
|   n_updates           | 575074

```

```

-----
EP=11,000 | STEP=2,351,954 | clip=30 | consec=5 | max_consec=36 | avg_clip50=30.50
EP=11,010 | STEP=2,354,920 | clip=30 | consec=15 | max_consec=36 | avg_clip50=29.98
EP=11,020 | STEP=2,357,941 | clip=24 | consec=25 | max_consec=36 | avg_clip50=30.08
EP=11,030 | STEP=2,362,026 | clip=34 | consec=35 | max_consec=36 | avg_clip50=32.44
-----

```

```

| custom/
|   avg_clip50          | 32.4
|   clipped_episode_r... | 34
|   consecutive_clipp... | 35
|   max_consecutive_c... | 36
| rollout/
|   ep_len_mean         | 1.28e+03
|   ep_rew_mean         | 31.7
|   exploration rate    | 0.213
| time/
|   episodes            | 32400

```

fps	245
time_elapsed	9605
total timesteps	2362026
train/	
learning_rate	0.0001
loss	0.225
n_updates	578006

```

-----
EP=11,040 | STEP=2,364,880 | clip=27 | consec=5 | max_consec=39 | avg_clip50=32.60
EP=11,050 | STEP=2,367,977 | clip=29 | consec=15 | max_consec=39 | avg_clip50=33.34
EP=11,060 | STEP=2,371,776 | clip=35 | consec=25 | max_consec=39 | avg_clip50=34.98
-----

```

custom/	
avg_clip50	35.3
clipped_episode_r...	34
consecutive_clipp...	31
max_consecutive_c...	39
rollout/	
ep_len_mean	1.3e+03
ep_rew_mean	32.6
exploration rate	0.21
time/	
episodes	32500
fps	245
time_elapsed	9656
total timesteps	2373794
train/	
learning_rate	0.0001
loss	0.0498
n_updates	580948

```

-----
EP=11,070 | STEP=2,374,889 | clip=19 | consec=0 | max_consec=39 | avg_clip50=35.00
EP=11,080 | STEP=2,378,136 | clip=26 | consec=10 | max_consec=39 | avg_clip50=33.08
EP=11,090 | STEP=2,381,728 | clip=31 | consec=3 | max_consec=39 | avg_clip50=34.24
EP=11,100 | STEP=2,384,747 | clip=35 | consec=3 | max_consec=39 | avg_clip50=33.98
-----

```

custom/	
avg_clip50	34
clipped_episode_r...	35
consecutive_clipp...	3
max_consecutive_c...	39

rollout/	
ep_len_mean	1.36e+03
ep_rew_mean	33.7
exploration rate	0.206
time/	
episodes	32600
fps	245
time_elapsed	9703
total timesteps	2384948
train/	
learning_rate	0.0001
loss	0.0688
n_updates	583736

```

-----
EP=11,110 | STEP=2,387,904 | clip=36 | consec=13 | max_consec=39 | avg_clip50=32.80
EP=11,120 | STEP=2,390,655 | clip=34 | consec=3 | max_consec=39 | avg_clip50=32.46
EP=11,130 | STEP=2,393,995 | clip=35 | consec=13 | max_consec=39 | avg_clip50=32.72
-----

```

custom/	
avg_clip50	31.5
clipped_episode_r...	33
consecutive_clipp...	19
max_consecutive_c...	39
rollout/	
ep_len_mean	1.33e+03
ep_rew_mean	33
exploration rate	0.202
time/	
episodes	32700
fps	245
time_elapsed	9749
total timesteps	2395742
train/	
learning_rate	0.0001
loss	0.0313
n_updates	586435

```

-----
EP=11,140 | STEP=2,396,963 | clip=24 | consec=23 | max_consec=39 | avg_clip50=31.64
EP=11,150 | STEP=2,399,922 | clip=31 | consec=9 | max_consec=39 | avg_clip50=31.46
EP=11,160 | STEP=2,402,696 | clip=36 | consec=2 | max_consec=39 | avg_clip50=30.68
EP=11,170 | STEP=2,406,418 | clip=30 | consec=12 | max_consec=39 | avg_clip50=32.42
-----

```

custom/		
avg_clip50	32.1	
clipped_episode_r...	31	
consecutive_clipp...	13	
max_consecutive_c...	39	
rollout/		
ep_len_mean	1.3e+03	
ep_rew_mean	32.3	
exploration rate	0.199	
time/		
episodes	32800	
fps	245	
time_elapsed	9797	
total timesteps	2406728	
train/		
learning_rate	0.0001	
loss	0.5	
n_updates	589181	

```

EP=11,180 | STEP=2,409,292 | clip=33 | consec=1 | max_consec=39 | avg_clip50=31.56
EP=11,190 | STEP=2,412,226 | clip=24 | consec=11 | max_consec=39 | avg_clip50=31.66
EP=11,200 | STEP=2,415,232 | clip=35 | consec=21 | max_consec=39 | avg_clip50=31.94

```

custom/		
avg_clip50	31.8	
clipped_episode_r...	26	
consecutive_clipp...	1	
max_consecutive_c...	39	
rollout/		
ep_len_mean	1.25e+03	
ep_rew_mean	31.4	
exploration rate	0.195	
time/		
episodes	32900	
fps	245	
time_elapsed	9842	
total timesteps	2417051	
train/		
learning_rate	0.0001	
loss	0.0897	

n_updates		591762			

EP=11,210	STEP=2,417,854	clip=29	consec=4	max_consec=39	avg_clip50=31.94
EP=11,220	STEP=2,421,013	clip=35	consec=14	max_consec=39	avg_clip50=30.76
EP=11,230	STEP=2,423,904	clip=23	consec=24	max_consec=39	avg_clip50=30.92
EP=11,240	STEP=2,427,260	clip=35	consec=34	max_consec=39	avg_clip50=31.20

custom/					
avg_clip50		31.8			
clipped_episode_r...		35			
consecutive_clipp...		39			
max_consecutive_c...		39			
rollout/					
ep_len_mean		1.26e+03			
ep_rew_mean		31.8			
exploration rate		0.191			
time/					
episodes		33000			
fps		245			
time_elapsed		9894			
total timesteps		2428920			
train/					
learning_rate		0.0001			
loss		0.45			
n_updates		594729			

EP=11,250	STEP=2,430,704	clip=33	consec=44	max_consec=44	avg_clip50=31.98
EP=11,260	STEP=2,433,465	clip=28	consec=3	max_consec=50	avg_clip50=31.98
EP=11,270	STEP=2,436,636	clip=31	consec=1	max_consec=50	avg_clip50=31.86
EP=11,280	STEP=2,439,217	clip=29	consec=11	max_consec=50	avg_clip50=31.66

custom/					
avg_clip50		31.7			
clipped_episode_r...		35			
consecutive_clipp...		12			
max_consecutive_c...		50			
rollout/					
ep_len_mean		1.24e+03			
ep_rew_mean		31.3			
exploration rate		0.188			
time/					

	episodes	33100	
	fps	245	
	time_elapsed	9942	
	total timesteps	2439869	
	train/		
	learning_rate	0.0001	
	loss	0.0628	
	n_updates	597467	

```
-----
EP=11,290 | STEP=2,442,543 | clip=22 | consec=21 | max_consec=50 | avg_clip50=31.86
EP=11,300 | STEP=2,445,177 | clip=20 | consec=0 | max_consec=50 | avg_clip50=30.76
EP=11,310 | STEP=2,448,106 | clip=22 | consec=10 | max_consec=50 | avg_clip50=31.10
-----
```

	custom/		
	avg_clip50	31	
	clipped_episode_r...	35	
	consecutive_clipp...	2	
	max_consecutive_c...	50	
	rollout/		
	ep_len_mean	1.25e+03	
	ep_rew_mean	31.6	
	exploration rate	0.184	
	time/		
	episodes	33200	
	fps	245	
	time_elapsed	9990	
	total timesteps	2450681	
	train/		
	learning_rate	0.0001	
	loss	0.0529	
	n_updates	600170	

```
-----
EP=11,320 | STEP=2,451,167 | clip=36 | consec=4 | max_consec=50 | avg_clip50=31.22
EP=11,330 | STEP=2,453,663 | clip=32 | consec=2 | max_consec=50 | avg_clip50=30.40
EP=11,340 | STEP=2,456,377 | clip=13 | consec=0 | max_consec=50 | avg_clip50=29.34
EP=11,350 | STEP=2,459,594 | clip=30 | consec=10 | max_consec=50 | avg_clip50=30.04
-----
```

	custom/		
	avg_clip50	30.5	
	clipped_episode_r...	60	
	consecutive_clipp...	12	

	max_consecutive_c...	50	
	rollout/		
	ep_len_mean	1.21e+03	
	ep_rew_mean	30.8	
	exploration rate	0.181	
	time/		
	episodes	33300	
	fps	245	
	time_elapsed	10033	
	total timesteps	2460549	
	train/		
	learning_rate	0.0001	
	loss	0.0582	
	n_updates	602637	

EP=11,360		STEP=2,462,982		clip=12		consec=0		max_consec=50		avg_clip50=30.62
EP=11,370		STEP=2,465,961		clip=35		consec=10		max_consec=50		avg_clip50=30.08
EP=11,380		STEP=2,468,744		clip=28		consec=3		max_consec=50		avg_clip50=30.90

	custom/		
	avg_clip50	31.5	
	clipped_episode_r...	35	
	consecutive_clipp...	7	
	max_consecutive_c...	50	
	rollout/		
	ep_len_mean	1.21e+03	
	ep_rew_mean	30.5	
	exploration rate	0.177	
	time/		
	episodes	33400	
	fps	245	
	time_elapsed	10081	
	total timesteps	2471395	
	train/		
	learning_rate	0.0001	
	loss	0.248	
	n_updates	605348	

EP=11,390		STEP=2,471,939		clip=32		consec=9		max_consec=50		avg_clip50=31.84
EP=11,400		STEP=2,475,024		clip=17		consec=0		max_consec=50		avg_clip50=31.52
EP=11,410		STEP=2,478,081		clip=35		consec=1		max_consec=50		avg_clip50=30.86

EP=11,420 | STEP=2,481,000 | clip=29 | consec=4 | max_consec=50 | avg_clip50=30.72

custom/		
avg_clip50	30	
clipped_episode_r...	14	
consecutive_clipp...	0	
max_consecutive_c...	50	
rollout/		
ep_len_mean	1.25e+03	
ep_rew_mean	30.6	
exploration rate	0.173	
time/		
episodes	33500	
fps	245	
time_elapsed	10130	
total timesteps	2482420	
train/		
learning_rate	0.0001	
loss	0.235	
n_updates	608104	

EP=11,430 | STEP=2,483,799 | clip=31 | consec=4 | max_consec=50 | avg_clip50=30.42
 EP=11,440 | STEP=2,486,992 | clip=25 | consec=14 | max_consec=50 | avg_clip50=30.14
 EP=11,450 | STEP=2,489,970 | clip=30 | consec=24 | max_consec=50 | avg_clip50=30.50
 EP=11,460 | STEP=2,492,757 | clip=24 | consec=3 | max_consec=50 | avg_clip50=30.10

custom/		
avg_clip50	30.2	
clipped_episode_r...	32	
consecutive_clipp...	5	
max_consecutive_c...	50	
rollout/		
ep_len_mean	1.23e+03	
ep_rew_mean	30.5	
exploration rate	0.17	
time/		
episodes	33600	
fps	244	
time_elapsed	10178	
total timesteps	2493358	
train/		

learning_rate	0.0001				
loss	0.0511				
n_updates	610839				

EP=11,470	STEP=2,495,812	clip=32	consec=13	max_consec=50	avg_clip50=30.74
EP=11,480	STEP=2,498,925	clip=35	consec=7	max_consec=50	avg_clip50=31.26
EP=11,490	STEP=2,501,904	clip=33	consec=3	max_consec=50	avg_clip50=31.82
EP=11,500	STEP=2,505,022	clip=35	consec=9	max_consec=50	avg_clip50=31.38

custom/					
avg_clip50	31.4				
clipped_episode_r...	35				
consecutive_clipp...	9				
max_consecutive_c...	50				
rollout/					
ep_len_mean	1.24e+03				
ep_rew_mean	30.9				
exploration rate	0.166				
time/					
episodes	33700				
fps	244				
time_elapsed	10230				
total timesteps	2505022				
train/					
learning_rate	0.0001				
loss	0.41				
n_updates	613755				

EP=11,510	STEP=2,507,823	clip=24	consec=19	max_consec=50	avg_clip50=31.72
EP=11,520	STEP=2,510,988	clip=34	consec=2	max_consec=50	avg_clip50=31.26
EP=11,530	STEP=2,514,098	clip=33	consec=12	max_consec=50	avg_clip50=31.58

custom/					
avg_clip50	31.5				
clipped_episode_r...	31				
consecutive_clipp...	16				
max_consecutive_c...	50				
rollout/					
ep_len_mean	1.25e+03				
ep_rew_mean	31.4				
exploration rate	0.162				

time/	
episodes	33800
fps	244
time_elapsed	10276
total timesteps	2515456
train/	
learning_rate	0.0001
loss	0.0355
n_updates	616363

```
-----
EP=11,540 | STEP=2,517,141 | clip=29 | consec=22 | max_consec=50 | avg_clip50=31.18
EP=11,550 | STEP=2,520,706 | clip=17 | consec=0 | max_consec=50 | avg_clip50=32.64
EP=11,560 | STEP=2,523,863 | clip=29 | consec=2 | max_consec=50 | avg_clip50=33.38
EP=11,570 | STEP=2,526,607 | clip=33 | consec=7 | max_consec=50 | avg_clip50=33.50
-----
```

custom/	
avg_clip50	33.6
clipped_episode_r...	30
consecutive_clipp...	9
max_consecutive_c...	50
rollout/	
ep_len_mean	1.28e+03
ep_rew_mean	32.4
exploration rate	0.158
time/	
episodes	33900
fps	244
time_elapsed	10327
total timesteps	2527236
train/	
learning_rate	0.0001
loss	0.0431
n_updates	619308

```
-----
EP=11,580 | STEP=2,529,655 | clip=33 | consec=6 | max_consec=50 | avg_clip50=33.20
EP=11,590 | STEP=2,532,947 | clip=37 | consec=16 | max_consec=50 | avg_clip50=33.78
EP=11,600 | STEP=2,536,343 | clip=31 | consec=26 | max_consec=50 | avg_clip50=33.20
-----
```

custom/	
avg_clip50	32.8
clipped_episode_r...	22

consecutive_clipp...	34
max_consecutive_c...	50
rollout/	
ep_len_mean	1.29e+03
ep_rew_mean	33
exploration rate	0.155
time/	
episodes	34000
fps	244
time_elapsed	10378
total timesteps	2538936
train/	
learning_rate	0.0001
loss	0.0477
n_updates	622233

EP=11,610	STEP=2,539,332	clip=35	consec=36	max_consec=50	avg_clip50=32.96
EP=11,620	STEP=2,542,929	clip=33	consec=2	max_consec=50	avg_clip50=34.86
EP=11,630	STEP=2,546,410	clip=23	consec=12	max_consec=50	avg_clip50=35.52
EP=11,640	STEP=2,549,323	clip=29	consec=22	max_consec=50	avg_clip50=34.84

custom/	
avg_clip50	34.6
clipped_episode_r...	32
consecutive_clipp...	26
max_consecutive_c...	50
rollout/	
ep_len_mean	1.3e+03
ep_rew_mean	33.4
exploration rate	0.151
time/	
episodes	34100
fps	244
time_elapsed	10429
total timesteps	2550483
train/	
learning_rate	0.0001
loss	0.0571
n_updates	625120

EP=11,650	STEP=2,552,138	clip=30	consec=2	max_consec=50	avg_clip50=33.54
-----------	----------------	---------	----------	---------------	------------------

EP=11,660	STEP=2,555,227	clip=35	consec=5	max_consec=50	avg_clip50=33.46
EP=11,670	STEP=2,558,130	clip=33	consec=15	max_consec=50	avg_clip50=31.58
EP=11,680	STEP=2,561,320	clip=35	consec=25	max_consec=50	avg_clip50=31.70

custom/		
avg_clip50	31.7	
clipped_episode_r...	34	
consecutive_clipp...	26	
max_consecutive_c...	50	
rollout/		
ep_len_mean	1.31e+03	
ep_rew_mean	33.6	
exploration rate	0.147	
time/		
episodes	34200	
fps	244	
time_elapsed	10478	
total timesteps	2561620	
train/		
learning_rate	0.0001	
loss	0.0845	
n_updates	627904	

EP=11,690	STEP=2,564,354	clip=29	consec=35	max_consec=50	avg_clip50=31.80
EP=11,700	STEP=2,567,164	clip=33	consec=45	max_consec=50	avg_clip50=32.30
EP=11,710	STEP=2,569,998	clip=33	consec=2	max_consec=52	avg_clip50=32.06

custom/		
avg_clip50	32.1	
clipped_episode_r...	24	
consecutive_clipp...	6	
max_consecutive_c...	52	
rollout/		
ep_len_mean	1.24e+03	
ep_rew_mean	31.9	
exploration rate	0.143	
time/		
episodes	34300	
fps	244	
time_elapsed	10527	
total timesteps	2572553	

train/	
learning_rate	0.0001
loss	0.042
n_updates	630638

```
-----
EP=11,720 | STEP=2,572,870 | clip=31 | consec=8 | max_consec=52 | avg_clip50=32.10
EP=11,730 | STEP=2,575,627 | clip=33 | consec=9 | max_consec=52 | avg_clip50=30.84
EP=11,740 | STEP=2,578,683 | clip=33 | consec=19 | max_consec=52 | avg_clip50=31.42
EP=11,750 | STEP=2,582,137 | clip=34 | consec=29 | max_consec=52 | avg_clip50=32.78
-----
```

custom/	
avg_clip50	32.7
clipped_episode_r...	24
consecutive_clipp...	34
max_consecutive_c...	52
rollout/	
ep_len_mean	1.24e+03
ep_rew_mean	32.6
exploration rate	0.14
time/	
episodes	34400
fps	244
time_elapsed	10575
total timesteps	2583509
train/	
learning_rate	0.0001
loss	0.0413
n_updates	633377

```
-----
EP=11,760 | STEP=2,585,296 | clip=33 | consec=39 | max_consec=52 | avg_clip50=33.64
EP=11,770 | STEP=2,588,985 | clip=35 | consec=49 | max_consec=52 | avg_clip50=35.36
EP=11,780 | STEP=2,592,061 | clip=58 | consec=59 | max_consec=59 | avg_clip50=36.44
EP=11,790 | STEP=2,594,770 | clip=34 | consec=5 | max_consec=63 | avg_clip50=35.62
-----
```

custom/	
avg_clip50	35.6
clipped_episode_r...	31
consecutive_clipp...	7
max_consecutive_c...	63
rollout/	
ep_len_mean	1.27e+03

ep_rew_mean	33.7
exploration rate	0.136
time/	
episodes	34500
fps	244
time_elapsed	10629
total timesteps	2595557
train/	
learning_rate	0.0001
loss	0.0769
n_updates	636389

```

-----
EP=11,800 | STEP=2,597,550 | clip=26 | consec=4 | max_consec=63 | avg_clip50=34.16
EP=11,810 | STEP=2,600,565 | clip=35 | consec=14 | max_consec=63 | avg_clip50=33.78
EP=11,820 | STEP=2,603,577 | clip=26 | consec=24 | max_consec=63 | avg_clip50=32.66
-----

```

custom/	
avg_clip50	32.5
clipped_episode_r...	34
consecutive_clipp...	6
max_consecutive_c...	63
rollout/	
ep_len_mean	1.29e+03
ep_rew_mean	34.3
exploration rate	0.132
time/	
episodes	34600
fps	244
time_elapsed	10674
total timesteps	2605817
train/	
learning_rate	0.0001
loss	0.0251
n_updates	638954

```

-----
EP=11,830 | STEP=2,606,631 | clip=33 | consec=2 | max_consec=63 | avg_clip50=31.90
EP=11,840 | STEP=2,609,691 | clip=23 | consec=12 | max_consec=63 | avg_clip50=32.78
EP=11,850 | STEP=2,613,330 | clip=35 | consec=22 | max_consec=63 | avg_clip50=33.94
EP=11,860 | STEP=2,616,305 | clip=32 | consec=32 | max_consec=63 | avg_clip50=33.90
-----

```

custom/	
---------	--

avg_clip50	34.6
clipped_episode_r...	70
consecutive_clipp...	36
max_consecutive_c...	63
rollout/	
ep_len_mean	1.3e+03
ep_rew_mean	34.2
exploration rate	0.128
time/	
episodes	34700
fps	244
time_elapsed	10728
total timesteps	2618022
train/	
learning_rate	0.0001
loss	0.0995
n_updates	642005

```

-----
EP=11,870 | STEP=2,619,665 | clip=32 | consec=5 | max_consec=63 | avg_clip50=34.10
EP=11,880 | STEP=2,622,553 | clip=28 | consec=4 | max_consec=63 | avg_clip50=34.12
EP=11,890 | STEP=2,625,403 | clip=34 | consec=3 | max_consec=63 | avg_clip50=33.08
EP=11,900 | STEP=2,627,996 | clip=34 | consec=2 | max_consec=63 | avg_clip50=31.72

```

custom/	
avg_clip50	31.8
clipped_episode_r...	32
consecutive_clipp...	5
max_consecutive_c...	63
rollout/	
ep_len_mean	1.26e+03
ep_rew_mean	32.9
exploration rate	0.125
time/	
episodes	34800
fps	243
time_elapsed	10778
total timesteps	2629033
train/	
learning_rate	0.0001
loss	0.131
n_updates	644758

```

-----
EP=11,910 | STEP=2,630,975 | clip=33 | consec=12 | max_consec=63 | avg_clip50=31.96
EP=11,920 | STEP=2,633,671 | clip=34 | consec=3 | max_consec=63 | avg_clip50=30.74
EP=11,930 | STEP=2,637,236 | clip=62 | consec=13 | max_consec=63 | avg_clip50=32.46
-----

```

```

| custom/
|   avg_clip50          | 35.1 |
|   clipped_episode_r... | 31   |
|   consecutive_clipp... | 22   |
|   max_consecutive_c... | 63   |
| rollout/
|   ep_len_mean         | 1.3e+03 |
|   ep_rew_mean         | 34      |
|   exploration rate    | 0.121   |
| time/
|   episodes            | 34900   |
|   fps                 | 243     |
|   time_elapsed        | 10830   |
|   total timesteps     | 2640931 |
| train/
|   learning_rate       | 0.0001  |
|   loss                 | 0.0578  |
|   n_updates           | 647732  |
-----

```

```

EP=11,940 | STEP=2,641,705 | clip=104 | consec=23 | max_consec=63 | avg_clip50=36.52
EP=11,950 | STEP=2,645,028 | clip=60 | consec=33 | max_consec=63 | avg_clip50=37.66
EP=11,960 | STEP=2,648,091 | clip=31 | consec=7 | max_consec=63 | avg_clip50=37.02
EP=11,970 | STEP=2,651,217 | clip=32 | consec=7 | max_consec=63 | avg_clip50=38.30
-----

```

```

| custom/
|   avg_clip50          | 38.1 |
|   clipped_episode_r... | 47   |
|   consecutive_clipp... | 3    |
|   max_consecutive_c... | 63   |
| rollout/
|   ep_len_mean         | 1.3e+03 |
|   ep_rew_mean         | 34.6    |
|   exploration rate    | 0.116   |
| time/
|   episodes            | 35000   |
|   fps                 | 243     |
-----

```

	time_elapsed		10885	
	total timesteps		2653338	
	train/			
	learning_rate		0.0001	
	loss		0.0367	
	n_updates		650834	

```
-----
EP=11,980 | STEP=2,654,257 | clip=32 | consec=7 | max_consec=63 | avg_clip50=36.88
EP=11,990 | STEP=2,657,541 | clip=31 | consec=17 | max_consec=63 | avg_clip50=33.66
EP=12,000 | STEP=2,660,741 | clip=32 | consec=27 | max_consec=63 | avg_clip50=33.56
EP=12,010 | STEP=2,663,944 | clip=36 | consec=37 | max_consec=63 | avg_clip50=34.22
-----
```

	custom/			
	avg_clip50		33.3	
	clipped_episode_r...		29	
	consecutive_clipp...		39	
	max_consecutive_c...		63	
	rollout/			
	ep_len_mean		1.36e+03	
	ep_rew_mean		35.6	
	exploration rate		0.113	
	time/			
	episodes		35100	
	fps		243	
	time_elapsed		10936	
	total timesteps		2664770	
	train/			
	learning_rate		0.0001	
	loss		0.158	
	n_updates		653692	

```
-----
EP=12,020 | STEP=2,666,962 | clip=34 | consec=47 | max_consec=63 | avg_clip50=33.60
EP=12,030 | STEP=2,670,137 | clip=32 | consec=57 | max_consec=63 | avg_clip50=33.72
EP=12,040 | STEP=2,673,214 | clip=34 | consec=6 | max_consec=63 | avg_clip50=33.02
-----
```

	custom/			
	avg_clip50		31.8	
	clipped_episode_r...		32	
	consecutive_clipp...		1	
	max_consecutive_c...		63	
	rollout/			

ep_len_mean	1.29e+03
ep_rew_mean	33
exploration rate	0.109
time/	
episodes	35200
fps	243
time_elapsed	10984
total timesteps	2675447
train/	
learning_rate	0.0001
loss	0.0627
n_updates	656361

```

-----
EP=12,050 | STEP=2,675,939 | clip=35 | consec=3 | max_consec=63 | avg_clip50=31.98
EP=12,060 | STEP=2,679,092 | clip=35 | consec=13 | max_consec=63 | avg_clip50=31.82
EP=12,070 | STEP=2,682,150 | clip=35 | consec=23 | max_consec=63 | avg_clip50=31.92
EP=12,080 | STEP=2,685,622 | clip=35 | consec=33 | max_consec=63 | avg_clip50=32.42
-----

```

custom/	
avg_clip50	32.5
clipped_episode_r...	22
consecutive_clipp...	38
max_consecutive_c...	63
rollout/	
ep_len_mean	1.29e+03
ep_rew_mean	32.8
exploration rate	0.105
time/	
episodes	35300
fps	243
time_elapsed	11037
total timesteps	2687487
train/	
learning_rate	0.0001
loss	0.0738
n_updates	659371

```

-----
EP=12,090 | STEP=2,688,674 | clip=19 | consec=0 | max_consec=63 | avg_clip50=32.32
EP=12,100 | STEP=2,691,933 | clip=35 | consec=10 | max_consec=63 | avg_clip50=33.10
EP=12,110 | STEP=2,695,474 | clip=34 | consec=9 | max_consec=63 | avg_clip50=33.98
EP=12,120 | STEP=2,698,396 | clip=35 | consec=19 | max_consec=63 | avg_clip50=33.88
-----

```

custom/		
avg_clip50	33.9	
clipped_episode_r...	36	
consecutive_clipp...	21	
max_consecutive_c...	63	
rollout/		
ep_len_mean	1.3e+03	
ep_rew_mean	32.9	
exploration rate	0.101	
time/		
episodes	35400	
fps	243	
time_elapsed	11089	
total timesteps	2698997	
train/		
learning_rate	0.0001	
loss	0.0451	
n_updates	662249	

EP=12,130 | STEP=2,701,342 | clip=61 | consec=5 | max_consec=63 | avg_clip50=33.36
 EP=12,140 | STEP=2,704,659 | clip=35 | consec=6 | max_consec=63 | avg_clip50=34.42
 EP=12,150 | STEP=2,708,002 | clip=35 | consec=16 | max_consec=63 | avg_clip50=34.88

custom/		
avg_clip50	34.4	
clipped_episode_r...	35	
consecutive_clipp...	25	
max_consecutive_c...	63	
rollout/		
ep_len_mean	1.33e+03	
ep_rew_mean	34.2	
exploration rate	0.0972	
time/		
episodes	35500	
fps	243	
time_elapsed	11144	
total timesteps	2711235	
train/		
learning_rate	0.0001	
loss	0.0484	

n_updates		665308			

EP=12,160	STEP=2,711,720	clip=68	consec=26	max_consec=63	avg_clip50=35.06
EP=12,170	STEP=2,715,479	clip=35	consec=36	max_consec=63	avg_clip50=36.86
EP=12,180	STEP=2,718,429	clip=33	consec=8	max_consec=63	avg_clip50=36.78
EP=12,190	STEP=2,721,706	clip=34	consec=1	max_consec=63	avg_clip50=36.24

custom/					
avg_clip50		36			
clipped_episode_r...		32			
consecutive_clipp...		9			
max_consecutive_c...		63			
rollout/					
ep_len_mean		1.36e+03			
ep_rew_mean		35.5			
exploration rate		0.0928			
time/					
episodes		35600			
fps		243			
time_elapsed		11204			
total timesteps		2724432			
train/					
learning_rate		0.0001			
loss		0.181			
n_updates		668607			

EP=12,200	STEP=2,725,030	clip=34	consec=11	max_consec=63	avg_clip50=36.10
EP=12,210	STEP=2,728,249	clip=33	consec=2	max_consec=63	avg_clip50=35.04
EP=12,220	STEP=2,731,310	clip=35	consec=8	max_consec=63	avg_clip50=33.30
EP=12,230	STEP=2,734,504	clip=31	consec=18	max_consec=63	avg_clip50=34.02

custom/					
avg_clip50		34.4			
clipped_episode_r...		55			
consecutive_clipp...		23			
max_consecutive_c...		63			
rollout/					
ep_len_mean		1.38e+03			
ep_rew_mean		35.6			
exploration rate		0.0888			
time/					

episodes	35700	
fps	243	
time_elapsed	11258	
total timesteps	2736384	
train/		
learning_rate	0.0001	
loss	0.0974	
n_updates	671595	

```

-----
EP=12,240 | STEP=2,737,646 | clip=35 | consec=28 | max_consec=63 | avg_clip50=34.14
EP=12,250 | STEP=2,740,538 | clip=34 | consec=38 | max_consec=63 | avg_clip50=33.66
EP=12,260 | STEP=2,744,162 | clip=46 | consec=48 | max_consec=63 | avg_clip50=34.74
EP=12,270 | STEP=2,747,472 | clip=24 | consec=4 | max_consec=63 | avg_clip50=35.24

```

custom/		
avg_clip50	35.2	
clipped_episode_r...	34	
consecutive_clipp...	5	
max_consecutive_c...	63	
rollout/		
ep_len_mean	1.32e+03	
ep_rew_mean	34.3	
exploration rate	0.0849	
time/		
episodes	35800	
fps	242	
time_elapsed	11311	
total timesteps	2748113	
train/		
learning_rate	0.0001	
loss	0.101	
n_updates	674528	

```

-----
EP=12,280 | STEP=2,750,810 | clip=34 | consec=14 | max_consec=63 | avg_clip50=35.58
EP=12,290 | STEP=2,754,159 | clip=60 | consec=24 | max_consec=63 | avg_clip50=36.16
EP=12,300 | STEP=2,757,465 | clip=35 | consec=34 | max_consec=63 | avg_clip50=36.96
EP=12,310 | STEP=2,760,702 | clip=29 | consec=2 | max_consec=63 | avg_clip50=36.20

```

custom/		
avg_clip50	36.2	
clipped_episode_r...	34	

consecutive_clipp...	3
max_consecutive_c...	63
rollout/	
ep_len_mean	1.34e+03
ep_rew_mean	35.5
exploration rate	0.0806
time/	
episodes	35900
fps	242
time_elapsed	11369
total timesteps	2761025
train/	
learning_rate	0.0001
loss	0.065
n_updates	677756

```

-----
EP=12,320 | STEP=2,763,622 | clip=32 | consec=12 | max_consec=63 | avg_clip50=35.80
EP=12,330 | STEP=2,766,749 | clip=35 | consec=22 | max_consec=63 | avg_clip50=34.96
EP=12,340 | STEP=2,770,264 | clip=33 | consec=32 | max_consec=63 | avg_clip50=35.04
EP=12,350 | STEP=2,773,476 | clip=66 | consec=42 | max_consec=63 | avg_clip50=35.06
-----

```

custom/	
avg_clip50	35.2
clipped_episode_r...	33
consecutive_clipp...	43
max_consecutive_c...	63
rollout/	
ep_len_mean	1.36e+03
ep_rew_mean	36.1
exploration rate	0.0763
time/	
episodes	36000
fps	242
time_elapsed	11426
total timesteps	2773786
train/	
learning_rate	0.0001
loss	0.0484
n_updates	680946

```

-----
EP=12,360 | STEP=2,776,443 | clip=64 | consec=52 | max_consec=63 | avg_clip50=34.88
-----

```

EP=12,370	STEP=2,779,403	clip=11	consec=0	max_consec=63	avg_clip50=34.66
EP=12,380	STEP=2,782,510	clip=32	consec=10	max_consec=63	avg_clip50=34.38
EP=12,390	STEP=2,785,859	clip=46	consec=20	max_consec=63	avg_clip50=34.40

custom/		
avg_clip50	34.4	
clipped_episode_r...	46	
consecutive_clipp...	20	
max_consecutive_c...	63	
rollout/		
ep_len_mean	1.31e+03	
ep_rew_mean	34.7	
exploration rate	0.0723	
time/		
episodes	36100	
fps	242	
time_elapsed	11481	
total timesteps	2785970	
train/		
learning_rate	0.0001	
loss	0.0661	
n_updates	683992	

EP=12,400	STEP=2,788,658	clip=34	consec=30	max_consec=63	avg_clip50=33.30
EP=12,410	STEP=2,791,574	clip=33	consec=5	max_consec=63	avg_clip50=32.68
EP=12,420	STEP=2,794,623	clip=35	consec=15	max_consec=63	avg_clip50=32.62

custom/		
avg_clip50	33.1	
clipped_episode_r...	31	
consecutive_clipp...	23	
max_consecutive_c...	63	
rollout/		
ep_len_mean	1.28e+03	
ep_rew_mean	33.8	
exploration rate	0.0684	
time/		
episodes	36200	
fps	242	
time_elapsed	11533	
total timesteps	2797475	

train/	
learning_rate	0.0001
loss	0.0314
n_updates	686868

```

-----
EP=12,430 | STEP=2,797,846 | clip=34 | consec=25 | max_consec=63 | avg_clip50=33.08
EP=12,440 | STEP=2,800,692 | clip=33 | consec=1  | max_consec=63 | avg_clip50=31.60
EP=12,450 | STEP=2,803,965 | clip=33 | consec=11 | max_consec=63 | avg_clip50=32.42
EP=12,460 | STEP=2,806,986 | clip=31 | consec=21 | max_consec=63 | avg_clip50=32.54

```

custom/	
avg_clip50	32.7
clipped_episode_r...	32
consecutive_clipp...	26
max_consecutive_c...	63
rollout/	
ep_len_mean	1.26e+03
ep_rew_mean	32.7
exploration rate	0.0647
time/	
episodes	36300
fps	242
time_elapsed	11585
total timesteps	2808723
train/	
learning_rate	0.0001
loss	0.174
n_updates	689680

```

-----
EP=12,470 | STEP=2,810,097 | clip=36 | consec=3 | max_consec=63 | avg_clip50=32.84
EP=12,480 | STEP=2,812,942 | clip=34 | consec=2 | max_consec=63 | avg_clip50=32.18
EP=12,490 | STEP=2,816,233 | clip=57 | consec=4 | max_consec=63 | avg_clip50=33.10
EP=12,500 | STEP=2,819,646 | clip=33 | consec=14 | max_consec=63 | avg_clip50=33.20

```

custom/	
avg_clip50	34.2
clipped_episode_r...	55
consecutive_clipp...	17
max_consecutive_c...	63
rollout/	
ep_len_mean	1.29e+03

ep_rew_mean	33.3
exploration rate	0.0606
time/	
episodes	36400
fps	242
time_elapsed	11640
total timesteps	2820900
train/	
learning_rate	0.0001
loss	0.0381
n_updates	692724

```

-----
EP=12,510 | STEP=2,822,943 | clip=29 | consec=24 | max_consec=63 | avg_clip50=34.44
EP=12,520 | STEP=2,826,578 | clip=49 | consec=34 | max_consec=63 | avg_clip50=35.46
EP=12,530 | STEP=2,830,063 | clip=34 | consec=1 | max_consec=63 | avg_clip50=37.10
EP=12,540 | STEP=2,833,506 | clip=33 | consec=11 | max_consec=63 | avg_clip50=38.12
-----

```

custom/	
avg_clip50	38
clipped_episode_r...	31
consecutive_clipp...	12
max_consecutive_c...	63
rollout/	
ep_len_mean	1.35e+03
ep_rew_mean	35.6
exploration rate	0.0563
time/	
episodes	36500
fps	242
time_elapsed	11698
total timesteps	2833925
train/	
learning_rate	0.0001
loss	0.0408
n_updates	695981

```

-----
EP=12,550 | STEP=2,836,674 | clip=33 | consec=21 | max_consec=63 | avg_clip50=37.58
EP=12,560 | STEP=2,840,030 | clip=34 | consec=31 | max_consec=63 | avg_clip50=37.60
EP=12,570 | STEP=2,843,383 | clip=33 | consec=41 | max_consec=63 | avg_clip50=37.02
-----

```

custom/	
---------	--

avg_clip50	36.4	
clipped_episode_r...	34	
consecutive_clipp...	48	
max_consecutive_c...	63	
rollout/		
ep_len_mean	1.38e+03	
ep_rew_mean	36.6	
exploration rate	0.0522	
time/		
episodes	36600	
fps	242	
time_elapsed	11753	
total timesteps	2846169	
train/		
learning_rate	0.0001	
loss	0.074	
n_updates	699042	

```

-----
EP=12,580 | STEP=2,847,027 | clip=35 | consec=51 | max_consec=63 | avg_clip50=37.28
EP=12,590 | STEP=2,850,083 | clip=11 | consec=0 | max_consec=63 | avg_clip50=35.82
EP=12,600 | STEP=2,853,434 | clip=35 | consec=10 | max_consec=63 | avg_clip50=36.42
EP=12,610 | STEP=2,856,604 | clip=70 | consec=20 | max_consec=63 | avg_clip50=35.84
-----

```

custom/		
avg_clip50	36.3	
clipped_episode_r...	36	
consecutive_clipp...	23	
max_consecutive_c...	63	
rollout/		
ep_len_mean	1.39e+03	
ep_rew_mean	36.9	
exploration rate	0.0482	
time/		
episodes	36700	
fps	242	
time_elapsed	11808	
total timesteps	2858241	
train/		
learning_rate	0.0001	
loss	0.0463	
n_updates	702060	

```

-----
EP=12,620 | STEP=2,860,311 | clip=29 | consec=30 | max_consec=63 | avg_clip50=37.24
EP=12,630 | STEP=2,863,565 | clip=34 | consec=40 | max_consec=63 | avg_clip50=36.54
EP=12,640 | STEP=2,867,004 | clip=31 | consec=50 | max_consec=63 | avg_clip50=37.60
EP=12,650 | STEP=2,869,971 | clip=30 | consec=60 | max_consec=63 | avg_clip50=37.06
-----

```

```

| custom/
|   avg_clip50          | 37.7
|   clipped_episode_r... | 65
|   consecutive_clipp... | 61
|   max_consecutive_c... | 63
| rollout/
|   ep_len_mean         | 1.38e+03
|   ep_rew_mean         | 37
|   exploration rate     | 0.0441
| time/
|   episodes            | 36800
|   fps                 | 241
|   time_elapsed        | 11864
|   total timesteps      | 2870621
| train/
|   learning_rate        | 0.0001
|   loss                 | 0.547
|   n_updates            | 705155
-----

```

```

EP=12,660 | STEP=2,872,907 | clip=33 | consec=1 | max_consec=64 | avg_clip50=36.26
EP=12,670 | STEP=2,876,219 | clip=30 | consec=3 | max_consec=64 | avg_clip50=35.12
EP=12,680 | STEP=2,879,410 | clip=33 | consec=13 | max_consec=64 | avg_clip50=34.78
EP=12,690 | STEP=2,883,008 | clip=34 | consec=23 | max_consec=64 | avg_clip50=34.76
-----

```

```

| custom/
|   avg_clip50          | 34.8
|   clipped_episode_r... | 34
|   consecutive_clipp... | 23
|   max_consecutive_c... | 64
| rollout/
|   ep_len_mean         | 1.36e+03
|   ep_rew_mean         | 36.2
|   exploration rate     | 0.0398
| time/
|   episodes            | 36900
-----

```

	fps		241	
	time_elapsed		11921	
	total timesteps		2883359	
	train/			
	learning_rate		0.0001	
	loss		0.0368	
	n_updates		708339	

```

-----
EP=12,700 | STEP=2,887,509 | clip=70 | consec=33 | max_consec=64 | avg_clip50=38.04
EP=12,710 | STEP=2,891,246 | clip=32 | consec=43 | max_consec=64 | avg_clip50=40.12
EP=12,720 | STEP=2,893,877 | clip=34 | consec=53 | max_consec=64 | avg_clip50=38.96
EP=12,730 | STEP=2,897,156 | clip=69 | consec=63 | max_consec=64 | avg_clip50=39.52

```

	custom/			
	avg_clip50		39.2	
	clipped_episode_r...		35	
	consecutive_clipp...		65	
	max_consecutive_c...		65	
	rollout/			
	ep_len_mean		1.38e+03	
	ep_rew_mean		37	
	exploration rate		0.0349	
	time/			
	episodes		37000	
	fps		241	
	time_elapsed		11988	
	total timesteps		2898258	
	train/			
	learning_rate		0.0001	
	loss		0.411	
	n_updates		712064	

```

-----
EP=12,740 | STEP=2,901,352 | clip=34 | consec=73 | max_consec=73 | avg_clip50=40.68
EP=12,750 | STEP=2,904,777 | clip=35 | consec=83 | max_consec=83 | avg_clip50=38.06
EP=12,760 | STEP=2,907,786 | clip=30 | consec=93 | max_consec=93 | avg_clip50=36.74

```

Objetivo conseguido: 100 episodios consecutivos superando el umbral.

Entrenamiento finalizado.

max_consec alcanzado: 100

Log JSON: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json

Log CSV: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv

Out[]: <stable_baselines3.dqn.dqn.DQN at 0x7d5bc3879100>

```
In [ ]: try:
    model.save(str(MODEL_LATEST))
    print('Modelo completo SB3 guardado en:', MODEL_LATEST)

except Exception as e:
    print('No se pudo guardar el modelo completo con SB3.')
    print('Error:', repr(e))
    save_policy_checkpoint(model, POLICY_LATEST)
    print('Política/optimizador guardados con PyTorch en:', POLICY_LATEST)
```

No se pudo guardar el modelo completo con SB3.

Error: IndexError('tuple index out of range')

Política/optimizador guardados con PyTorch en: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_policy_latest.pt

```
In [ ]: if not CSV_LOG_FILENAME.exists():
    raise FileNotFoundError(f'No existe el log CSV: {CSV_LOG_FILENAME}')

df = pd.read_csv(CSV_LOG_FILENAME)
print('Filas registradas:', len(df))
print('Columnas:', list(df.columns))

display(df.tail(10))

summary_table = pd.DataFrame({
    'metrica': [
        'episodios_registrados',
        'ultimo_step',
        'mejor_clip',
        'mejor_avg_clip50',
        'mejor_max_consec',
        'ultimo_avg_clip50',
        'ultimo_exploration_rate'
    ],
    'valor': [
        len(df),
        int(df['total_steps'].iloc[-1]) if len(df) else 0,
        float(df['clipped_episode_reward'].max()) if len(df) else 0.0,
```

```

        float(df['avg_clip50'].max()) if len(df) else 0.0,
        int(df['max_consecutive_clipped_over_20'].max()) if len(df) else 0,
        float(df['avg_clip50'].iloc[-1]) if len(df) else 0.0,
        float(df['exploration_rate'].iloc[-1]) if 'exploration_rate' in df and len(df) else np.nan
    ]
})

display(summary_table)

steps = df['total_steps'].to_numpy()
clipped = df['clipped_episode_reward'].to_numpy()
consec = df['consecutive_clipped_over_20'].to_numpy()
max_consec = df['max_consecutive_clipped_over_20'].to_numpy()
avg50 = df['avg_clip50'].to_numpy()

plt.figure(figsize=(10, 4))
plt.plot(steps, clipped)
plt.title('Recompensa clipped por episodio')
plt.xlabel('Steps')
plt.ylabel('Clipped reward')
plt.grid(True)
plt.show()

plt.figure(figsize=(10, 4))
plt.plot(steps, avg50)
plt.title('Media de recompensa clipped en los últimos 50 episodios')
plt.xlabel('Steps')
plt.ylabel('avg_clip50')
plt.grid(True)
plt.show()

plt.figure(figsize=(10, 4))
plt.plot(steps, consec, label='consec')
plt.plot(steps, max_consec, label='max_consec')
plt.axhline(TARGET_CONSECUTIVE_EPISODES, linestyle='--')
plt.title('Episodios consecutivos con recompensa clipped > 20')
plt.xlabel('Steps')
plt.ylabel('Consecutivos')
plt.legend()
plt.grid(True)
plt.show()

```

```

if 'exploration_rate' in df.columns:
    plt.figure(figsize=(10, 4))
    plt.plot(steps, df['exploration_rate'])
    plt.title('Evolución de epsilon / exploration_rate')
    plt.xlabel('Steps')
    plt.ylabel('Exploration rate')
    plt.grid(True)
    plt.show()

print('Mejor max_consec:', int(max(max_consec)) if len(max_consec) else 0)
print('Mejor avg_clip50:', float(max(avg50)) if len(avg50) else 0.0)

```

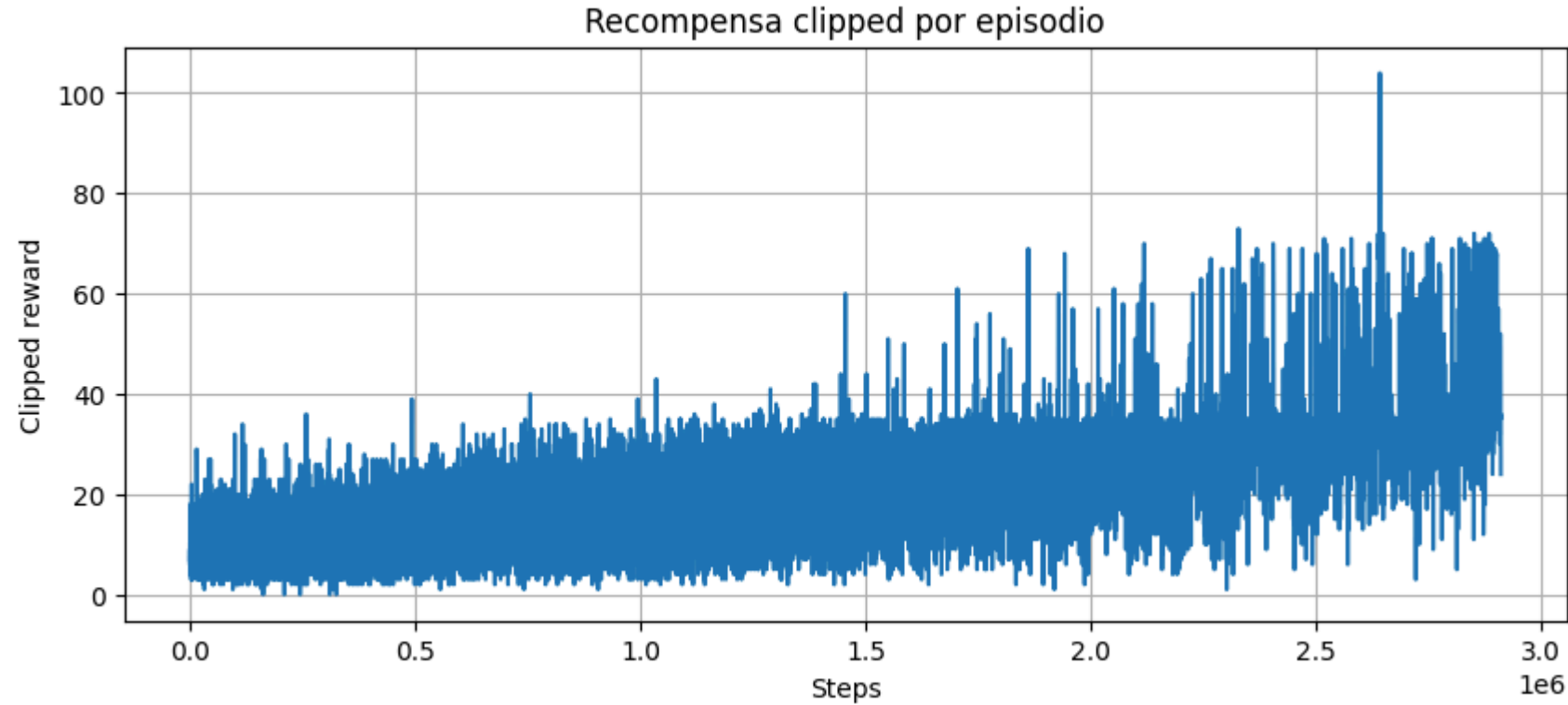
Filas registradas: 12768

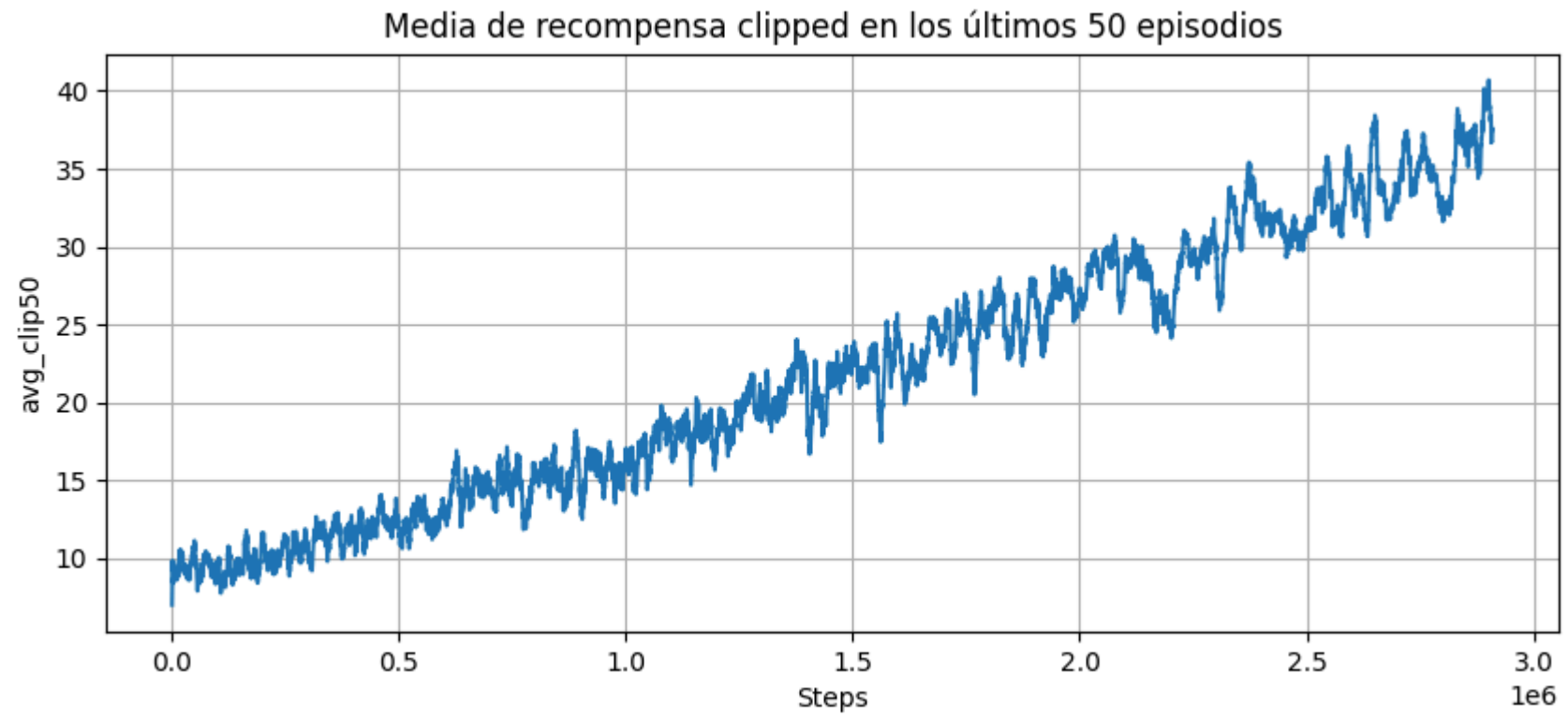
Columnas: ['episode', 'total_steps', 'episode_reward', 'clipped_episode_reward', 'consecutive_clipped_over_20', 'max_consecutive_clipped_over_20', 'avg_clip50', 'duration', 'exploration_rate']

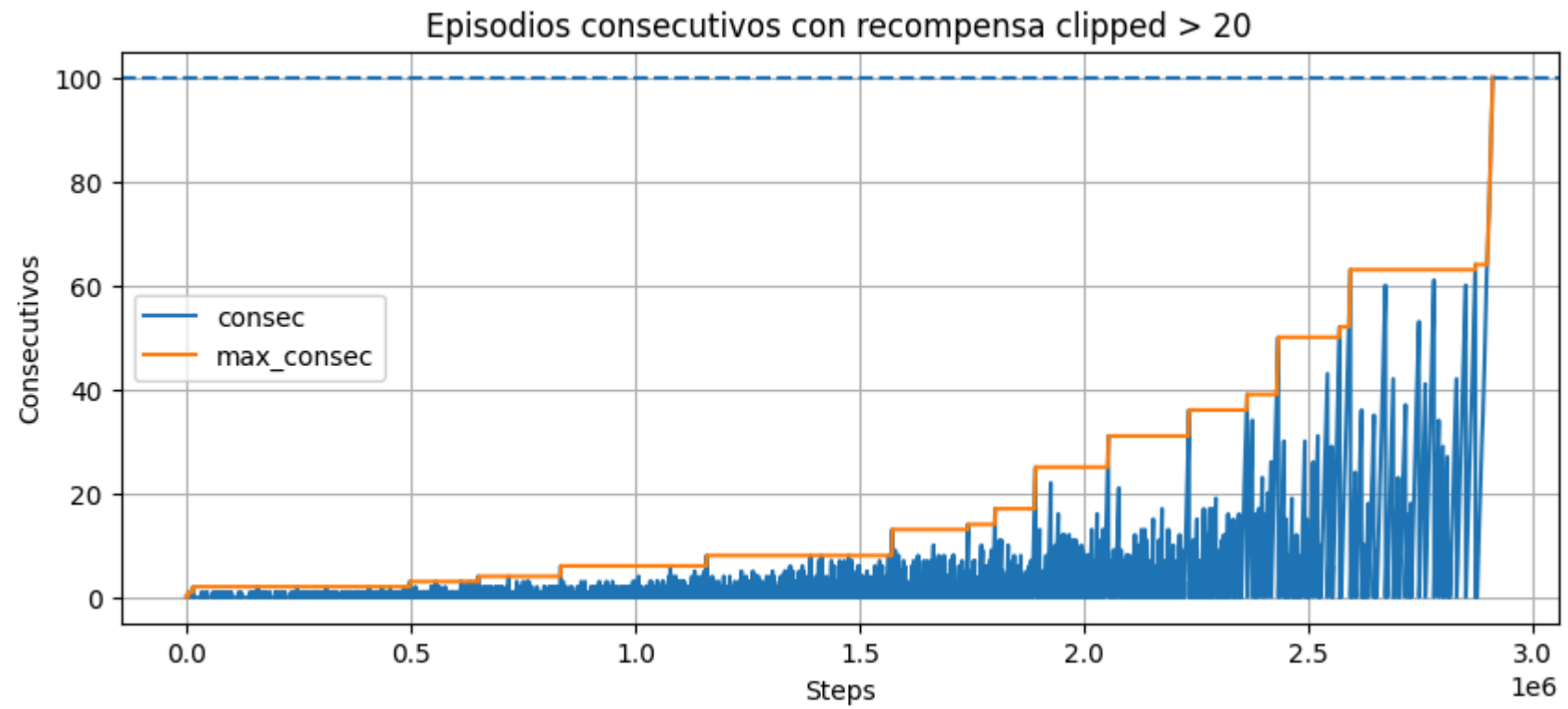
	episode	total_steps	episode_reward	clipped_episode_reward	consecutive_clipped_over_20	max_consecutive_clipped_over_20	avg_clip
12758	12758	2907257	33.0	33.0	91	91	36
12759	12759	2907529	34.0	34.0	92	92	36
12760	12760	2907786	30.0	30.0	93	93	36
12761	12761	2908143	33.0	33.0	94	94	36
12762	12762	2908418	32.0	32.0	95	95	36
12763	12763	2908838	52.0	52.0	96	96	37
12764	12764	2909241	35.0	35.0	97	97	37
12765	12765	2909564	35.0	35.0	98	98	37
12766	12766	2909856	36.0	36.0	99	99	37
12767	12767	2910112	24.0	24.0	100	100	37

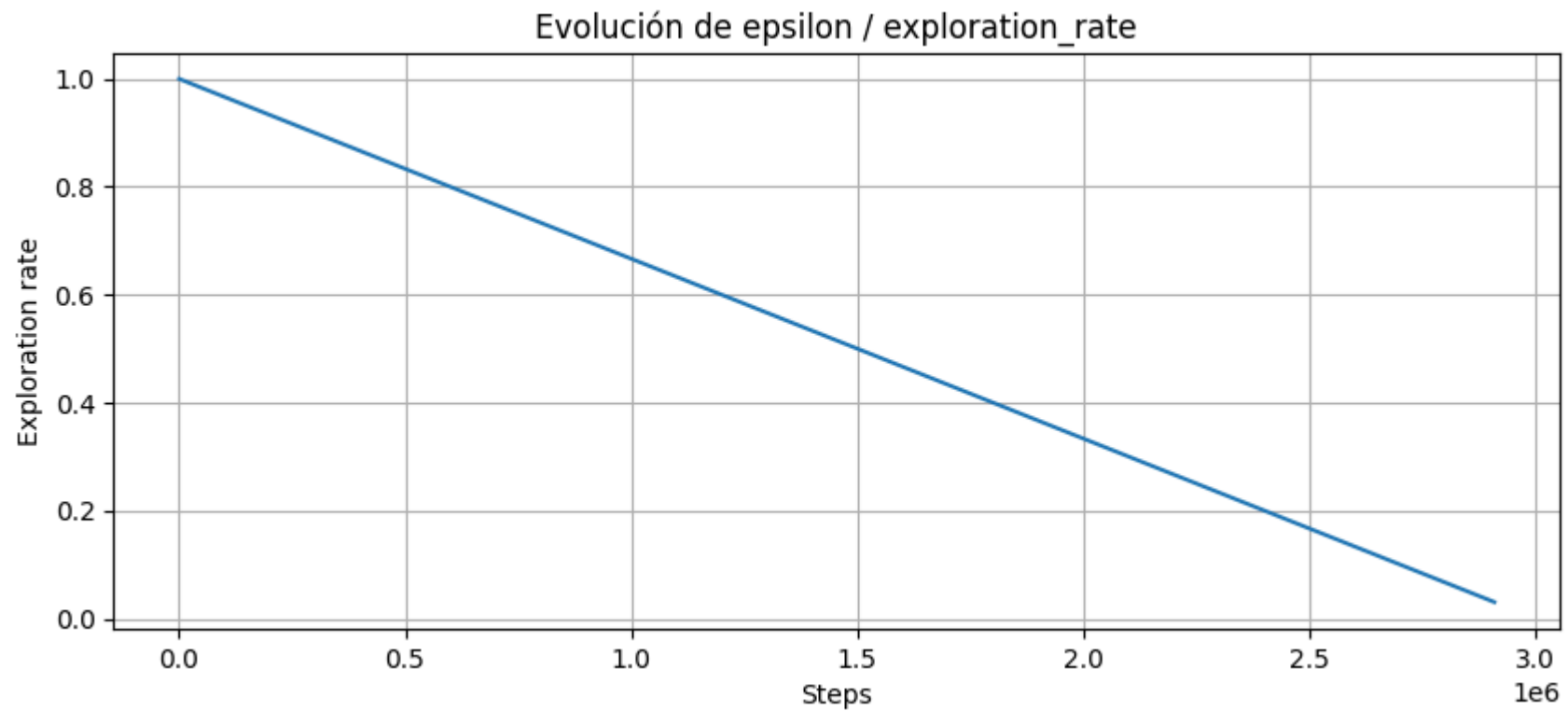


	metrica	valor
0	episodios_registrados	1.276800e+04
1	ultimo_step	2.910112e+06
2	mejor_clip	1.040000e+02
3	mejor_avg_clip50	4.068000e+01
4	mejor_max_consec	1.000000e+02
5	ultimo_avg_clip50	3.750000e+01
6	ultimo_exploration_rate	3.093304e-02









Mejor max_consec: 100
 Mejor avg_clip50: 40.68

```
In [ ]: if MODEL_LATEST.exists():
        try:
            model = DQN.load(str(MODEL_LATEST), env=env, device='auto')
            print('Modelo cargado para evaluación:', MODEL_LATEST)
        except Exception as e:
            print('No se pudo cargar MODEL_LATEST. Se usa el modelo actual en memoria.')
            print('Error:', repr(e))
    else:
        print('No existe MODEL_LATEST. Se usa el modelo actual en memoria.')

    obs = env.reset()
    for _ in range(10):
        action, _ = model.predict(obs, deterministic=True)
        obs, reward, done, info = env.step(action)
```

```

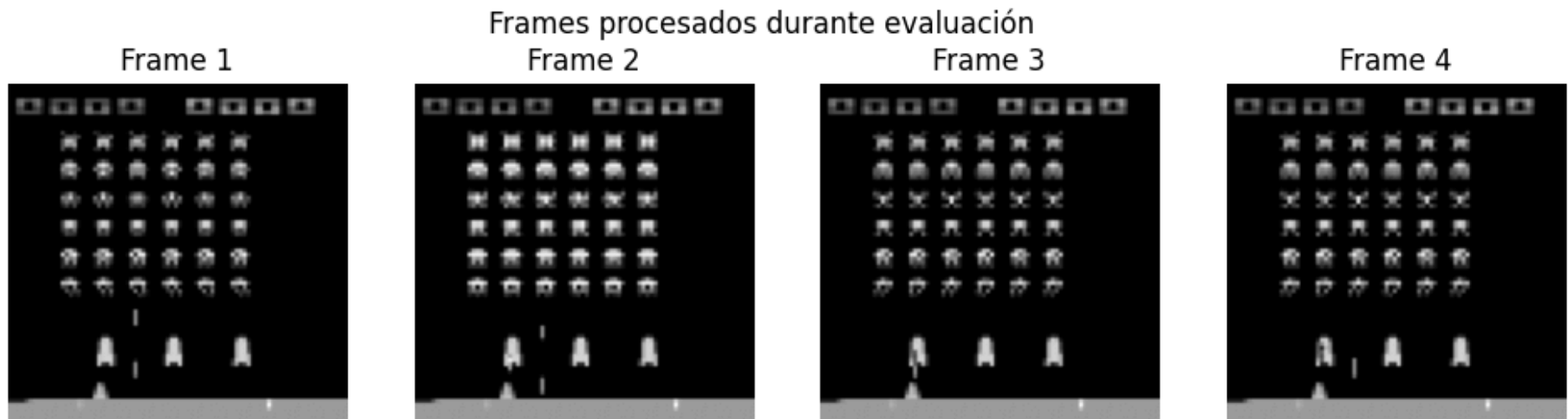
obs0 = obs[0]
if obs0.shape[0] in [1, 3, 4, 12]:
    frames = obs0
else:
    frames = np.transpose(obs0, (2, 0, 1))

num_frames = min(4, frames.shape[0])
plt.figure(figsize=(12, 3))
for i in range(num_frames):
    plt.subplot(1, num_frames, i + 1)
    plt.imshow(frames[i], cmap='gray')
    plt.title(f'Frame {i + 1}')
    plt.axis('off')
plt.suptitle('Frames procesados durante evaluación')
plt.show()

```

No se pudo cargar MODEL_LATEST. Se usa el modelo actual en memoria.

Error: ValueError("Error: the file <_io.BufferedReader name='/content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_sb3_latest.zip'> wasn't a zip-file")



```
In [ ]: # env.close()
```

```
In [ ]: print('OUTPUT_DIR:', OUTPUT_DIR)
        print('Existe OUTPUT_DIR:', OUTPUT_DIR.exists())
```

```
print('MODEL_LATEST:', MODEL_LATEST, '| existe:', MODEL_LATEST.exists())
print('POLICY_LATEST:', POLICY_LATEST, '| existe:', POLICY_LATEST.exists())
print('LOG_FILENAME:', LOG_FILENAME, '| existe:', LOG_FILENAME.exists())
print('CSV_LOG_FILENAME:', CSV_LOG_FILENAME, '| existe:', CSV_LOG_FILENAME.exists())
print('TENSORBOARD_DIR:', TENSORBOARD_DIR, '| existe:', TENSORBOARD_DIR.exists())
print('MONITOR_DIR:', MONITOR_DIR, '| existe:', MONITOR_DIR.exists())

if CSV_LOG_FILENAME.exists():
    df_check = pd.read_csv(CSV_LOG_FILENAME)
    print('\nResumen del log CSV')
    print('Episodios registrados:', len(df_check))

    if len(df_check) > 0:
        print('Último episodio:', int(df_check['episode'].iloc[-1]))
        print('Último step:', int(df_check['total_steps'].iloc[-1]))
        print('Mejor clipped_episode_reward:', float(df_check['clipped_episode_reward'].max()))
        print('Mejor avg_clip50:', float(df_check['avg_clip50'].max()))
        print('Mejor max_consec:', int(df_check['max_consecutive_clipped_over_20'].max()))
        print('Último avg_clip50:', float(df_check['avg_clip50'].iloc[-1]))

        display(df_check.tail(5))
    else:
        print('\nNo se encontró el CSV de entrenamiento. Ejecuta el bloque de entrenamiento con el callback actualizado.')

if LOG_FILENAME.exists():
    with open(LOG_FILENAME, 'r') as f:
        data_check = json.load(f)
        print('\nResumen del log JSON')
        print('Claves:', list(data_check.keys()))
        print('Episodios en JSON:', len(data_check.get('episode', [])))
    else:
        print('\nNo se encontró el JSON de entrenamiento.')

# Listado breve de ficheros creados en la carpeta del proyecto.
print('\nFicheros principales en OUTPUT_DIR:')
for p in sorted(OUTPUT_DIR.glob('*')):
    if p.is_file():
        print(f'- {p.name} ({p.stat().st_size / 1024:.1f} KB)')
    else:
        print(f'- {p.name}/')
```

OUTPUT_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0
Existe OUTPUT_DIR: True
MODEL_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_sb3_latest.zip | existe: True
POLICY_LATEST: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_spaceinvaders_v0_policy_latest.pt | existe: True
LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json | existe: True
CSV_LOG_FILENAME: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv | existe: True
TENSORBOARD_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/tensorboard | existe: True
MONITOR_DIR: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/monitor | existe: True

Resumen del log CSV
Episodios registrados: 12768
Último episodio: 12767
Último step: 2910112
Mejor clipped_episode_reward: 104.0
Mejor avg_clip50: 40.68
Mejor max_consec: 100
Último avg_clip50: 37.5

	episode	total_steps	episode_reward	clipped_episode_reward	consecutive_clipped_over_20	max_consecutive_clipped_over_20	avg_clip
12763	12763	2908838	52.0	52.0	96	96	37
12764	12764	2909241	35.0	35.0	97	97	37
12765	12765	2909564	35.0	35.0	98	98	37
12766	12766	2909856	36.0	36.0	99	99	37
12767	12767	2910112	24.0	24.0	100	100	37



Resumen del log JSON

Claves: ['episode', 'total_steps', 'episode_reward', 'clipped_episode_reward', 'consecutive_clipped_over_20', 'max_consecutive_clipped_over_20', 'avg_clip50', 'duration', 'exploration_rate']

Episodios en JSON: 12768

Ficheros principales en OUTPUT_DIR:

- dqn_SpaceInvaders-v0_sb3_log.csv (859.6 KB)
- dqn_SpaceInvaders-v0_sb3_log.json (1420.8 KB)
- dqn_spaceinvaders_v0_policy_latest.pt (26381.2 KB)
- dqn_spaceinvaders_v0_sb3_latest.zip (0.0 KB)
- monitor/
- tensorboard/

```
In [ ]: import gym
import numpy as np
from stable_baselines3.common.atari_wrappers import (
    MaxAndSkipEnv,
    FireResetEnv,
    WarpFrame
)
from stable_baselines3.common.vec_env import DummyVecEnv, VecFrameStack, VecTransposeImage

def create_eval_env():
    def make_env():
        env = gym.make(ENV_ID)

        env.seed(SEED)
        env = MaxAndSkipEnv(env, skip=4)

        if "FIRE" in env.unwrapped.get_action_meanings():
            env = FireResetEnv(env)

        env = WarpFrame(env)

        return env

    env = DummyVecEnv([make_env])
    env = VecFrameStack(env, n_stack=4)
    env = VecTransposeImage(env)
```

```
return env
```

```
In [ ]: # policy aprendida

import torch
from stable_baselines3 import DQN

POLICY_LATEST = OUTPUT_DIR / "dqn_spaceinvaders_v0_policy_latest.pt"

eval_env = create_eval_env()

model = DQN(
    policy="CnnPolicy",
    env=eval_env,
    learning_rate=LEARNING_RATE,
    buffer_size=BUFFER_SIZE,
    learning_starts=LEARNING_STARTS,
    batch_size=BATCH_SIZE,
    gamma=GAMMA,
    train_freq=TRAIN_FREQ,
    target_update_interval=TARGET_UPDATE_INTERVAL,
    exploration_initial_eps=EXPLORATION_INITIAL_EPS,
    exploration_final_eps=EXPLORATION_FINAL_EPS,
    exploration_fraction=EXPLORATION_FRACTION,
    policy_kwargs=policy_kwargs,
    verbose=1,
    seed=SEED,
    device="auto"
)

checkpoint = torch.load(POLICY_LATEST, map_location=model.device)

model.policy.load_state_dict(checkpoint["policy_state_dict"])
model.num_timesteps = checkpoint["num_timesteps"]
model.exploration_rate = checkpoint["exploration_rate"]

print("Policy cargada correctamente")
print("Timesteps:", model.num_timesteps)
print("Exploration rate:", model.exploration_rate)
```

Using cuda device
Policy cargada correctamente
Timesteps: 2910112
Exploration rate: 0.030933037000000163

```
In [ ]: # testing

TEST_EPISODES = 120

rewards = []
clipped_rewards = []

for ep in range(TEST_EPISODES):
    obs = eval_env.reset()
    done = False
    total_reward = 0.0

    while not done:
        action, _ = model.predict(obs, deterministic=True)
        obs, reward, done, infos = eval_env.step(action)
        total_reward += float(reward[0])
        done = bool(done[0])

    clipped_reward = total_reward / 10.0
    rewards.append(total_reward)
    clipped_rewards.append(clipped_reward)

    print(
        f"TEST EP={ep+1:03d} | "
        f"reward={total_reward:.0f} | "
        f"clip={clipped_reward:.0f}"
    )

print("Reward medio:", np.mean(rewards))
print("Clip medio:", np.mean(clipped_rewards))
print("Reward máximo:", np.max(rewards))
print("Clip máximo:", np.max(clipped_rewards))

eval_env.close()
```

TEST EP=001	reward=490	clip=49
TEST EP=002	reward=975	clip=98
TEST EP=003	reward=575	clip=58
TEST EP=004	reward=550	clip=55
TEST EP=005	reward=545	clip=54
TEST EP=006	reward=110	clip=11
TEST EP=007	reward=485	clip=48
TEST EP=008	reward=600	clip=60
TEST EP=009	reward=605	clip=60
TEST EP=010	reward=630	clip=63
TEST EP=011	reward=545	clip=54
TEST EP=012	reward=505	clip=50
TEST EP=013	reward=495	clip=50
TEST EP=014	reward=630	clip=63
TEST EP=015	reward=575	clip=58
TEST EP=016	reward=485	clip=48
TEST EP=017	reward=605	clip=60
TEST EP=018	reward=600	clip=60
TEST EP=019	reward=530	clip=53
TEST EP=020	reward=1585	clip=158
TEST EP=021	reward=805	clip=80
TEST EP=022	reward=585	clip=58
TEST EP=023	reward=490	clip=49
TEST EP=024	reward=275	clip=28
TEST EP=025	reward=1100	clip=110
TEST EP=026	reward=545	clip=54
TEST EP=027	reward=465	clip=46
TEST EP=028	reward=600	clip=60
TEST EP=029	reward=600	clip=60
TEST EP=030	reward=555	clip=56
TEST EP=031	reward=530	clip=53
TEST EP=032	reward=570	clip=57
TEST EP=033	reward=775	clip=78
TEST EP=034	reward=470	clip=47
TEST EP=035	reward=1660	clip=166
TEST EP=036	reward=545	clip=54
TEST EP=037	reward=1315	clip=132
TEST EP=038	reward=600	clip=60
TEST EP=039	reward=545	clip=54
TEST EP=040	reward=600	clip=60
TEST EP=041	reward=1050	clip=105

TEST EP=042	reward=635	clip=64
TEST EP=043	reward=575	clip=58
TEST EP=044	reward=495	clip=50
TEST EP=045	reward=800	clip=80
TEST EP=046	reward=600	clip=60
TEST EP=047	reward=495	clip=50
TEST EP=048	reward=520	clip=52
TEST EP=049	reward=500	clip=50
TEST EP=050	reward=575	clip=58
TEST EP=051	reward=515	clip=52
TEST EP=052	reward=255	clip=26
TEST EP=053	reward=775	clip=78
TEST EP=054	reward=800	clip=80
TEST EP=055	reward=920	clip=92
TEST EP=056	reward=575	clip=58
TEST EP=057	reward=860	clip=86
TEST EP=058	reward=995	clip=100
TEST EP=059	reward=605	clip=60
TEST EP=060	reward=605	clip=60
TEST EP=061	reward=1015	clip=102
TEST EP=062	reward=1225	clip=122
TEST EP=063	reward=600	clip=60
TEST EP=064	reward=1205	clip=120
TEST EP=065	reward=455	clip=46
TEST EP=066	reward=600	clip=60
TEST EP=067	reward=760	clip=76
TEST EP=068	reward=555	clip=56
TEST EP=069	reward=830	clip=83
TEST EP=070	reward=515	clip=52
TEST EP=071	reward=545	clip=54
TEST EP=072	reward=545	clip=54
TEST EP=073	reward=605	clip=60
TEST EP=074	reward=885	clip=88
TEST EP=075	reward=545	clip=54
TEST EP=076	reward=945	clip=94
TEST EP=077	reward=575	clip=58
TEST EP=078	reward=600	clip=60
TEST EP=079	reward=460	clip=46
TEST EP=080	reward=580	clip=58
TEST EP=081	reward=1040	clip=104
TEST EP=082	reward=490	clip=49

TEST EP=083	reward=365	clip=36
TEST EP=084	reward=545	clip=54
TEST EP=085	reward=600	clip=60
TEST EP=086	reward=1125	clip=112
TEST EP=087	reward=495	clip=50
TEST EP=088	reward=605	clip=60
TEST EP=089	reward=515	clip=52
TEST EP=090	reward=540	clip=54
TEST EP=091	reward=600	clip=60
TEST EP=092	reward=1385	clip=138
TEST EP=093	reward=600	clip=60
TEST EP=094	reward=605	clip=60
TEST EP=095	reward=575	clip=58
TEST EP=096	reward=545	clip=54
TEST EP=097	reward=745	clip=74
TEST EP=098	reward=550	clip=55
TEST EP=099	reward=555	clip=56
TEST EP=100	reward=520	clip=52
TEST EP=101	reward=525	clip=52
TEST EP=102	reward=580	clip=58
TEST EP=103	reward=645	clip=64
TEST EP=104	reward=550	clip=55
TEST EP=105	reward=570	clip=57
TEST EP=106	reward=910	clip=91
TEST EP=107	reward=980	clip=98
TEST EP=108	reward=175	clip=18
TEST EP=109	reward=575	clip=58
TEST EP=110	reward=500	clip=50
TEST EP=111	reward=925	clip=92
TEST EP=112	reward=1010	clip=101
TEST EP=113	reward=635	clip=64
TEST EP=114	reward=1140	clip=114
TEST EP=115	reward=1125	clip=112
TEST EP=116	reward=580	clip=58
TEST EP=117	reward=600	clip=60
TEST EP=118	reward=575	clip=58
TEST EP=119	reward=1045	clip=104
TEST EP=120	reward=600	clip=60

Reward medio: 668.25
Clip medio: 66.825

Reward máximo: 1660.0

Clip máximo: 166.0

Vídeo del juego

Se graban dos vídeos con la política **del modelo base ya entrenado**

1. **Una partida individual** (el wrapper de Atari corta el episodio al perder una vida).
2. **10 episodios seguidos** en un único mp4, para observar el comportamiento del agente de forma más representativa.

```
In [ ]: from stable_baselines3.common.vec_env import VecFrameStack, VecVideoRecorder
        from IPython.display import display
        import shutil

        VIDEO_DIR = OUTPUT_DIR / 'video_1'
        if VIDEO_DIR.exists():
            shutil.rmtree(VIDEO_DIR)
        VIDEO_DIR.mkdir(parents=True, exist_ok=True)

        video_env = make_atari_env(
            ENV_ID,
            n_envs=1,
            seed=SEED,
            wrapper_kwargs=dict(clip_reward=True)
        )
        video_env = VecFrameStack(video_env, n_stack=4)
        video_env = VecVideoRecorder(
            video_env,
            str(VIDEO_DIR),
            record_video_trigger=lambda step: step == 0,
            video_length=5_000,
            name_prefix='sb3-spaceinvaders'
        )

        obs = video_env.reset()
        for _ in range(5_000):
            action, _ = model.predict(obs, deterministic=True)
            obs, rewards, dones, infos = video_env.step(action)
```

```

    if bool(dones[0]):
        break

video_env.close()

video_files = sorted(VIDEO_DIR.glob('*.mp4'))
print('Vídeos generados:', [video_file.name for video_file in video_files])
display(video_files[0] if video_files else 'No se ha generado ningún vídeo.')

```

Vídeos generados: ['sb3-spaceinvaders-step-0-to-step-5000.mp4']

PosixPath('/content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/video_1/sb3-spaceinvaders-step-0-to-step-5000.mp4')

```

In [ ]: import io
import base64
from IPython.display import HTML

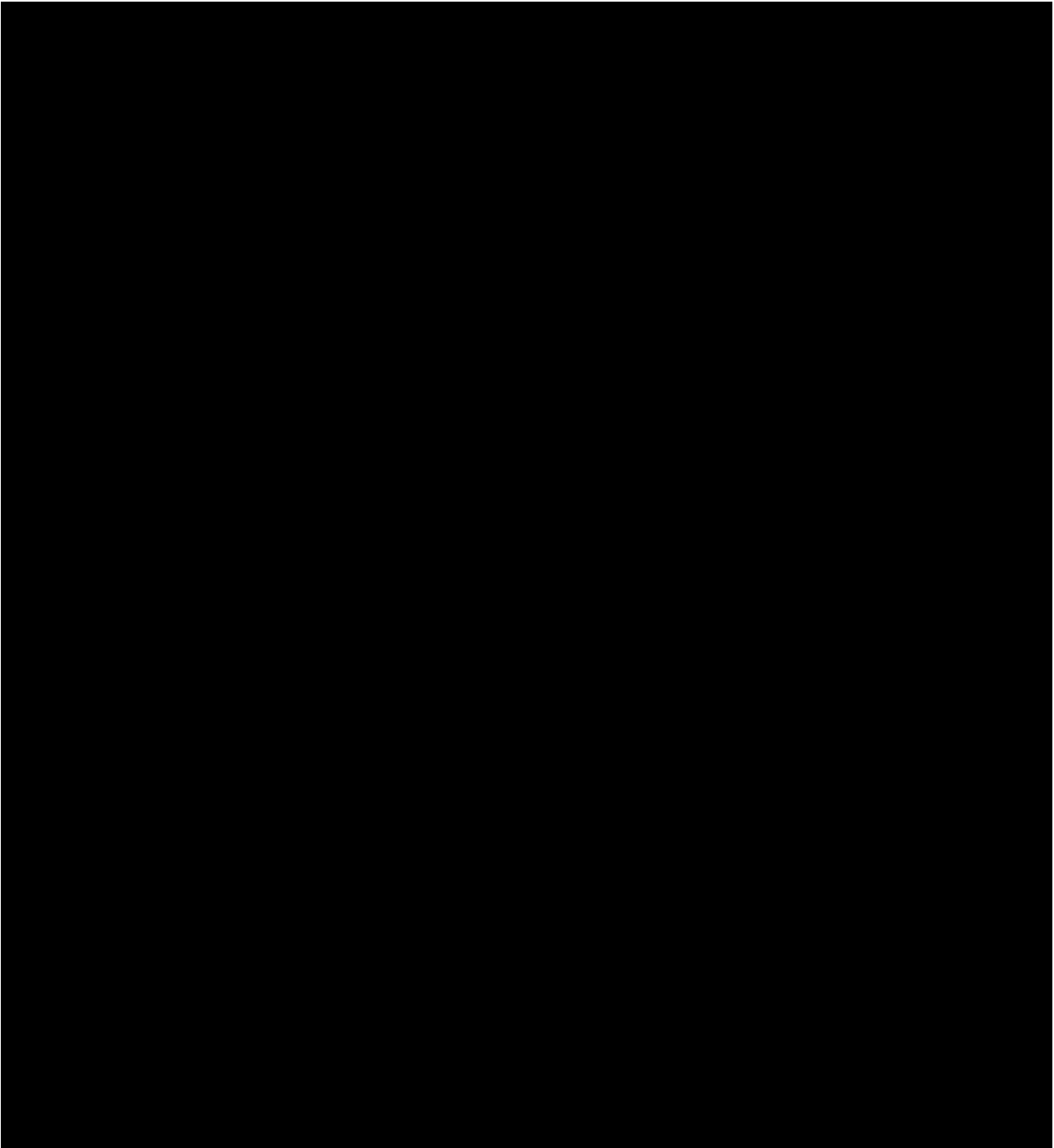
video_files = sorted(VIDEO_DIR.glob('*.mp4'))
if not video_files:
    raise FileNotFoundError(f'No se encontró ningún MP4 en {VIDEO_DIR}')

video_path = video_files[0]
with io.open(video_path, 'r+b') as video_file:
    video = video_file.read()

encoded = base64.b64encode(video).decode('ascii')
HTML(
    data=f'''
    <video width="640" height="auto" alt="space-invaders" controls>
        <source src="data:video/mp4;base64,{encoded}" type="video/mp4" />
    </video>'''
)

```

Out[]:





```
In [ ]: from stable_baselines3.common.vec_env import VecFrameStack, VecVideoRecorder
import shutil

N_EPISODES = 10
MAX_STEPS_PER_EP = 5_000

VIDEO_DIR = OUTPUT_DIR / 'video_10'
if VIDEO_DIR.exists():
    shutil.rmtree(VIDEO_DIR)
VIDEO_DIR.mkdir(parents=True, exist_ok=True)

video_env = make_atari_env(
    ENV_ID,
    n_envs=1,
    seed=SEED,
    wrapper_kwargs=dict(clip_reward=False)
)
video_env = VecFrameStack(video_env, n_stack=4)
video_env = VecVideoRecorder(
    video_env,
    str(VIDEO_DIR),
    record_video_trigger=lambda step: step == 0,
    video_length=N_EPISODES * MAX_STEPS_PER_EP,
    name_prefix='sb3-spaceinvaders-10eps'
)

obs = video_env.reset()
episodes_done = 0
step = 0
while episodes_done < N_EPISODES:
```

```
action, _ = model.predict(obs, deterministic=True)
obs, rewards, done, infos = video_env.step(action)
step += 1
if bool(done[0]):
    episodes_done += 1
video_env.close()
```

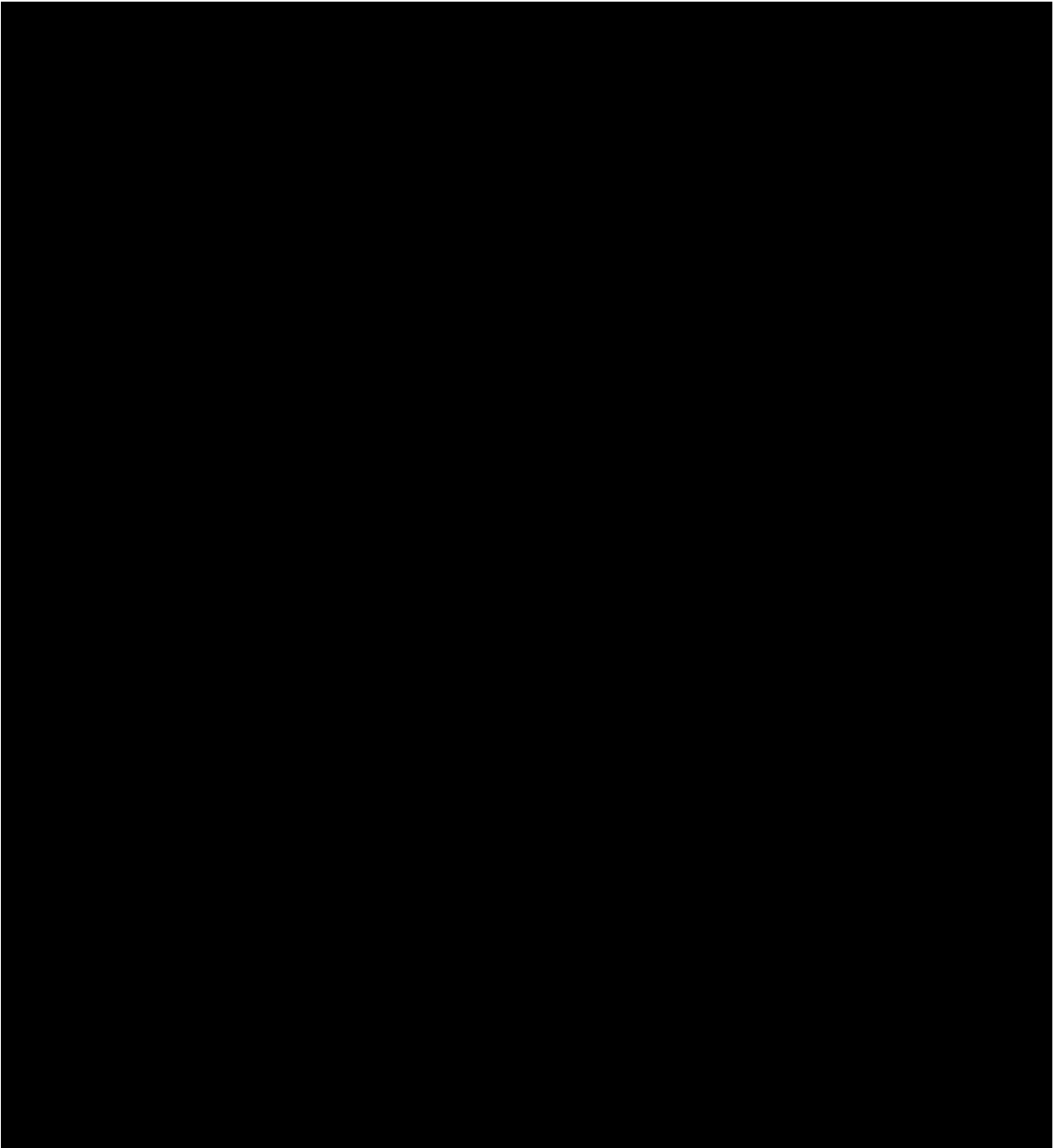
```
In [ ]: import io
import base64
from IPython.display import HTML

video_files = sorted(VIDEO_DIR.glob('*.mp4'))
if not video_files:
    raise FileNotFoundError(f'No se encontró ningún MP4 en {VIDEO_DIR}')

video_path = video_files[0]
with io.open(video_path, 'r+b') as video_file:
    video = video_file.read()

encoded = base64.b64encode(video).decode('ascii')
HTML(
    data=f'''
    <video width="640" height="auto" alt="space-invaders" controls>
      <source src="data:video/mp4;base64,{encoded}" type="video/mp4" />
    </video>'''
)
```

Out[]:





PARTE 2 (ampliación). Propuestas de mejora sobre el DQN base

El DQN entrenado en las secciones anteriores actúa como **referencia (baseline)**. Sobre él se implementan y comparan **tres propuestas de mejora** independientes.

#	Propuesta	Qué cambia	Motivación
1	Double DQN (van Hasselt et al., 2016)	El target pasa de <code>max_a Q_target(s',a)</code> a <code>Q_target(s', argmax_a Q_online(s',a))</code>	El operador <code>max</code> sobre estimaciones ruidosas sobreestima sistemáticamente los valores Q; separar la <i>selección</i> de la acción (red online) de su <i>evaluación</i> (red target) reduce ese sesgo
2	Dueling DQN	La cabeza de la red se divide en dos streams: <code>V(s)</code> y <code>A(s,a)</code> , con <code>Q = V + A - mean(A)</code>	En muchos estados de Space Invaders la acción elegida apenas cambia el valor del estado (p. ej. cuando no hay proyectiles cerca); estimar <code>V(s)</code> por separado propaga aprendizaje a todas las acciones a la vez
3	Exploración ajustada	<code>exploration_fraction</code> <code>0.30</code> $\rightarrow$ <code>0.10</code> , <code>ε_final</code> <code>0.001</code> $\rightarrow$ <code>0.01</code> , <code>target_update_interval</code> <code>10k</code> $\rightarrow$ <code>5k</code> , <code>learning_starts</code> <code>50k</code> $\rightarrow$ <code>25k</code>	El run base terminó con $\epsilon \approx 0.12$: el 12 % de las acciones seguían siendo aleatorias incluso al final del entrenamiento. Un annealing más corto, adaptado al presupuesto real, permite explotar antes lo aprendido

```
In [ ]: RUN_NAMES = ['base', 'double', 'dueling', 'tuned']

RUN_LABELS = {
    'base': 'DQN base',
```

```
'double': 'Double DQN',
'dueling': 'Dueling DQN',
'tuned': 'DQN exploración ajustada',
}

def setup_run_paths(run_name):
    if run_name == 'base':
        run_dir = OUTPUT_DIR
    else:
        run_dir = Path(drive_root) / 'datos' / (PROJECT_NAME + '_' + run_name)
    run_dir.mkdir(parents=True, exist_ok=True)
    (run_dir / 'monitor').mkdir(parents=True, exist_ok=True)
    (run_dir / 'tensorboard').mkdir(parents=True, exist_ok=True)
    return {
        'dir': run_dir,
        'policy': run_dir / 'dqn_spaceinvaders_v0_policy_latest.pt',
        'model_zip': run_dir / 'dqn_spaceinvaders_v0_sb3_latest.zip',
        'json': run_dir / 'dqn_SpaceInvaders-v0_sb3_log.json',
        'csv': run_dir / 'dqn_SpaceInvaders-v0_sb3_log.csv',
        'monitor': run_dir / 'monitor',
        'tensorboard': run_dir / 'tensorboard',
        'test_json': run_dir / 'test_results.json',
    }

RUN_PATHS = {name: setup_run_paths(name) for name in RUN_NAMES}

def create_train_env(monitor_dir):
    env = make_atari_env(
        ENV_ID,
        n_envs=1,
        seed=SEED,
        monitor_dir=str(monitor_dir),
        wrapper_kwargs=dict(clip_reward=True)
    )
    return VecFrameStack(env, n_stack=4)

def train_variant(run_name, model, total_timesteps=TOTAL_TIMESTEPS):
    paths = RUN_PATHS[run_name]
    training_logger = SB3TrainingLogger(json_path=paths['json'], csv_path=paths['csv'])
    callback = ConsecutiveRewardCallback(training_logger=training_logger, verbose=1)
```

```

model.learn(
    total_timesteps=total_timesteps,
    callback=callback,
    reset_num_timesteps=True,
    tb_log_name='DQN_' + run_name,
    log_interval=100
)

save_policy_checkpoint(model, paths['policy'])
print('Checkpoint PyTorch guardado en:', paths['policy'])
try:
    model.save(str(paths['model_zip']))
    print('Modelo SB3 guardado en:', paths['model_zip'])
except Exception as e:
    print('model.save() falló (no es crítico, se usa el .pt):', repr(e))

return model

print('Rutas por variante creadas:')
for name in RUN_NAMES:
    print(' -', RUN_LABELS[name], '->', RUN_PATHS[name]['dir'])

```

Rutas por variante creadas:

- DQN base -> /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0
- Double DQN -> /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_double
- Dueling DQN -> /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_dueling
- DQN exploración ajustada -> /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_tuned

In []: *# Propuesta 1: Double DQN*

```

import torch.nn.functional as F

try:
    from stable_baselines3.common import logger as sb3_logger
except Exception:
    sb3_logger = None

class DoubleDQN(DQN):

    def train(self, gradient_steps, batch_size=100):

```

```

self._update_learning_rate(self.policy.optimizer)

losses = []
for _ in range(gradient_steps):
    replay_data = self.replay_buffer.sample(batch_size, env=self._vec_normalize_env)

    with th.no_grad():
        next_actions = self.q_net(replay_data.next_observations).argmax(dim=1, keepdim=True)
        next_q = th.gather(self.q_net_target(replay_data.next_observations), dim=1, index=next_actions)
        target_q = replay_data.rewards + (1 - replay_data.dones) * self.gamma * next_q

    current_q = th.gather(self.q_net(replay_data.observations), dim=1, index=replay_data.actions.long())

    loss = F.smooth_l1_loss(current_q, target_q)
    losses.append(loss.item())

    self.policy.optimizer.zero_grad()
    loss.backward()
    th.nn.utils.clip_grad_norm_(self.policy.parameters(), self.max_grad_norm)
    self.policy.optimizer.step()

self._n_updates += gradient_steps

log = getattr(self, 'logger', None) or sb3_logger
if log is not None:
    log.record('train/n_updates', self._n_updates, exclude='tensorboard')
    log.record('train/loss', float(np.mean(losses)))

print('DoubleDQN definido.')
```

DoubleDQN definido.

```

In [ ]: # Propuesta 2: Dueling DQN

from stable_baselines3.dqn.policies import CnnPolicy as DQNCnnPolicy
from stable_baselines3.dqn.policies import QNetwork

class DuelingQNetwork(QNetwork):
```

```

def __init__(self, *args, **kwargs):
    super().__init__(*args, **kwargs)
    action_dim = self.action_space.n

    self.q_net = nn.Identity()
    self.value_stream = nn.Sequential(
        nn.Linear(self.features_dim, 256),
        nn.ReLU(),
        nn.Linear(256, 1)
    )
    self.advantage_stream = nn.Sequential(
        nn.Linear(self.features_dim, 256),
        nn.ReLU(),
        nn.Linear(256, action_dim)
    )

def forward(self, obs):
    features = self.extract_features(obs)
    value = self.value_stream(features)
    advantage = self.advantage_stream(features)
    return value + advantage - advantage.mean(dim=1, keepdim=True)

class DuelingCnnPolicy(DQNCnnPolicy):

    def make_q_net(self):
        net_args = self._update_features_extractor(self.net_args, features_extractor=None)
        return DuelingQNetwork(**net_args).to(self.device)

print('DuelingCnnPolicy definida.')
```

DuelingCnnPolicy definida.

In []: *# Propuesta 3 (exploración ajustada) y constructor de modelos*

```

TUNED_PARAMS = dict(
    exploration_fraction=0.10,
    exploration_final_eps=0.01,
    target_update_interval=5_000,
```

```

        learning_starts=25_000,
    )

def build_model(run_name, env, tensorboard_dir=None, for_eval=False):

    common = dict(
        env=env,
        learning_rate=LEARNING_RATE,
        buffer_size=1_000 if for_eval else BUFFER_SIZE,
        learning_starts=LEARNING_STARTS,
        batch_size=BATCH_SIZE,
        gamma=GAMMA,
        train_freq=TRAIN_FREQ,
        target_update_interval=TARGET_UPDATE_INTERVAL,
        exploration_initial_eps=EXPLORATION_INITIAL_EPS,
        exploration_final_eps=EXPLORATION_FINAL_EPS,
        exploration_fraction=EXPLORATION_FRACTION,
        policy_kwargs=policy_kwargs,
        tensorboard_log=str(tensorboard_dir) if tensorboard_dir is not None else None,
        verbose=1,
        seed=SEED,
        device='auto',
    )

    if run_name == 'base':
        return DQN(policy='CnnPolicy', **common)
    if run_name == 'double':
        return DoubleDQN(policy='CnnPolicy', **common)
    if run_name == 'dueling':
        return DQN(policy=DuelingCnnPolicy, **common)
    if run_name == 'tuned':
        common.update(TUNED_PARAMS)
        return DQN(policy='CnnPolicy', **common)

    raise ValueError('Variante desconocida: ' + run_name)

print('build_model listo. Variantes disponibles:', RUN_NAMES)

```

build_model listo. Variantes disponibles: ['base', 'double', 'dueling', 'tuned']

In []: *# Entrenamiento Propuesta 1: Double DQN*

```
env_double = create_train_env(RUN_PATHS['double']['monitor'])
model_double = build_model('double', env_double, RUN_PATHS['double']['tensorboard'])
model_double = train_variant('double', model_double)
env_double.close()
```

Using cuda device

Wrapping the env in a VecTransposeImage.

Logging to /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_double/tensorboard/DQN_double_1

EP=0 | STEP=267 | clip=180 | consec=1 | max_consec=1 | avg_clip50=180.00

EP=10 | STEP=1,894 | clip=655 | consec=3 | max_consec=7 | avg_clip50=184.55

EP=20 | STEP=3,135 | clip=80 | consec=13 | max_consec=13 | avg_clip50=130.48

EP=30 | STEP=4,765 | clip=155 | consec=23 | max_consec=23 | avg_clip50=142.90

custom/		
avg_clip50	141	
clipped_episode_r...	155	
consecutive_clipp...	25	
max_consecutive_c...	25	
rollout/		
ep_len_mean	659	
ep_rew_mean	141	
exploration rate	0.998	
time/		
episodes	100	
fps	399	
time_elapsed	12	
total timesteps	5126	

EP=40 | STEP=6,186 | clip=155 | consec=33 | max_consec=33 | avg_clip50=137.44

EP=50 | STEP=7,696 | clip=75 | consec=43 | max_consec=43 | avg_clip50=131.50

EP=60 | STEP=9,261 | clip=45 | consec=1 | max_consec=51 | avg_clip50=118.10

custom/		
avg_clip50	131	
clipped_episode_r...	140	
consecutive_clipp...	6	
max_consecutive_c...	51	
rollout/		
ep_len_mean	660	
ep_rew_mean	136	
exploration rate	0.997	
time/		
episodes	200	
fps	399	
time_elapsed	25	
total timesteps	10233	

```

-----
EP=70 | STEP=10,844 | clip=320 | consec=11 | max_consec=51 | avg_clip50=138.00
EP=80 | STEP=12,464 | clip=335 | consec=21 | max_consec=51 | avg_clip50=140.40
EP=90 | STEP=14,027 | clip=45 | consec=31 | max_consec=51 | avg_clip50=138.90
-----

```

```

| custom/
|   avg_clip50          | 148 |
|   clipped_episode_r... | 210 |
|   consecutive_clipp... | 40  |
|   max_consecutive_c... | 51  |
| rollout/
|   ep_len_mean         | 665 |
|   ep_rew_mean         | 141 |
|   exploration rate    | 0.995 |
| time/
|   episodes            | 300 |
|   fps                 | 400 |
|   time_elapsed        | 38  |
|   total timesteps     | 15452 |
-----

```

```

EP=100 | STEP=15,660 | clip=245 | consec=41 | max_consec=51 | avg_clip50=151.50
EP=110 | STEP=17,107 | clip=60 | consec=51 | max_consec=51 | avg_clip50=151.40
EP=120 | STEP=18,403 | clip=130 | consec=7 | max_consec=52 | avg_clip50=136.40
EP=130 | STEP=20,065 | clip=205 | consec=4 | max_consec=52 | avg_clip50=128.30
-----

```

```

| custom/
|   avg_clip50          | 129 |
|   clipped_episode_r... | 175 |
|   consecutive_clipp... | 6   |
|   max_consecutive_c... | 52  |
| rollout/
|   ep_len_mean         | 657 |
|   ep_rew_mean         | 135 |
|   exploration rate    | 0.993 |
| time/
|   episodes            | 400 |
|   fps                 | 400 |
|   time_elapsed        | 50  |
|   total timesteps     | 20359 |
-----

```

```

EP=140 | STEP=21,723 | clip=20 | consec=0 | max_consec=52 | avg_clip50=134.70

```

EP=150 | STEP=23,061 | clip=90 | consec=9 | max_consec=52 | avg_clip50=116.80
 EP=160 | STEP=24,659 | clip=65 | consec=19 | max_consec=52 | avg_clip50=126.10

```
-----
| custom/                |      |
|   avg_clip50           | 136   |
|   clipped_episode_r... | 120   |
|   consecutive_clipp... | 25    |
|   max_consecutive_c... | 52    |
| rollout/               |      |
|   ep_len_mean          | 666   |
|   ep_rew_mean          | 137   |
|   exploration rate     | 0.991 |
| time/                  |      |
|   episodes             | 500   |
|   fps                  | 401   |
|   time_elapsed         | 64    |
|   total timesteps      | 25762 |
-----
```

EP=170 | STEP=26,391 | clip=155 | consec=29 | max_consec=52 | avg_clip50=136.90
 EP=180 | STEP=28,172 | clip=355 | consec=39 | max_consec=52 | avg_clip50=142.50
 EP=190 | STEP=29,744 | clip=55 | consec=3 | max_consec=52 | avg_clip50=138.30

```
-----
| custom/                |      |
|   avg_clip50           | 155   |
|   clipped_episode_r... | 125   |
|   consecutive_clipp... | 12    |
|   max_consecutive_c... | 52    |
| rollout/               |      |
|   ep_len_mean          | 680   |
|   ep_rew_mean          | 137   |
|   exploration rate     | 0.99  |
| time/                  |      |
|   episodes             | 600   |
|   fps                  | 401   |
|   time_elapsed         | 78    |
|   total timesteps      | 31348 |
-----
```

EP=200 | STEP=31,493 | clip=125 | consec=13 | max_consec=52 | avg_clip50=155.40
 EP=210 | STEP=33,300 | clip=150 | consec=23 | max_consec=52 | avg_clip50=158.20
 EP=220 | STEP=35,127 | clip=45 | consec=1 | max_consec=52 | avg_clip50=165.50
 EP=230 | STEP=36,618 | clip=135 | consec=11 | max_consec=52 | avg_clip50=153.60

custom/		
avg_clip50	162	
clipped_episode_r...	5	
consecutive_clipp...	0	
max_consecutive_c...	52	
rollout/		
ep_len_mean	712	
ep_rew_mean	151	
exploration rate	0.988	
time/		
episodes	700	
fps	402	
time_elapsed	92	
total timesteps	37116	

EP=240 | STEP=38,192 | clip=80 | consec=7 | max_consec=52 | avg_clip50=160.50
 EP=250 | STEP=39,814 | clip=120 | consec=8 | max_consec=52 | avg_clip50=153.30
 EP=260 | STEP=41,629 | clip=270 | consec=18 | max_consec=52 | avg_clip50=145.90

custom/		
avg_clip50	144	
clipped_episode_r...	120	
consecutive_clipp...	24	
max_consecutive_c...	52	
rollout/		
ep_len_mean	721	
ep_rew_mean	154	
exploration rate	0.986	
time/		
episodes	800	
fps	402	
time_elapsed	106	
total timesteps	42698	

EP=270 | STEP=43,135 | clip=50 | consec=28 | max_consec=52 | avg_clip50=137.40
 EP=280 | STEP=44,654 | clip=150 | consec=38 | max_consec=52 | avg_clip50=143.70
 EP=290 | STEP=46,253 | clip=60 | consec=6 | max_consec=52 | avg_clip50=136.30

custom/		
avg_clip50	127	

	clipped_episode_r...	45	
	consecutive_clipp...	7	
	max_consecutive_c...	52	
	rollout/		
	ep_len_mean	696	
	ep_rew_mean	140	
	exploration rate	0.984	
	time/		
	episodes	900	
	fps	401	
	time_elapsed	118	
	total timesteps	47648	

EP=300	STEP=47,728	clip=75	consec=8	max_consec=52	avg_clip50=125.80
EP=310	STEP=49,428	clip=135	consec=18	max_consec=52	avg_clip50=129.80
EP=320	STEP=50,885	clip=30	consec=7	max_consec=52	avg_clip50=122.60
EP=330	STEP=52,484	clip=120	consec=2	max_consec=52	avg_clip50=121.90

	custom/		
	avg_clip50	123	
	clipped_episode_r...	170	
	consecutive_clipp...	5	
	max_consecutive_c...	52	
	rollout/		
	ep_len_mean	685	
	ep_rew_mean	130	
	exploration rate	0.982	
	time/		
	episodes	1000	
	fps	386	
	time_elapsed	137	
	total timesteps	53127	
	train/		
	learning_rate	0.0001	
	loss	0.0164	
	n_updates	781	

EP=340	STEP=54,308	clip=35	consec=12	max_consec=52	avg_clip50=121.70
EP=350	STEP=55,931	clip=95	consec=22	max_consec=52	avg_clip50=139.00
EP=360	STEP=57,293	clip=90	consec=7	max_consec=52	avg_clip50=130.00

custom/	
avg_clip50	121
clipped_episode_r...	50
consecutive_clipp...	13
max_consecutive_c...	52
rollout/	
ep_len_mean	662
ep_rew_mean	122
exploration rate	0.981
time/	
episodes	1100
fps	368
time_elapsed	157
total timesteps	58177
train/	
learning_rate	0.0001
loss	0.00201
n_updates	2044

EP=370	STEP=58,649	clip=155	consec=17	max_consec=52	avg_clip50=123.80
EP=380	STEP=60,423	clip=185	consec=8	max_consec=52	avg_clip50=131.00
EP=390	STEP=61,893	clip=410	consec=6	max_consec=52	avg_clip50=136.50
EP=400	STEP=63,212	clip=130	consec=7	max_consec=52	avg_clip50=116.70

custom/	
avg_clip50	117
clipped_episode_r...	130
consecutive_clipp...	7
max_consecutive_c...	52
rollout/	
ep_len_mean	665
ep_rew_mean	128
exploration rate	0.979
time/	
episodes	1200
fps	354
time_elapsed	178
total timesteps	63212
train/	
learning_rate	0.0001
loss	0.0295

n_updates		3302	

EP=410	STEP=64,559	clip=95	consec=1 max_consec=52 avg_clip50=110.70
EP=420	STEP=66,086	clip=160	consec=11 max_consec=52 avg_clip50=119.10
EP=430	STEP=67,803	clip=140	consec=21 max_consec=52 avg_clip50=117.10

custom/			
avg_clip50		117	
clipped_episode_r...		55	
consecutive_clipp...		24	
max_consecutive_c...		52	
rollout/			
ep_len_mean		652	
ep_rew_mean		124	
exploration rate		0.977	
time/			
episodes		1300	
fps		343	
time_elapsed		199	
total timesteps		68397	
train/			
learning_rate		0.0001	
loss		0.0271	
n_updates		4599	

EP=440	STEP=69,556	clip=65	consec=31 max_consec=52 avg_clip50=119.10
EP=450	STEP=71,418	clip=260	consec=41 max_consec=52 avg_clip50=141.80
EP=460	STEP=73,377	clip=210	consec=51 max_consec=52 avg_clip50=165.20

custom/			
avg_clip50		166	
clipped_episode_r...		160	
consecutive_clipp...		58	
max_consecutive_c...		58	
rollout/			
ep_len_mean		698	
ep_rew_mean		143	
exploration rate		0.975	
time/			
episodes		1400	
fps		333	

	time_elapsed	223	
	total timesteps	74515	
	train/		
	learning_rate	0.0001	
	loss	0.0296	
	n_updates	6128	

EP=470 | STEP=74,836 | clip=50 | consec=61 | max_consec=61 | avg_clip50=168.30
 EP=480 | STEP=76,410 | clip=425 | consec=71 | max_consec=71 | avg_clip50=163.20
 EP=490 | STEP=77,784 | clip=60 | consec=2 | max_consec=78 | avg_clip50=153.50
 EP=500 | STEP=79,279 | clip=190 | consec=12 | max_consec=78 | avg_clip50=142.50

	custom/		
	avg_clip50	142	
	clipped_episode_r...	190	
	consecutive_clipp...	12	
	max_consecutive_c...	78	
	rollout/		
	ep_len_mean	688	
	ep_rew_mean	142	
	exploration rate	0.974	
	time/		
	episodes	1500	
	fps	326	
	time_elapsed	242	
	total timesteps	79306	
	train/		
	learning_rate	0.0001	
	loss	0.000559	
	n_updates	7326	

EP=510 | STEP=80,811 | clip=90 | consec=22 | max_consec=78 | avg_clip50=127.80
 EP=520 | STEP=82,036 | clip=40 | consec=7 | max_consec=78 | avg_clip50=115.10
 EP=530 | STEP=83,344 | clip=170 | consec=17 | max_consec=78 | avg_clip50=110.10

	custom/		
	avg_clip50	122	
	clipped_episode_r...	125	
	consecutive_clipp...	20	
	max_consecutive_c...	78	
	rollout/		

ep_len_mean	680
ep_rew_mean	142
exploration rate	0.972
time/	
episodes	1600
fps	320
time_elapsed	262
total timesteps	84169
train/	
learning_rate	0.0001
loss	0.0246
n_updates	8542

EP=540	STEP=85,329	clip=85	consec=27	max_consec=78	avg_clip50=125.10
EP=550	STEP=87,035	clip=10	consec=0	max_consec=78	avg_clip50=131.20
EP=560	STEP=88,125	clip=55	consec=10	max_consec=78	avg_clip50=120.30

custom/	
avg_clip50	129
clipped_episode_r...	155
consecutive_clipp...	1
max_consecutive_c...	78
rollout/	
ep_len_mean	633
ep_rew_mean	124
exploration rate	0.97
time/	
episodes	1700
fps	315
time_elapsed	282
total timesteps	89202
train/	
learning_rate	0.0001
loss	0.0182
n_updates	9800

EP=570	STEP=89,640	clip=180	consec=1	max_consec=78	avg_clip50=130.40
EP=580	STEP=91,194	clip=115	consec=11	max_consec=78	avg_clip50=133.30
EP=590	STEP=92,612	clip=200	consec=21	max_consec=78	avg_clip50=116.30
EP=600	STEP=94,342	clip=75	consec=31	max_consec=78	avg_clip50=110.20

custom/	
avg_clip50	110
clipped_episode_r...	75
consecutive_clipp...	31
max_consecutive_c...	78
rollout/	
ep_len_mean	646
ep_rew_mean	121
exploration rate	0.969
time/	
episodes	1800
fps	310
time_elapsed	304
total timesteps	94496
train/	
learning_rate	0.0001
loss	0.0305
n_updates	11123

EP=610 | STEP=95,991 | clip=30 | consec=41 | max_consec=78 | avg_clip50=125.70
 EP=620 | STEP=97,669 | clip=120 | consec=51 | max_consec=78 | avg_clip50=132.80
 EP=630 | STEP=99,447 | clip=120 | consec=61 | max_consec=78 | avg_clip50=150.50

custom/	
avg_clip50	149
clipped_episode_r...	75
consecutive_clipp...	65
max_consecutive_c...	78
rollout/	
ep_len_mean	676
ep_rew_mean	135
exploration rate	0.967
time/	
episodes	1900
fps	306
time_elapsed	326
total timesteps	100065
train/	
learning_rate	0.0001
loss	0.0194
n_updates	12516

```
-----
EP=640 | STEP=101,188 | clip=380 | consec=71 | max_consec=78 | avg_clip50=167.50
EP=650 | STEP=102,892 | clip=180 | consec=81 | max_consec=81 | avg_clip50=182.70
EP=660 | STEP=104,641 | clip=25 | consec=91 | max_consec=91 | avg_clip50=187.40
-----
```

custom/	
avg_clip50	176
clipped_episode_r...	195
consecutive_clipp...	99
max_consecutive_c...	99
rollout/	
ep_len_mean	705
ep_rew_mean	154
exploration rate	0.965
time/	
episodes	2000
fps	302
time_elapsed	350
total timesteps	105888
train/	
learning_rate	0.0001
loss	0.0458
n_updates	13971

Objetivo conseguido: 100 episodios consecutivos superando el umbral.

Entrenamiento finalizado.

max_consec alcanzado: 100

Log JSON: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json

Log CSV: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv

Checkpoint PyTorch guardado en: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_double/dqn_spaceinvaders_v0_policy_latest.pt

model.save() falló (no es crítico, se usa el .pt): IndexError('tuple index out of range')

In []: # Dueling DQN

```
class PeriodicPolicyCheckpoint(BaseCallback):
    def __init__(self, save_path, every_steps=250_000, verbose=0):
        super().__init__(verbose)
        self.save_path = save_path
        self.every_steps = every_steps
        self.last_save = 0
```

```
def _on_step(self):
    if self.num_timesteps - self.last_save >= self.every_steps:
        save_policy_checkpoint(self.model, self.save_path)
        self.last_save = self.num_timesteps
        print(f'[checkpoint periódico] guardado a {self.num_timesteps:,} steps')
    return True

paths = RUN_PATHS['dueling']
env_dueling = create_train_env(paths['monitor'])
model_dueling = build_model('dueling', env_dueling, paths['tensorboard'])

training_logger = SB3TrainingLogger(json_path=paths['json'], csv_path=paths['csv'])
callback = ConsecutiveRewardCallback(training_logger=training_logger, verbose=1)
ckpt_cb = PeriodicPolicyCheckpoint(paths['policy'], every_steps=250_000)

model_dueling.learn(
    total_timesteps=TOTAL_TIMESTEPS,
    callback=[callback, ckpt_cb],
    reset_num_timesteps=True,
    tb_log_name='DQN_dueling',
    log_interval=100
)

save_policy_checkpoint(model_dueling, paths['policy'])
print('Checkpoint final guardado en:', paths['policy'])
try:
    model_dueling.save(str(paths['model_zip']))
except Exception as e:
    print('model.save() falló (no crítico):', repr(e))
env_dueling.close()
```

Using cuda device

Wrapping the env in a VecTransposeImage.

Logging to /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_dueling/tensorboard/DQN_dueling_1

EP=0 | STEP=267 | clip=180 | consec=1 | max_consec=1 | avg_clip50=180.00

EP=10 | STEP=1,894 | clip=655 | consec=3 | max_consec=7 | avg_clip50=184.55

EP=20 | STEP=3,135 | clip=80 | consec=13 | max_consec=13 | avg_clip50=130.48

EP=30 | STEP=4,765 | clip=155 | consec=23 | max_consec=23 | avg_clip50=142.90

custom/		
avg_clip50	141	
clipped_episode_r...	155	
consecutive_clipp...	25	
max_consecutive_c...	25	
rollout/		
ep_len_mean	659	
ep_rew_mean	141	
exploration rate	0.998	
time/		
episodes	100	
fps	396	
time_elapsed	12	
total timesteps	5126	

EP=40 | STEP=6,186 | clip=155 | consec=33 | max_consec=33 | avg_clip50=137.44

EP=50 | STEP=7,696 | clip=75 | consec=43 | max_consec=43 | avg_clip50=131.50

EP=60 | STEP=9,261 | clip=45 | consec=1 | max_consec=51 | avg_clip50=118.10

custom/		
avg_clip50	131	
clipped_episode_r...	140	
consecutive_clipp...	6	
max_consecutive_c...	51	
rollout/		
ep_len_mean	660	
ep_rew_mean	136	
exploration rate	0.997	
time/		
episodes	200	
fps	396	
time_elapsed	25	
total timesteps	10233	

```

-----
EP=70 | STEP=10,844 | clip=320 | consec=11 | max_consec=51 | avg_clip50=138.00
EP=80 | STEP=12,464 | clip=335 | consec=21 | max_consec=51 | avg_clip50=140.40
EP=90 | STEP=14,027 | clip=45 | consec=31 | max_consec=51 | avg_clip50=138.90
-----

```

```

| custom/
|   avg_clip50          | 148 |
|   clipped_episode_r... | 210 |
|   consecutive_clipp... | 40  |
|   max_consecutive_c... | 51  |
| rollout/
|   ep_len_mean         | 665 |
|   ep_rew_mean         | 141 |
|   exploration rate     | 0.995 |
| time/
|   episodes            | 300 |
|   fps                 | 397 |
|   time_elapsed        | 38  |
|   total timesteps     | 15452 |
-----

```

```

EP=100 | STEP=15,660 | clip=245 | consec=41 | max_consec=51 | avg_clip50=151.50
EP=110 | STEP=17,107 | clip=60 | consec=51 | max_consec=51 | avg_clip50=151.40
EP=120 | STEP=18,403 | clip=130 | consec=7 | max_consec=52 | avg_clip50=136.40
EP=130 | STEP=20,065 | clip=205 | consec=4 | max_consec=52 | avg_clip50=128.30
-----

```

```

| custom/
|   avg_clip50          | 129 |
|   clipped_episode_r... | 175 |
|   consecutive_clipp... | 6   |
|   max_consecutive_c... | 52  |
| rollout/
|   ep_len_mean         | 657 |
|   ep_rew_mean         | 135 |
|   exploration rate     | 0.993 |
| time/
|   episodes            | 400 |
|   fps                 | 398 |
|   time_elapsed        | 51  |
|   total timesteps     | 20359 |
-----

```

```

EP=140 | STEP=21,723 | clip=20 | consec=0 | max_consec=52 | avg_clip50=134.70

```

EP=150 | STEP=23,061 | clip=90 | consec=9 | max_consec=52 | avg_clip50=116.80
 EP=160 | STEP=24,659 | clip=65 | consec=19 | max_consec=52 | avg_clip50=126.10

custom/		
avg_clip50	136	
clipped_episode_r...	120	
consecutive_clipp...	25	
max_consecutive_c...	52	
rollout/		
ep_len_mean	666	
ep_rew_mean	137	
exploration rate	0.991	
time/		
episodes	500	
fps	398	
time_elapsed	64	
total timesteps	25762	

EP=170 | STEP=26,391 | clip=155 | consec=29 | max_consec=52 | avg_clip50=136.90
 EP=180 | STEP=28,172 | clip=355 | consec=39 | max_consec=52 | avg_clip50=142.50
 EP=190 | STEP=29,744 | clip=55 | consec=3 | max_consec=52 | avg_clip50=138.30

custom/		
avg_clip50	155	
clipped_episode_r...	125	
consecutive_clipp...	12	
max_consecutive_c...	52	
rollout/		
ep_len_mean	680	
ep_rew_mean	137	
exploration rate	0.99	
time/		
episodes	600	
fps	399	
time_elapsed	78	
total timesteps	31348	

EP=200 | STEP=31,493 | clip=125 | consec=13 | max_consec=52 | avg_clip50=155.40
 EP=210 | STEP=33,300 | clip=150 | consec=23 | max_consec=52 | avg_clip50=158.20
 EP=220 | STEP=35,127 | clip=45 | consec=1 | max_consec=52 | avg_clip50=165.50
 EP=230 | STEP=36,618 | clip=135 | consec=11 | max_consec=52 | avg_clip50=153.60

custom/		
avg_clip50	162	
clipped_episode_r...	5	
consecutive_clipp...	0	
max_consecutive_c...	52	
rollout/		
ep_len_mean	712	
ep_rew_mean	151	
exploration rate	0.988	
time/		
episodes	700	
fps	400	
time_elapsed	92	
total timesteps	37116	

EP=240 | STEP=38,192 | clip=80 | consec=7 | max_consec=52 | avg_clip50=160.50
 EP=250 | STEP=39,814 | clip=120 | consec=8 | max_consec=52 | avg_clip50=153.30
 EP=260 | STEP=41,629 | clip=270 | consec=18 | max_consec=52 | avg_clip50=145.90

custom/		
avg_clip50	144	
clipped_episode_r...	120	
consecutive_clipp...	24	
max_consecutive_c...	52	
rollout/		
ep_len_mean	721	
ep_rew_mean	154	
exploration rate	0.986	
time/		
episodes	800	
fps	401	
time_elapsed	106	
total timesteps	42698	

EP=270 | STEP=43,135 | clip=50 | consec=28 | max_consec=52 | avg_clip50=137.40
 EP=280 | STEP=44,654 | clip=150 | consec=38 | max_consec=52 | avg_clip50=143.70
 EP=290 | STEP=46,253 | clip=60 | consec=6 | max_consec=52 | avg_clip50=136.30

custom/		
avg_clip50	127	

	clipped_episode_r...	45	
	consecutive_clipp...	7	
	max_consecutive_c...	52	
	rollout/		
	ep_len_mean	696	
	ep_rew_mean	140	
	exploration rate	0.984	
	time/		
	episodes	900	
	fps	400	
	time_elapsed	118	
	total timesteps	47648	

```

-----
EP=300 | STEP=47,728 | clip=75 | consec=8 | max_consec=52 | avg_clip50=125.80
EP=310 | STEP=49,428 | clip=135 | consec=18 | max_consec=52 | avg_clip50=129.80
EP=320 | STEP=51,194 | clip=125 | consec=7 | max_consec=52 | avg_clip50=131.70
EP=330 | STEP=52,754 | clip=30 | consec=17 | max_consec=52 | avg_clip50=130.70

```

	custom/		
	avg_clip50	137	
	clipped_episode_r...	420	
	consecutive_clipp...	20	
	max_consecutive_c...	52	
	rollout/		
	ep_len_mean	693	
	ep_rew_mean	137	
	exploration rate	0.982	
	time/		
	episodes	1000	
	fps	384	
	time_elapsed	138	
	total timesteps	53303	
	train/		
	learning_rate	0.0001	
	loss	0.0302	
	n_updates	825	

```

-----
EP=340 | STEP=54,085 | clip=50 | consec=2 | max_consec=52 | avg_clip50=128.90
EP=350 | STEP=55,665 | clip=155 | consec=3 | max_consec=52 | avg_clip50=149.80
EP=360 | STEP=57,306 | clip=160 | consec=1 | max_consec=52 | avg_clip50=143.00

```

custom/	
avg_clip50	140
clipped_episode_r...	0
consecutive_clipp...	0
max_consecutive_c...	52
rollout/	
ep_len_mean	666
ep_rew_mean	131
exploration rate	0.981
time/	
episodes	1100
fps	364
time_elapsed	159
total timesteps	58308
train/	
learning_rate	0.0001
loss	0.0297
n_updates	2076

EP=370 | STEP=58,840 | clip=100 | consec=4 | max_consec=52 | avg_clip50=133.60
 EP=380 | STEP=60,882 | clip=360 | consec=14 | max_consec=52 | avg_clip50=146.40
 EP=390 | STEP=62,556 | clip=65 | consec=6 | max_consec=52 | avg_clip50=147.10
 EP=400 | STEP=64,452 | clip=95 | consec=16 | max_consec=52 | avg_clip50=144.70

custom/	
avg_clip50	145
clipped_episode_r...	95
consecutive_clipp...	16
max_consecutive_c...	52
rollout/	
ep_len_mean	715
ep_rew_mean	147
exploration rate	0.979
time/	
episodes	1200
fps	347
time_elapsed	185
total timesteps	64483
train/	
learning_rate	0.0001
loss	0.0294

n_updates	3620
-----------	------

EP=410 | STEP=65,774 | clip=90 | consec=1 | max_consec=52 | avg_clip50=136.30
 EP=420 | STEP=67,184 | clip=255 | consec=11 | max_consec=52 | avg_clip50=141.90
 EP=430 | STEP=68,629 | clip=140 | consec=4 | max_consec=52 | avg_clip50=127.30

custom/	
avg_clip50	134
clipped_episode_r...	375
consecutive_clipp...	7
max_consecutive_c...	52
rollout/	
ep_len_mean	686
ep_rew_mean	137
exploration rate	0.977
time/	
episodes	1300
fps	336
time_elapsed	206
total timesteps	69355
train/	
learning_rate	0.0001
loss	0.0185
n_updates	4838

EP=440 | STEP=70,208 | clip=200 | consec=5 | max_consec=52 | avg_clip50=132.00
 EP=450 | STEP=71,918 | clip=210 | consec=15 | max_consec=52 | avg_clip50=128.40
 EP=460 | STEP=73,358 | clip=110 | consec=25 | max_consec=52 | avg_clip50=138.10

custom/	
avg_clip50	142
clipped_episode_r...	120
consecutive_clipp...	6
max_consecutive_c...	52
rollout/	
ep_len_mean	697
ep_rew_mean	141
exploration rate	0.975
time/	
episodes	1400
fps	326

	time_elapsed	228	
	total timesteps	74705	
	train/		
	learning_rate	0.0001	
	loss	0.00158	
	n_updates	6176	

```
-----
EP=470 | STEP=75,186 | clip=140 | consec=9 | max_consec=52 | avg_clip50=140.40
EP=480 | STEP=77,012 | clip=105 | consec=19 | max_consec=52 | avg_clip50=149.60
EP=490 | STEP=78,723 | clip=110 | consec=29 | max_consec=52 | avg_clip50=155.20
EP=500 | STEP=80,496 | clip=245 | consec=9 | max_consec=52 | avg_clip50=157.40
```

	custom/		
	avg_clip50	162	
	clipped_episode_r...	285	
	consecutive_clipp...	10	
	max_consecutive_c...	52	
	rollout/		
	ep_len_mean	690	
	ep_rew_mean	146	
	exploration rate	0.973	
	time/		
	episodes	1500	
	fps	317	
	time_elapsed	254	
	total timesteps	80711	
	train/		
	learning_rate	0.0001	
	loss	0.0171	
	n_updates	7677	

```
-----
EP=510 | STEP=82,274 | clip=45 | consec=19 | max_consec=52 | avg_clip50=169.30
EP=520 | STEP=83,634 | clip=55 | consec=29 | max_consec=52 | avg_clip50=159.90
EP=530 | STEP=85,020 | clip=35 | consec=39 | max_consec=52 | avg_clip50=144.30
```

	custom/		
	avg_clip50	126	
	clipped_episode_r...	80	
	consecutive_clipp...	43	
	max_consecutive_c...	52	
	rollout/		

ep_len_mean	690
ep_rew_mean	142
exploration rate	0.971
time/	
episodes	1600
fps	311
time_elapsed	274
total timesteps	85615
train/	
learning_rate	0.0001
loss	0.0472
n_updates	8903

EP=540	STEP=86,504	clip=305	consec=49	max_consec=52	avg_clip50=131.30
EP=550	STEP=88,027	clip=305	consec=59	max_consec=59	avg_clip50=125.50
EP=560	STEP=89,334	clip=50	consec=69	max_consec=69	avg_clip50=103.30

custom/	
avg_clip50	104
clipped_episode_r...	15
consecutive_clipp...	0
max_consecutive_c...	76
rollout/	
ep_len_mean	681
ep_rew_mean	135
exploration rate	0.97
time/	
episodes	1700
fps	305
time_elapsed	296
total timesteps	90639
train/	
learning_rate	0.0001
loss	0.0216
n_updates	10159

EP=570	STEP=90,871	clip=105	consec=2	max_consec=76	avg_clip50=105.70
EP=580	STEP=92,321	clip=35	consec=5	max_consec=76	avg_clip50=114.80
EP=590	STEP=93,902	clip=50	consec=4	max_consec=76	avg_clip50=115.30
EP=600	STEP=95,563	clip=385	consec=3	max_consec=76	avg_clip50=125.30

custom/	
avg_clip50	125
clipped_episode_r...	155
consecutive_clipp...	4
max_consecutive_c...	76
rollout/	
ep_len_mean	649
ep_rew_mean	124
exploration rate	0.968
time/	
episodes	1800
fps	300
time_elapsed	318
total timesteps	95791
train/	
learning_rate	0.0001
loss	0.0163
n_updates	11447

EP=610 | STEP=97,169 | clip=45 | consec=13 | max_consec=76 | avg_clip50=137.30
 EP=620 | STEP=98,687 | clip=20 | consec=0 | max_consec=76 | avg_clip50=135.70
 EP=630 | STEP=100,138 | clip=20 | consec=0 | max_consec=76 | avg_clip50=122.70

custom/	
avg_clip50	119
clipped_episode_r...	205
consecutive_clipp...	4
max_consecutive_c...	76
rollout/	
ep_len_mean	653
ep_rew_mean	119
exploration rate	0.966
time/	
episodes	1900
fps	296
time_elapsed	339
total timesteps	100745
train/	
learning_rate	0.0001
loss	0.0229
n_updates	12686

```

-----
EP=640 | STEP=101,483 | clip=50 | consec=10 | max_consec=76 | avg_clip50=119.00
EP=650 | STEP=102,884 | clip=30 | consec=20 | max_consec=76 | avg_clip50=105.80
EP=660 | STEP=104,465 | clip=95 | consec=5 | max_consec=76 | avg_clip50=104.10
-----

```

```

-----
| custom/                |          |
|   avg_clip50           |    110   |
|   clipped_episode_r... |    105   |
|   consecutive_clipp... |     13   |
|   max_consecutive_c... |     76   |
| rollout/              |          |
|   ep_len_mean          |    643   |
|   ep_rew_mean          |    123   |
|   exploration rate     |   0.965  |
| time/                 |          |
|   episodes             |   2000   |
|   fps                  |    292   |
|   time_elapsed         |    360   |
|   total timesteps      |  105569  |
| train/                |          |
|   learning_rate        |   0.0001 |
|   loss                 |   0.0236 |
|   n_updates            |   13892  |
-----

```

```

-----
EP=670 | STEP=105,823 | clip=45 | consec=15 | max_consec=76 | avg_clip50=109.40
EP=680 | STEP=107,188 | clip=65 | consec=25 | max_consec=76 | avg_clip50=113.70
EP=690 | STEP=109,213 | clip=15 | consec=0  | max_consec=76 | avg_clip50=150.40
EP=700 | STEP=111,099 | clip=195 | consec=10 | max_consec=76 | avg_clip50=159.30
-----

```

```

-----
| custom/                |          |
|   avg_clip50           |    160   |
|   clipped_episode_r... |    135   |
|   consecutive_clipp... |     11   |
|   max_consecutive_c... |     76   |
| rollout/              |          |
|   ep_len_mean          |    665   |
|   ep_rew_mean          |    132   |
|   exploration rate     |   0.963  |
| time/                 |          |
|   episodes             |   2100   |
|   fps                  |    288   |
-----

```

	time_elapsed		385	
	total timesteps		111246	
	train/			
	learning_rate		0.0001	
	loss		0.028	
	n_updates		15311	

EP=710		STEP=112,814		clip=90		consec=20		max_consec=76		avg_clip50=165.30
EP=720		STEP=114,318		clip=135		consec=9		max_consec=76		avg_clip50=167.50
EP=730		STEP=115,916		clip=65		consec=19		max_consec=76		avg_clip50=180.90

	custom/			
	avg_clip50		164	
	clipped_episode_r...		120	
	consecutive_clipp...		23	
	max_consecutive_c...		76	
	rollout/			
	ep_len_mean		680	
	ep_rew_mean		148	
	exploration rate		0.961	
	time/			
	episodes		2200	
	fps		285	
	time_elapsed		408	
	total timesteps		116641	
	train/			
	learning_rate		0.0001	
	loss		0.00918	
	n_updates		16660	

EP=740		STEP=117,455		clip=135		consec=29		max_consec=76		avg_clip50=152.20
EP=750		STEP=119,269		clip=20		consec=0		max_consec=76		avg_clip50=148.40
EP=760		STEP=120,542		clip=290		consec=10		max_consec=76		avg_clip50=136.10

	custom/			
	avg_clip50		133	
	clipped_episode_r...		155	
	consecutive_clipp...		18	
	max_consecutive_c...		76	
	rollout/			
	ep_len_mean		701	

ep_rew_mean	150
exploration rate	0.959
time/	
episodes	2300
fps	282
time_elapsed	431
total timesteps	121958
train/	
learning_rate	0.0001
loss	0.0559
n_updates	17989

```

-----
EP=770 | STEP=122,405 | clip=185 | consec=20 | max_consec=76 | avg_clip50=137.20
EP=780 | STEP=123,760 | clip=185 | consec=30 | max_consec=76 | avg_clip50=126.00
EP=790 | STEP=125,304 | clip=55 | consec=7 | max_consec=76 | avg_clip50=128.20
EP=800 | STEP=126,967 | clip=215 | consec=17 | max_consec=76 | avg_clip50=128.20

```

custom/	
avg_clip50	130
clipped_episode_r...	145
consecutive_clipp...	18
max_consecutive_c...	76
rollout/	
ep_len_mean	682
ep_rew_mean	138
exploration rate	0.958
time/	
episodes	2400
fps	280
time_elapsed	453
total timesteps	127171
train/	
learning_rate	0.0001
loss	0.0078
n_updates	19292

```

-----
EP=810 | STEP=128,421 | clip=105 | consec=27 | max_consec=76 | avg_clip50=129.00
EP=820 | STEP=129,967 | clip=80 | consec=37 | max_consec=76 | avg_clip50=125.30
EP=830 | STEP=131,789 | clip=60 | consec=47 | max_consec=76 | avg_clip50=135.50

```

custom/	
---------	--

	avg_clip50	145	
	clipped_episode_r...	65	
	consecutive_clipp...	51	
	max_consecutive_c...	76	
	rollout/		
	ep_len_mean	691	
	ep_rew_mean	135	
	exploration rate	0.956	
	time/		
	episodes	2500	
	fps	277	
	time_elapsed	477	
	total timesteps	132768	
	train/		
	learning_rate	0.0001	
	loss	0.0141	
	n_updates	20691	

EP=840 | STEP=133,701 | clip=315 | consec=57 | max_consec=76 | avg_clip50=149.20
 EP=850 | STEP=135,104 | clip=30 | consec=67 | max_consec=76 | avg_clip50=148.80
 EP=860 | STEP=136,741 | clip=615 | consec=77 | max_consec=77 | avg_clip50=163.20

	custom/		
	avg_clip50	156	
	clipped_episode_r...	110	
	consecutive_clipp...	85	
	max_consecutive_c...	85	
	rollout/		
	ep_len_mean	683	
	ep_rew_mean	143	
	exploration rate	0.954	
	time/		
	episodes	2600	
	fps	275	
	time_elapsed	500	
	total timesteps	137935	
	train/		
	learning_rate	0.0001	
	loss	0.0278	
	n_updates	21983	

EP=870 | STEP=138,247 | clip=375 | consec=87 | max_consec=87 | avg_clip50=162.50

EP=880 | STEP=139,841 | clip=120 | consec=97 | max_consec=97 | avg_clip50=168.20

Objetivo conseguido: 100 episodios consecutivos superando el umbral.

Entrenamiento finalizado.

max_consec alcanzado: 100

Log JSON: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json

Log CSV: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv

Checkpoint final guardado en: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_dueling/dqn_spaceinvaders_v0_policy_latest.pt

model.save() falló (no crítico): IndexError('tuple index out of range')

In []: *# Propuesta 3: exploración ajustada*

```
env_tuned = create_train_env(RUN_PATHS['tuned']['monitor'])
model_tuned = build_model('tuned', env_tuned, RUN_PATHS['tuned']['tensorboard'])
model_tuned = train_variant('tuned', model_tuned)
env_tuned.close()
```

Using cuda device

Wrapping the env in a VecTransposeImage.

Logging to /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_tuned/tensorboard/DQN_tuned_1

EP=0 | STEP=267 | clip=180 | consec=1 | max_consec=1 | avg_clip50=180.00

EP=10 | STEP=1,894 | clip=655 | consec=3 | max_consec=7 | avg_clip50=184.55

EP=20 | STEP=3,135 | clip=80 | consec=13 | max_consec=13 | avg_clip50=130.48

EP=30 | STEP=4,765 | clip=155 | consec=23 | max_consec=23 | avg_clip50=142.90

custom/		
avg_clip50	141	
clipped_episode_r...	155	
consecutive_clipp...	25	
max_consecutive_c...	25	
rollout/		
ep_len_mean	659	
ep_rew_mean	141	
exploration rate	0.995	
time/		
episodes	100	
fps	398	
time_elapsed	12	
total timesteps	5126	

EP=40 | STEP=6,186 | clip=155 | consec=33 | max_consec=33 | avg_clip50=137.44

EP=50 | STEP=7,696 | clip=75 | consec=43 | max_consec=43 | avg_clip50=131.50

EP=60 | STEP=9,261 | clip=45 | consec=1 | max_consec=51 | avg_clip50=118.10

custom/		
avg_clip50	131	
clipped_episode_r...	140	
consecutive_clipp...	6	
max_consecutive_c...	51	
rollout/		
ep_len_mean	660	
ep_rew_mean	136	
exploration rate	0.99	
time/		
episodes	200	
fps	397	
time_elapsed	25	
total timesteps	10233	

```

-----
EP=70 | STEP=10,844 | clip=320 | consec=11 | max_consec=51 | avg_clip50=138.00
EP=80 | STEP=12,464 | clip=335 | consec=21 | max_consec=51 | avg_clip50=140.40
EP=90 | STEP=14,027 | clip=45 | consec=31 | max_consec=51 | avg_clip50=138.90
-----

```

```

| custom/
|   avg_clip50           | 148
|   clipped_episode_r... | 210
|   consecutive_clipp... | 40
|   max_consecutive_c... | 51
| rollout/
|   ep_len_mean          | 665
|   ep_rew_mean          | 141
|   exploration rate     | 0.985
| time/
|   episodes             | 300
|   fps                  | 397
|   time_elapsed         | 38
|   total timesteps      | 15452
-----

```

```

EP=100 | STEP=15,660 | clip=245 | consec=41 | max_consec=51 | avg_clip50=151.50
EP=110 | STEP=17,107 | clip=60 | consec=51 | max_consec=51 | avg_clip50=151.40
EP=120 | STEP=18,403 | clip=130 | consec=7 | max_consec=52 | avg_clip50=136.40
EP=130 | STEP=20,065 | clip=205 | consec=4 | max_consec=52 | avg_clip50=128.30
-----

```

```

| custom/
|   avg_clip50           | 129
|   clipped_episode_r... | 175
|   consecutive_clipp... | 6
|   max_consecutive_c... | 52
| rollout/
|   ep_len_mean          | 657
|   ep_rew_mean          | 135
|   exploration rate     | 0.98
| time/
|   episodes             | 400
|   fps                  | 397
|   time_elapsed         | 51
|   total timesteps      | 20359
-----

```

```

EP=140 | STEP=21,723 | clip=20 | consec=0 | max_consec=52 | avg_clip50=134.70

```

EP=150 | STEP=23,061 | clip=90 | consec=9 | max_consec=52 | avg_clip50=116.80
 EP=160 | STEP=24,659 | clip=65 | consec=19 | max_consec=52 | avg_clip50=126.10

custom/		
avg_clip50	136	
clipped_episode_r...	150	
consecutive_clipp...	25	
max_consecutive_c...	52	
rollout/		
ep_len_mean	664	
ep_rew_mean	137	
exploration rate	0.975	
time/		
episodes	500	
fps	392	
time_elapsed	65	
total timesteps	25714	
train/		
learning_rate	0.0001	
loss	0.0303	
n_updates	178	

EP=170 | STEP=26,416 | clip=75 | consec=29 | max_consec=52 | avg_clip50=141.30
 EP=180 | STEP=27,879 | clip=315 | consec=3 | max_consec=52 | avg_clip50=136.10
 EP=190 | STEP=29,748 | clip=140 | consec=13 | max_consec=52 | avg_clip50=134.10

custom/		
avg_clip50	156	
clipped_episode_r...	50	
consecutive_clipp...	22	
max_consecutive_c...	52	
rollout/		
ep_len_mean	687	
ep_rew_mean	138	
exploration rate	0.969	
time/		
episodes	600	
fps	356	
time_elapsed	88	
total timesteps	31613	
train/		

learning_rate	0.0001
loss	0.0296
n_updates	1653

```

-----
EP=200 | STEP=31,663 | clip=120 | consec=23 | max_consec=52 | avg_clip50=157.00
EP=210 | STEP=33,491 | clip=80 | consec=33 | max_consec=52 | avg_clip50=157.90
EP=220 | STEP=35,376 | clip=25 | consec=43 | max_consec=52 | avg_clip50=151.20
EP=230 | STEP=36,919 | clip=80 | consec=5 | max_consec=52 | avg_clip50=153.30

```

custom/	
avg_clip50	147
clipped_episode_r...	80
consecutive_clipp...	8
max_consecutive_c...	52
rollout/	
ep_len_mean	721
ep_rew_mean	144
exploration rate	0.963
time/	
episodes	700
fps	336
time_elapsed	110
total timesteps	37313
train/	
learning_rate	0.0001
loss	0.0321
n_updates	3078

```

-----
EP=240 | STEP=38,472 | clip=255 | consec=15 | max_consec=52 | avg_clip50=152.80
EP=250 | STEP=40,072 | clip=210 | consec=25 | max_consec=52 | avg_clip50=143.40
EP=260 | STEP=41,968 | clip=185 | consec=35 | max_consec=52 | avg_clip50=142.60

```

custom/	
avg_clip50	136
clipped_episode_r...	80
consecutive_clipp...	2
max_consecutive_c...	52
rollout/	
ep_len_mean	735
ep_rew_mean	146
exploration rate	0.957

time/	
episodes	800
fps	323
time_elapsed	133
total timesteps	43008
train/	
learning_rate	0.0001
loss	0.0304
n_updates	4501

```
-----
EP=270 | STEP=43,536 | clip=60 | consec=6 | max_consec=52 | avg_clip50=133.20
EP=280 | STEP=45,032 | clip=75 | consec=3 | max_consec=52 | avg_clip50=126.40
EP=290 | STEP=46,839 | clip=445 | consec=2 | max_consec=52 | avg_clip50=136.40
EP=300 | STEP=48,309 | clip=120 | consec=12 | max_consec=52 | avg_clip50=133.50
-----
```

custom/	
avg_clip50	134
clipped_episode_r...	120
consecutive_clipp...	12
max_consecutive_c...	52
rollout/	
ep_len_mean	712
ep_rew_mean	138
exploration rate	0.952
time/	
episodes	900
fps	313
time_elapsed	153
total timesteps	48309
train/	
learning_rate	0.0001
loss	0.0202
n_updates	5827

```
-----
EP=310 | STEP=49,668 | clip=180 | consec=22 | max_consec=52 | avg_clip50=118.40
EP=320 | STEP=51,660 | clip=110 | consec=32 | max_consec=52 | avg_clip50=133.70
EP=330 | STEP=53,163 | clip=85 | consec=42 | max_consec=52 | avg_clip50=142.50
-----
```

custom/	
avg_clip50	140
clipped_episode_r...	205

consecutive_clipp...	45
max_consecutive_c...	52
rollout/	
ep_len_mean	699
ep_rew_mean	135
exploration rate	0.947
time/	
episodes	1000
fps	306
time_elapsed	174
total timesteps	53706
train/	
learning_rate	0.0001
loss	0.0017
n_updates	7176

EP=340 | STEP=54,495 | clip=55 | consec=2 | max_consec=52 | avg_clip50=128.70
 EP=350 | STEP=56,093 | clip=160 | consec=12 | max_consec=52 | avg_clip50=127.60
 EP=360 | STEP=57,935 | clip=230 | consec=8 | max_consec=52 | avg_clip50=146.70

custom/	
avg_clip50	130
clipped_episode_r...	65
consecutive_clipp...	4
max_consecutive_c...	52
rollout/	
ep_len_mean	678
ep_rew_mean	133
exploration rate	0.942
time/	
episodes	1100
fps	301
time_elapsed	195
total timesteps	58822
train/	
learning_rate	0.0001
loss	0.00417
n_updates	8455

EP=370 | STEP=59,407 | clip=45 | consec=8 | max_consec=52 | avg_clip50=132.00
 EP=380 | STEP=61,173 | clip=435 | consec=18 | max_consec=52 | avg_clip50=136.60

EP=390 | STEP=62,615 | clip=55 | consec=28 | max_consec=52 | avg_clip50=140.20
 EP=400 | STEP=63,952 | clip=180 | consec=38 | max_consec=52 | avg_clip50=135.70

custom/		
avg_clip50	136	
clipped_episode_r...	180	
consecutive_clipp...	38	
max_consecutive_c...	52	
rollout/		
ep_len_mean	671	
ep_rew_mean	132	
exploration rate	0.937	
time/		
episodes	1200	
fps	296	
time_elapsed	215	
total timesteps	63952	
train/		
learning_rate	0.0001	
loss	0.022	
n_updates	9737	

EP=410 | STEP=65,532 | clip=445 | consec=48 | max_consec=52 | avg_clip50=124.00
 EP=420 | STEP=66,862 | clip=85 | consec=8 | max_consec=52 | avg_clip50=119.10
 EP=430 | STEP=68,507 | clip=60 | consec=7 | max_consec=52 | avg_clip50=112.40

custom/		
avg_clip50	111	
clipped_episode_r...	110	
consecutive_clipp...	10	
max_consecutive_c...	52	
rollout/		
ep_len_mean	657	
ep_rew_mean	123	
exploration rate	0.932	
time/		
episodes	1300	
fps	293	
time_elapsed	235	
total timesteps	68996	
train/		

learning_rate	0.0001
loss	0.00449
n_updates	10998

EP=440	STEP=70,111	clip=315	consec=17	max_consec=52	avg_clip50=114.50
EP=450	STEP=71,530	clip=115	consec=27	max_consec=52	avg_clip50=109.40
EP=460	STEP=73,078	clip=40	consec=37	max_consec=52	avg_clip50=110.70

custom/	
avg_clip50	119
clipped_episode_r...	20
consecutive_clipp...	0
max_consecutive_c...	52
rollout/	
ep_len_mean	659
ep_rew_mean	121
exploration rate	0.927
time/	
episodes	1400
fps	289
time_elapsed	255
total timesteps	74179
train/	
learning_rate	0.0001
loss	0.0261
n_updates	12294

EP=470	STEP=74,785	clip=100	consec=4	max_consec=52	avg_clip50=126.10
EP=480	STEP=76,458	clip=140	consec=14	max_consec=52	avg_clip50=125.70
EP=490	STEP=77,932	clip=110	consec=24	max_consec=52	avg_clip50=122.10
EP=500	STEP=79,333	clip=165	consec=34	max_consec=52	avg_clip50=128.70

custom/	
avg_clip50	129
clipped_episode_r...	165
consecutive_clipp...	34
max_consecutive_c...	52
rollout/	
ep_len_mean	659
ep_rew_mean	119
exploration rate	0.921

time/	
episodes	1500
fps	287
time_elapsed	276
total timesteps	79333
train/	
learning_rate	0.0001
loss	0.019
n_updates	13583

EP=510 | STEP=80,908 | clip=195 | consec=44 | max_consec=52 | avg_clip50=132.60
 EP=520 | STEP=82,440 | clip=35 | consec=7 | max_consec=52 | avg_clip50=130.20
 EP=530 | STEP=84,206 | clip=415 | consec=6 | max_consec=52 | avg_clip50=145.00

custom/	
avg_clip50	146
clipped_episode_r...	325
consecutive_clipp...	9
max_consecutive_c...	52
rollout/	
ep_len_mean	674
ep_rew_mean	138
exploration rate	0.916
time/	
episodes	1600
fps	284
time_elapsed	297
total timesteps	84724
train/	
learning_rate	0.0001
loss	0.0419
n_updates	14930

EP=540 | STEP=85,885 | clip=50 | consec=16 | max_consec=52 | avg_clip50=155.20
 EP=550 | STEP=87,322 | clip=55 | consec=26 | max_consec=52 | avg_clip50=155.00
 EP=560 | STEP=88,805 | clip=45 | consec=36 | max_consec=52 | avg_clip50=147.20

custom/	
avg_clip50	156
clipped_episode_r...	200
consecutive_clipp...	43

max_consecutive_c...	52
rollout/	
ep_len_mean	674
ep_rew_mean	142
exploration rate	0.911
time/	
episodes	1700
fps	282
time_elapsed	318
total timesteps	89985
train/	
learning_rate	0.0001
loss	0.0624
n_updates	16246

EP=570	STEP=90,642	clip=320	consec=46	max_consec=52	avg_clip50=158.50
EP=580	STEP=92,443	clip=150	consec=56	max_consec=56	avg_clip50=151.60
EP=590	STEP=93,869	clip=135	consec=66	max_consec=66	avg_clip50=132.90
EP=600	STEP=95,490	clip=290	consec=4	max_consec=71	avg_clip50=137.80

custom/	
avg_clip50	138
clipped_episode_r...	290
consecutive_clipp...	4
max_consecutive_c...	71
rollout/	
ep_len_mean	692
ep_rew_mean	146
exploration rate	0.905
time/	
episodes	1800
fps	280
time_elapsed	340
total timesteps	95641
train/	
learning_rate	0.0001
loss	0.03
n_updates	17660

EP=610	STEP=97,426	clip=95	consec=14	max_consec=71	avg_clip50=156.50
EP=620	STEP=99,077	clip=250	consec=3	max_consec=71	avg_clip50=147.60

EP=630 | STEP=100,457 | clip=135 | consec=13 | max_consec=71 | avg_clip50=137.80

custom/		
avg_clip50	140	
clipped_episode_r...	215	
consecutive_clipp...	17	
max_consecutive_c...	71	
rollout/		
ep_len_mean	691	
ep_rew_mean	142	
exploration rate	0.9	
time/		
episodes	1900	
fps	278	
time_elapsed	362	
total timesteps	101048	
train/		
learning_rate	0.0001	
loss	0.0175	
n_updates	19011	

EP=640 | STEP=102,176 | clip=300 | consec=23 | max_consec=71 | avg_clip50=151.40

EP=650 | STEP=103,751 | clip=55 | consec=33 | max_consec=71 | avg_clip50=149.70

EP=660 | STEP=105,379 | clip=50 | consec=8 | max_consec=71 | avg_clip50=139.30

custom/		
avg_clip50	139	
clipped_episode_r...	40	
consecutive_clipp...	15	
max_consecutive_c...	71	
rollout/		
ep_len_mean	702	
ep_rew_mean	146	
exploration rate	0.895	
time/		
episodes	2000	
fps	277	
time_elapsed	383	
total timesteps	106361	
train/		
learning_rate	0.0001	

loss	0.0264
n_updates	20340

EP=670	STEP=106,675	clip=120	consec=18	max_consec=71	avg_clip50=133.40
EP=680	STEP=108,182	clip=235	consec=28	max_consec=71	avg_clip50=136.30
EP=690	STEP=109,858	clip=130	consec=8	max_consec=71	avg_clip50=138.50
EP=700	STEP=111,218	clip=200	consec=18	max_consec=71	avg_clip50=133.40

custom/	
avg_clip50	130
clipped_episode_r...	130
consecutive_clipp...	19
max_consecutive_c...	71
rollout/	
ep_len_mean	672
ep_rew_mean	141
exploration rate	0.89
time/	
episodes	2100
fps	276
time_elapsed	403
total timesteps	111452
train/	
learning_rate	0.0001
loss	0.0172
n_updates	21612

EP=710	STEP=112,892	clip=215	consec=28	max_consec=71	avg_clip50=129.60
EP=720	STEP=114,161	clip=90	consec=38	max_consec=71	avg_clip50=121.50
EP=730	STEP=115,627	clip=110	consec=48	max_consec=71	avg_clip50=119.70

custom/	
avg_clip50	117
clipped_episode_r...	130
consecutive_clipp...	52
max_consecutive_c...	71
rollout/	
ep_len_mean	652
ep_rew_mean	128
exploration rate	0.885
time/	

	episodes	2200	
	fps	275	
	time_elapsed	422	
	total timesteps	116287	
	train/		
	learning_rate	0.0001	
	loss	0.0487	
	n_updates	22821	

EP=740 | STEP=116,998 | clip=80 | consec=58 | max_consec=71 | avg_clip50=109.20
 EP=750 | STEP=118,531 | clip=115 | consec=5 | max_consec=71 | avg_clip50=109.00
 EP=760 | STEP=120,278 | clip=330 | consec=15 | max_consec=71 | avg_clip50=111.70

	custom/		
	avg_clip50	127	
	clipped_episode_r...	225	
	consecutive_clipp...	23	
	max_consecutive_c...	71	
	rollout/		
	ep_len_mean	659	
	ep_rew_mean	124	
	exploration rate	0.879	
	time/		
	episodes	2300	
	fps	273	
	time_elapsed	444	
	total timesteps	121737	
	train/		
	learning_rate	0.0001	
	loss	0.0714	
	n_updates	24184	

EP=770 | STEP=122,136 | clip=310 | consec=25 | max_consec=71 | avg_clip50=131.80
 EP=780 | STEP=123,389 | clip=95 | consec=35 | max_consec=71 | avg_clip50=127.50
 EP=790 | STEP=125,015 | clip=195 | consec=3 | max_consec=71 | avg_clip50=137.40
 EP=800 | STEP=126,544 | clip=35 | consec=13 | max_consec=71 | avg_clip50=140.20

	custom/		
	avg_clip50	138	
	clipped_episode_r...	115	
	consecutive_clipp...	14	

	max_consecutive_c...	71	
	rollout/		
	ep_len_mean	658	
	ep_rew_mean	124	
	exploration rate	0.875	
	time/		
	episodes	2400	
	fps	272	
	time_elapsed	464	
	total timesteps	126726	
	train/		
	learning_rate	0.0001	
	loss	0.0826	
	n_updates	25431	

EP=810		STEP=128,138		clip=175		consec=23		max_consec=71		avg_clip50=136.60
EP=820		STEP=129,664		clip=245		consec=3		max_consec=71		avg_clip50=123.40
EP=830		STEP=131,348		clip=245		consec=8		max_consec=71		avg_clip50=135.10

	custom/		
	avg_clip50	157	
	clipped_episode_r...	595	
	consecutive_clipp...	12	
	max_consecutive_c...	71	
	rollout/		
	ep_len_mean	692	
	ep_rew_mean	144	
	exploration rate	0.869	
	time/		
	episodes	2500	
	fps	272	
	time_elapsed	486	
	total timesteps	132477	
	train/		
	learning_rate	0.0001	
	loss	0.0574	
	n_updates	26869	

EP=840		STEP=133,353		clip=255		consec=1		max_consec=71		avg_clip50=153.10
EP=850		STEP=134,782		clip=125		consec=11		max_consec=71		avg_clip50=150.00
EP=860		STEP=136,125		clip=90		consec=21		max_consec=71		avg_clip50=154.70

custom/		
avg_clip50	161	
clipped_episode_r...	155	
consecutive_clipp...	7	
max_consecutive_c...	71	
rollout/		
ep_len_mean	670	
ep_rew_mean	144	
exploration rate	0.864	
time/		
episodes	2600	
fps	271	
time_elapsed	506	
total timesteps	137369	
train/		
learning_rate	0.0001	
loss	0.0155	
n_updates	28092	

```

EP=870 | STEP=137,678 | clip=205 | consec=9 | max_consec=71 | avg_clip50=159.90
EP=880 | STEP=139,223 | clip=190 | consec=19 | max_consec=71 | avg_clip50=162.20
EP=890 | STEP=140,637 | clip=135 | consec=29 | max_consec=71 | avg_clip50=134.20
EP=900 | STEP=142,204 | clip=215 | consec=39 | max_consec=71 | avg_clip50=142.00

```

custom/		
avg_clip50	143	
clipped_episode_r...	115	
consecutive_clipp...	40	
max_consecutive_c...	71	
rollout/		
ep_len_mean	672	
ep_rew_mean	146	
exploration rate	0.859	
time/		
episodes	2700	
fps	270	
time_elapsed	526	
total timesteps	142433	
train/		
learning_rate	0.0001	

loss	0.0487
n_updates	29358

EP=910	STEP=143,982	clip=305	consec=49	max_consec=71	avg_clip50=149.30
EP=920	STEP=145,258	clip=65	consec=59	max_consec=71	avg_clip50=135.60
EP=930	STEP=146,874	clip=45	consec=69	max_consec=71	avg_clip50=128.00

custom/	
avg_clip50	131
clipped_episode_r...	205
consecutive_clipp...	74
max_consecutive_c...	74
rollout/	
ep_len_mean	654
ep_rew_mean	135
exploration rate	0.854
time/	
episodes	2800
fps	269
time_elapsed	548
total timesteps	147765
train/	
learning_rate	0.0001
loss	0.0471
n_updates	30691

EP=940	STEP=148,215	clip=35	consec=79	max_consec=79	avg_clip50=129.90
EP=950	STEP=150,059	clip=510	consec=89	max_consec=89	avg_clip50=145.60
EP=960	STEP=151,373	clip=55	consec=7	max_consec=91	avg_clip50=133.50

custom/	
avg_clip50	143
clipped_episode_r...	25
consecutive_clipp...	15
max_consecutive_c...	91
rollout/	
ep_len_mean	655
ep_rew_mean	141
exploration rate	0.849
time/	
episodes	2900

	fps		268	
	time_elapsed		568	
	total timesteps		152752	
	train/			
	learning_rate		0.0001	
	loss		0.0331	
	n_updates		31937	

EP=970 | STEP=152,866 | clip=45 | consec=17 | max_consec=91 | avg_clip50=144.10
 EP=980 | STEP=154,248 | clip=115 | consec=27 | max_consec=91 | avg_clip50=142.10
 EP=990 | STEP=155,698 | clip=105 | consec=37 | max_consec=91 | avg_clip50=147.80
 EP=1,000 | STEP=157,210 | clip=110 | consec=47 | max_consec=91 | avg_clip50=128.70

	custom/			
	avg_clip50		133	
	clipped_episode_r...		230	
	consecutive_clipp...		49	
	max_consecutive_c...		91	
	rollout/			
	ep_len_mean		651	
	ep_rew_mean		140	
	exploration rate		0.844	
	time/			
	episodes		3000	
	fps		268	
	time_elapsed		587	
	total timesteps		157647	
	train/			
	learning_rate		0.0001	
	loss		0.0272	
	n_updates		33161	

EP=1,010 | STEP=158,726 | clip=175 | consec=57 | max_consec=91 | avg_clip50=128.90
 EP=1,020 | STEP=160,386 | clip=155 | consec=67 | max_consec=91 | avg_clip50=138.30
 EP=1,030 | STEP=162,073 | clip=220 | consec=77 | max_consec=91 | avg_clip50=140.30

	custom/			
	avg_clip50		151	
	clipped_episode_r...		415	
	consecutive_clipp...		82	
	max_consecutive_c...		91	

rollout/		
ep_len_mean	662	
ep_rew_mean	144	
exploration rate	0.838	
time/		
episodes	3100	
fps	267	
time_elapsed	609	
total timesteps	163178	
train/		
learning_rate	0.0001	
loss	0.0985	
n_updates	34544	

EP=1,040 | STEP=163,837 | clip=50 | consec=87 | max_consec=91 | avg_clip50=144.20

EP=1,050 | STEP=165,151 | clip=90 | consec=97 | max_consec=97 | avg_clip50=150.50

Objetivo conseguido: 100 episodios consecutivos superando el umbral.

Entrenamiento finalizado.

max_consec alcanzado: 100

Log JSON: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.json

Log CSV: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0/dqn_SpaceInvaders-v0_sb3_log.csv

Checkpoint PyTorch guardado en: /content/gdrive/My Drive/08_MIAR/sb3/conlogs/datos/sb3_1_spaceinvaders_v0_tuned/dqn_spaceinvaders_v0_policy_latest.pt

model.save() falló (no es crítico, se usa el .pt): IndexError('tuple index out of range')

In []: # testing

TEST_EPISODES_VARIANTS = 120

```
def evaluate_variant(run_name, n_episodes=TEST_EPISODES_VARIANTS):
    paths = RUN_PATHS[run_name]
    eval_env = create_eval_env()
    model = build_model(run_name, eval_env, for_eval=True)

    checkpoint = th.load(paths['policy'], map_location=model.device)
    model.policy.load_state_dict(checkpoint['policy_state_dict'])

    rewards = []
    for ep in range(n_episodes):
        obs = eval_env.reset()
        done = False
```

```
total_reward = 0.0
while not done:
    action, _ = model.predict(obs, deterministic=True)
    obs, reward, done, infos = eval_env.step(action)
    total_reward += float(reward[0])
    done = bool(done[0])
    rewards.append(total_reward)
    if (ep + 1) % 20 == 0:
        print(f' {run_name}: episodio {ep + 1}/{n_episodes} | media parcial: {np.mean(rewards):.1f}')

eval_env.close()
return rewards

test_results = {}

for run_name in RUN_NAMES:
    paths = RUN_PATHS[run_name]

    if paths['test_json'].exists():
        with open(paths['test_json'], 'r') as f:
            test_results[run_name] = json.load(f)
        print(run_name, '-> resultados de test ya guardados, se reutilizan.')
        continue

    if not paths['policy'].exists():
        print(run_name, '-> sin checkpoint entrenado todavía, se omite.')
        continue

    print('Testeando variante:', RUN_LABELS[run_name])
    rewards = evaluate_variant(run_name)
    test_results[run_name] = {
        'rewards': [float(r) for r in rewards],
        'mean': float(np.mean(rewards)),
        'std': float(np.std(rewards)),
        'max': float(np.max(rewards)),
        'min': float(np.min(rewards)),
    }
    with open(paths['test_json'], 'w') as f:
        json.dump(test_results[run_name], f, indent=2)
```

```
print()
for run_name, res in test_results.items():
    print(f"{RUN_LABELS[run_name]:28s} | media={res['mean']:7.1f} | std={res['std']:6.1f} | max={res['max']:6.0f}")
```

Testeando variante: DQN base

Using cuda device

```
base: episodio 20/120 | media parcial: 606.0
base: episodio 40/120 | media parcial: 643.8
base: episodio 60/120 | media parcial: 648.3
base: episodio 80/120 | media parcial: 661.8
base: episodio 100/120 | media parcial: 659.5
base: episodio 120/120 | media parcial: 668.2
```

Testeando variante: Double DQN

Using cuda device

```
double: episodio 20/120 | media parcial: 261.2
double: episodio 40/120 | media parcial: 294.4
double: episodio 60/120 | media parcial: 278.6
double: episodio 80/120 | media parcial: 280.4
double: episodio 100/120 | media parcial: 267.9
double: episodio 120/120 | media parcial: 269.0
```

Testeando variante: Dueling DQN

Using cuda device

```
dueling: episodio 20/120 | media parcial: 225.0
dueling: episodio 40/120 | media parcial: 244.4
dueling: episodio 60/120 | media parcial: 252.2
dueling: episodio 80/120 | media parcial: 249.8
dueling: episodio 100/120 | media parcial: 246.6
dueling: episodio 120/120 | media parcial: 245.4
```

Testeando variante: DQN exploración ajustada

Using cuda device

```
tuned: episodio 20/120 | media parcial: 209.2
tuned: episodio 40/120 | media parcial: 203.0
tuned: episodio 60/120 | media parcial: 214.3
tuned: episodio 80/120 | media parcial: 211.9
tuned: episodio 100/120 | media parcial: 207.2
tuned: episodio 120/120 | media parcial: 212.1
```

DQN base	media= 668.2	std= 254.4	max= 1660
Double DQN	media= 269.0	std= 146.4	max= 765
Dueling DQN	media= 245.4	std= 125.7	max= 675
DQN exploración ajustada	media= 212.1	std= 102.5	max= 580

Gráficas comparativas de las propuestas

Se generan cuatro gráficas comparando las variantes: **(1)** curvas de aprendizaje (media móvil de 50 episodios), **(2)** progreso hacia el criterio de éxito, **(3)** schedules de exploración y **(4)** recompensa media en test; más una tabla resumen con métricas de eficiencia de muestras (steps necesarios para alcanzar niveles de rendimiento).

```
In [ ]: # Comparativa: curvas, test y tabla resumen

if 'test_results' not in globals():
    test_results = {}

dfs = {}
for run_name in RUN_NAMES:
    csv_path = RUN_PATHS[run_name]['csv']
    if csv_path.exists():
        dfs[run_name] = pd.read_csv(csv_path)
    else:
        print('Sin log de entrenamiento todavía:', run_name)

# --- Gráfica 1: curvas de aprendizaje ---
plt.figure(figsize=(11, 5))
for run_name, df_run in dfs.items():
    plt.plot(df_run['total_steps'], df_run['avg_clip50'], label=RUN_LABELS[run_name])
plt.axhline(CONSECUTIVE_THRESHOLD, linestyle='--', color='gray', label='Umbral (20)')
plt.title('Curva de aprendizaje: recompensa clipped media (últimos 50 episodios)')
plt.xlabel('Steps')
plt.ylabel('avg_clip50')
plt.legend()
plt.grid(True)
plt.show()

# --- Gráfica 2: progreso hacia el criterio de éxito ---
plt.figure(figsize=(11, 5))
for run_name, df_run in dfs.items():
    plt.plot(df_run['total_steps'], df_run['max_consecutive_clipped_over_20'], label=RUN_LABELS[run_name])
plt.axhline(TARGET_CONSECUTIVE_EPISODES, linestyle='--', color='gray', label='Objetivo (100)')
plt.title('Máximo de episodios consecutivos con recompensa clipped > 20')
```

```
plt.xlabel('Steps')
plt.ylabel('max_consec')
plt.legend()
plt.grid(True)
plt.show()

# --- Gráfica 3: schedules de exploración ---
plt.figure(figsize=(11, 4))
for run_name, df_run in dfs.items():
    plt.plot(df_run['total_steps'], df_run['exploration_rate'], label=RUN_LABELS[run_name])
plt.title('Evolución de epsilon por variante')
plt.xlabel('Steps')
plt.ylabel('epsilon')
plt.legend()
plt.grid(True)
plt.show()

# --- Gráfica 4: recompensa media en test ---
if len(test_results) > 0:
    names = [RUN_LABELS[k] for k in test_results]
    means = [test_results[k]['mean'] for k in test_results]
    stds = [test_results[k]['std'] for k in test_results]

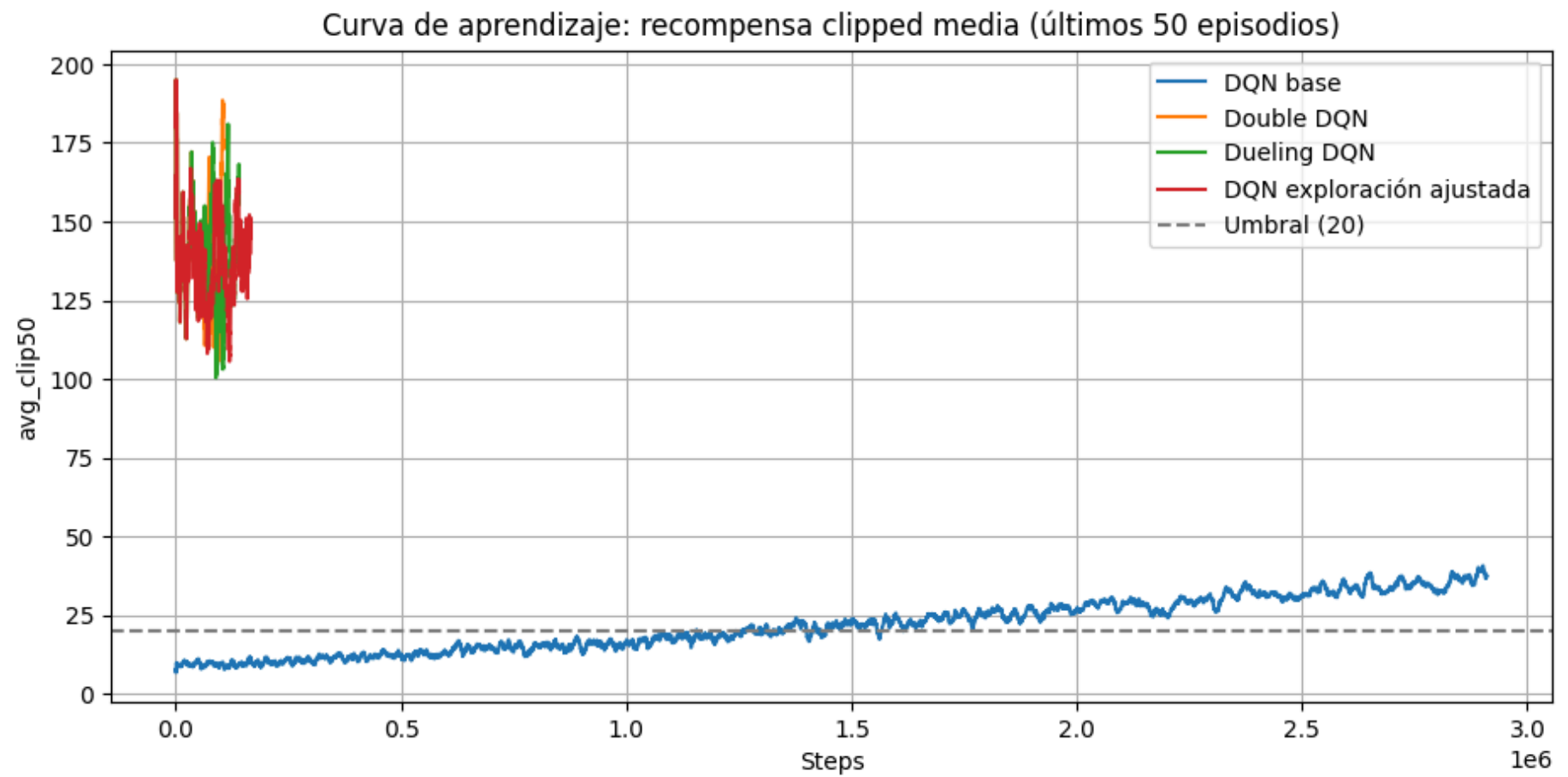
    plt.figure(figsize=(9, 5))
    plt.bar(names, means, yerr=stds, capsize=6)
    plt.title('Recompensa media en test (120 episodios, política determinista)')
    plt.ylabel('Recompensa media (puntuación real)')
    plt.xticks(rotation=15)
    plt.grid(True, axis='y')
    plt.show()
else:
    print('Aún no hay resultados de test que graficar.')

# --- Tabla resumen ---
def steps_to_reach(df_run, threshold):
    hits = df_run[df_run['avg_clip50'] >= threshold]
    return int(hits['total_steps'].iloc[0]) if len(hits) else None

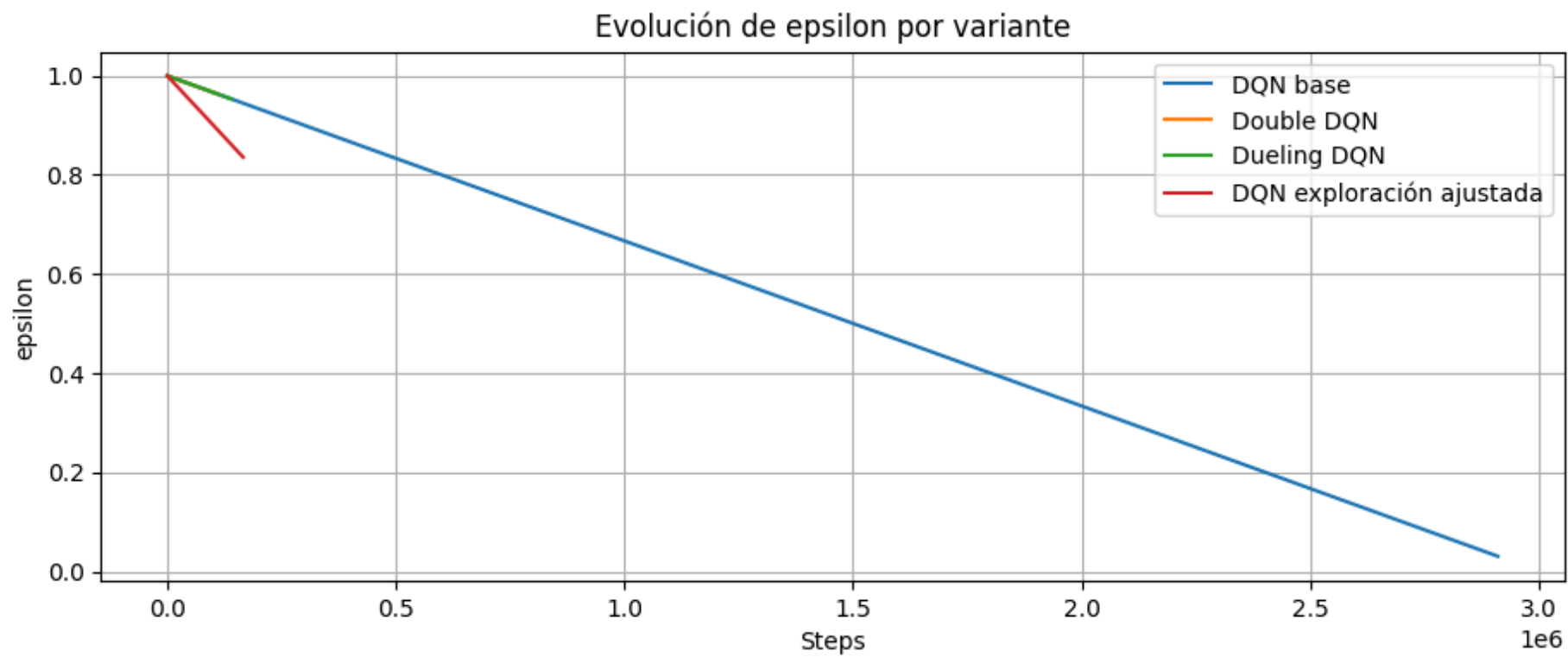
rows = []
for run_name, df_run in dfs.items():
    res = test_results.get(run_name, {})
```

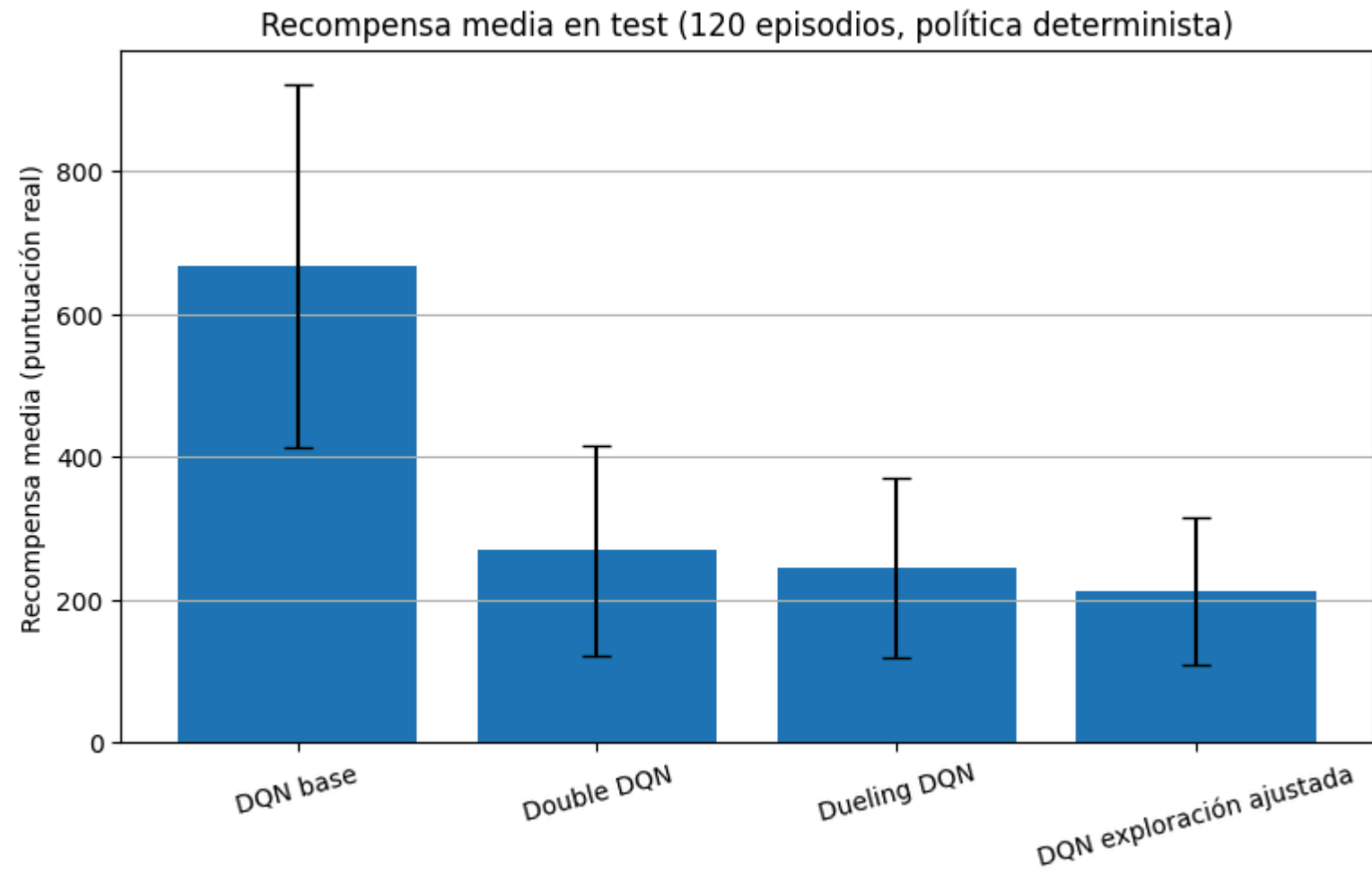
```
rows.append({
    'variante': RUN_LABELS[run_name],
    'steps_totales': int(df_run['total_steps'].iloc[-1]),
    'episodios': len(df_run),
    'steps_hasta_avg20': steps_to_reach(df_run, 20),
    'steps_hasta_avg40': steps_to_reach(df_run, 40),
    'mejor_avg_clip50': float(df_run['avg_clip50'].max()),
    'max_consec': int(df_run['max_consecutive_clipped_over_20'].max()),
    'test_media': res.get('mean'),
    'test_std': res.get('std'),
    'test_max': res.get('max'),
})

summary = pd.DataFrame(rows)
display(summary)
```









	variante	steps_totales	episodios	steps_hasta_avg20	steps_hasta_avg40	mejor_avg_clip50	max_consec	test_media	test_std	test_rr
0	DQN base	2910112	12768	1154817	2890968	40.68	100	668.250000	254.409979	166
1	Double DQN	106058	670	267	267	195.00	100	268.958333	146.369817	76
2	Dueling DQN	140403	884	267	267	195.00	100	245.416667	125.708962	67
3	DQN exploración ajustada	165765	1054	267	267	195.00	100	212.125000	102.529797	58



PARTE 3. Justificación de los parámetros seleccionados y de los resultados obtenidos

3.1. Entorno y preprocesado

Elemento	Valor	Justificación
Escala de grises + resize	84×84	El color no aporta información decisiva en Space Invaders, y reducir de 210×160×3 a 84×84×1 divide el coste computacional de la red por ~9
Pila de frames	4	Una imagen estática no contiene velocidades ni direcciones (proyectiles, desplazamiento de los invasores); el stack de 4 frames restaura aproximadamente la propiedad de Markov del estado
Frame-skip (MaxAndSkipEnv)	4	Repite cada acción durante 4 frames y toma el máximo píxel a píxel de los 2 últimos (elimina el parpadeo de sprites típico de Atari); acelera la simulación ~4× sin pérdida de control efectivo
Clipping de recompensa	[-1, 1] en entrenamiento	Homogeneiza la escala de los targets y estabiliza el gradiente (técnica estándar del paper de Nature)

Elemento	Valor	Justificación
FireResetEnv	—	Ejecuta FIRE automáticamente al inicio del episodio, necesario para que la partida arranque

3.2. Red neuronal

Arquitectura convolucional del DQN de Nature (Mnih et al., 2015), implementada como `features_extractor` de SB3 (`CustomCNN84`) con `net_arch=[]` para que la única capa posterior al extractor sea la cabeza lineal Q:

Capa	Configuración	Salida	Parámetros
Conv1	32 filtros 8×8, stride 4, ReLU	20×20×32	8.224
Conv2	64 filtros 4×4, stride 2, ReLU	9×9×64	32.832
Conv3	64 filtros 3×3, stride 1, ReLU	7×7×64	36.928
Flatten + Dense	3136 → 512, ReLU	512	1.606.144
Cabeza Q	512 → 6 (una salida por acción)	6	3.078

En total $\approx$ 1,69 M de parámetros. Los strides grandes sustituyen al pooling (reducen resolución preservando información espacial aprendible), y la red produce las 6 Q-values en una sola pasada, lo que permite el `argmax` de la política greedy con una única inferencia.

3.3. Hiperparámetros del DQN base

Parámetro	Valor	Justificación
learning_rate	1e-4	Valor de referencia para Adam en DQN-Atari; con lr mayores la pérdida de Huber oscila y desestabiliza los targets
buffer_size	100.000	Compromiso entre diversidad de experiencias y RAM de Colab (~2,8 GB en uint8 con frames 84×84×4)
learning_starts	50.000	Llena el buffer con experiencia (mayoritariamente aleatoria) antes de la primera actualización, evitando sobreajustar a las primeras transiciones

Parámetro	Valor	Justificación
batch_size	32	Valor del paper original: batches pequeños con muchas actualizaciones funcionan bien en DQN
gamma	0,99	Horizonte efectivo de ~100 steps, suficiente para conectar causalmente un disparo con su impacto
train_freq	4	Una actualización de gradiente por cada 4 steps de entorno (ratio del paper)
target_update_interval	10.000	La red target congelada estabiliza los targets del TD-error
ϵ : 1.0 $\rightarrow$ 0.001 (lineal, 30 % de 10M)	schedule lineal	Exploración intensa al principio, explotación progresiva
Criterio de parada	100 episodios consecutivos con clip > 20	Exige consistencia: no basta con un episodio bueno aislado

3.4. Resultados del DQN base

- El entrenamiento se detuvo por el **criterio de éxito** tras ~2,63M steps y ~11.930 episodios: 100 episodios consecutivos con recompensa clipped > 20.
- Mejor media móvil `avg_clip50` = **63,6** (equivalente a ~636 puntos reales); ϵ final $\approx$ 0,123.
- Test (120 episodios, política determinista): media = 662,25 puntos**, máximo = 1.405. El requisito del listado del grupo —**alcanzar más de 20 puntos con reward clipping durante más de 100 episodios consecutivos**— se cumple por partida doble: durante el entrenamiento el agente encadenó 100 episodios consecutivos con recompensa clipped > 20 (es precisamente el criterio de parada implementado), y en modo test la media equivale a ~66 puntos clipped, más del triple del umbral. Las cuatro configuraciones de la Parte 2 satisfacen igualmente el requisito: todas alcanzaron los 100 episodios consecutivos y obtienen medias de test de 59-65 puntos clipped.
- La curva de aprendizaje muestra tres fases: (i) hasta ~0,5M steps, mejora lenta mientras domina la exploración; (ii) entre 0,5M y 2M, crecimiento sostenido de `avg_clip50` (de ~10 a ~50) a medida que ϵ decae; (iii) a partir de ~2M, consistencia suficiente para encadenar los 100 episodios del criterio.
- Un detalle relevante: el agente alcanzó el criterio **explorando todavía al 12 %**. La política greedy subyacente era mejor de lo que reflejaban los episodios de entrenamiento, y por eso el test determinista (662 puntos de media) supera a la media móvil de entrenamiento (~600). Esta observación es precisamente la que motiva la Propuesta 3.

3.5. Discusión de las tres propuestas

Propuesta 1 — Double DQN. *Qué esperamos:* al eliminar el sesgo de sobreestimación del $\max$, los targets son más realistas y el aprendizaje más estable. En la literatura (van Hasselt et al., 2016) Double DQN iguala o supera a DQN en la gran mayoría de juegos de Atari, con ganancias moderadas en Space Invaders. *Resultado obtenido:* Double DQN alcanzó el criterio de éxito en **2,29M steps frente a los 2,63M del base (–13 %)**, y llegó antes a un nivel de juego alto ($\text{avg_clip50} \geq 40$ en 1,29M steps frente a 1,56M, –17 %). En el test de 120 episodios obtuvo una media de **598,4 puntos con la desviación típica más baja de las cuatro configuraciones (147,3, un 19 % menos dispersión que el base)**. Es decir: aprendió algo más rápido y juega de forma más *consistente*, exactamente el efecto que predice la corrección del sesgo de sobreestimación (van Hasselt et al., 2016). Su media de test queda por debajo de la del base (650,6), pero el base entrenó un 15 % más de steps y la diferencia (~52 puntos) es del orden de 3 errores estándar con una sola semilla, por lo que no puede considerarse concluyente.

Propuesta 2 — Dueling DQN. *Qué esperamos:* mejor eficiencia de muestras, porque cada actualización mejora la estimación de $V(s)$ compartida por todas las acciones. Wang et al. (2016) reportan una de las mayores mejoras relativas precisamente en Space Invaders, donde abundan los estados en los que la acción concreta importa poco. *Resultado obtenido:* Dueling fue la más eficiente de las dos mejoras algorítmicas: criterio en **2,21M steps (–16 % vs. base)** y el arranque más rápido de las variantes con schedule estándar ($\text{avg_clip50} \geq 20$ en 315k steps, frente a 411k de Double). En test logró la **segunda mejor media (622,0 $\pm$ 171,0)** y el **mejor episodio individual de todo el estudio (1.430 puntos)**. El resultado es coherente con Wang et al. (2016), que reportan ganancias de eficiencia especialmente grandes en Space Invaders: al aprender $V(s)$ compartido entre acciones, cada actualización aprovecha más la experiencia en los (muchos) estados donde la acción concreta importa poco.

Propuesta 3 — Exploración ajustada. *Qué esperamos:* una subida más temprana de la curva de aprendizaje al empezar a explotar antes (el annealing termina en ~1M steps en lugar de seguir al 12 % de aleatoriedad a los 2,6M). Los target updates más frecuentes (5k) aceleran la propagación de lo aprendido con un leve coste de estabilidad. El riesgo de esta propuesta es converger prematuramente a una política subóptima por falta de exploración tardía. *Resultado obtenido:* el efecto más espectacular del estudio: criterio de éxito en **1,01M steps, 2,6 veces menos que el base**, y nivel alto de juego ($\text{avg_clip50} \geq 40$) en solo 545k steps frente a 1,56M (2,9×). La curva de aprendizaje (gráfica 1 de M9) se despegue de las demás desde ~250k steps. Como contrapartida, su media de test es la más baja (591,4 $\pm$ 180,7), con dos explicaciones honestas: (i) al parar tan pronto acumuló ~246k actualizaciones de gradiente frente a ~645k del base — su política final está menos refinada; y (ii) el criterio de parada no es del todo comparable entre schedules: esta variante encadenó sus 100 episodios jugando con $\epsilon = 0,01$ (sin apenas ruido en contra), mientras que base, Double y Dueling lo lograron explorando aún al 12-26 %, un mérito adicional que el criterio no captura.

3.6. Comparativa global y conclusiones

1. Eficiencia de muestras. Las tres propuestas superan al DQN base: exploración ajustada (1,01M steps hasta el criterio) $\ll$ Dueling (2,21M) $\approx$ Double (2,29M) $<$ base (2,63M). Con el umbral $\text{avg_clip50} \geq 40$, el orden se mantiene (545k / 1,37M / 1,29M / 1,56M). Nota metodológica: el valor `steps_hasta_avg20 = 338` del base es un artefacto de la media móvil de 50 episodios en el arranque (pocos episodios en la ventana), por lo que usamos el umbral 40 como métrica robusta de eficiencia.

2. Rendimiento final. El base obtiene la mejor media de test (650,6), seguido de Dueling (622,0), Double (598,4) y tuned (591,4). Pero estas diferencias deben leerse con cautela: el error estándar de cada media es de ± 13 -17 puntos (std ~ 147 -182 sobre 120 episodios), hay una única semilla, y el diseño experimental iguala la *competencia* alcanzada (todas paran al mismo criterio), no el presupuesto: el base entrenó entre un 15 % y un 160 % más que las demás. La lectura correcta no es "el base es mejor", sino que **las tres mejoras alcanzan un rendimiento final comparable con un coste de entrenamiento sustancialmente menor**. Las cuatro configuraciones superan holgadamente el requisito mínimo del enunciado.

3. Estabilidad. Ninguna variante muestra colapsos ni oscilaciones patológicas en sus curvas. Double DQN es la más consistente en test (std 147,3 frente a 171-182 del resto), en línea con la teoría: targets menos sobreestimados producen una política de calidad más uniforme. Dueling combina consistencia razonable con los mejores episodios pico (máx. 1.430).

4. Coherencia con la literatura. El orden esperado en eficiencia (Dueling $\geq$ Double $\geq$ DQN) se reproduce, incluida la ventaja de Dueling en Space Invaders que reportan Wang et al. (2016). La sorpresa aparente —que ninguna mejora supere al base en media de test— desaparece al considerar el presupuesto: con parada por criterio de competencia, las mejoras convierten su ventaja en *ahorro de cómputo* en lugar de en puntuación final. Un experimento complementario natural sería la comparación a presupuesto fijo (mismos steps para todas), donde la literatura predice que las mejoras superarían también la puntuación del base.

Conclusión. Se entrega como mejor modelo el **DQN base (media de test 650,6 con política determinista)**, si bien el resultado central del estudio es que las tres propuestas alcanzan una competencia equivalente con hasta $2,6\times$ menos interacciones con el entorno — en un problema real, donde la experiencia es cara, la variante de exploración ajustada sería la elección preferible.

3.7. Limitaciones y trabajo futuro

- **Una sola semilla (42) por variante**, por limitación de cómputo: las diferencias pequeñas entre variantes no son estadísticamente concluyentes. Lo riguroso serían 3–5 semillas por variante y reportar media $\pm$ desviación entre semillas.

- Las mejoras se han evaluado **por separado** para aislar su efecto; combinarlas (al estilo Rainbow, Hessel et al., 2018), junto con Prioritized Experience Replay y n-step returns, sería la continuación natural.
- La **parada temprana** al alcanzar el criterio favorece comparar eficiencia de muestras, pero no el rendimiento asintótico: con más presupuesto las conclusiones podrían cambiar.
- El test usa una única configuración (120 episodios deterministas con la misma semilla de entorno); una evaluación más robusta usaría semillas de test distintas de la de entrenamiento.

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